

DEPARTMENT OF INFORMATION TECHNOLOGY

SMT. PARMESHWARIDEVI DURGADUTT TIBREWALA

LIONS JUHU COLLEGE

OF ARTS, COMMERE AND SCIENCE

Affiliated to University of Mumbai

(Autonomous)

J.B. NAGAR, ANDHERI (E), MUMBAI-400059



Academic Year 2025-2026

DATA SCIENCE

For

Semester I

Submitted By:

Mr. Ahmed Dilshad Shaikh

Msc.IT (Sem I)

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Certificate of Approval

This is to certify that practical entitled **“DATA SCIENCE”**, Undertaken at **SMT.PARMESHWARIDEVI DURGADUTT TIBREWALA LIONS JUHU COLLEGE OF ARTS, COMMERECE & SCIENCE.**

By **Mr. Ahmed Dilshad Shaikh** Seat No. **MIT251009** in partial fulfilment of **M.Sc. (IT) master degree (Semester I)** Examination had not been submitted for any other examination and does not form of any other course undergone by the candidate. It is further certified that he has completed all required phases of the practical.

Internal Examiner

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**Signature/
Principal/Stamp**

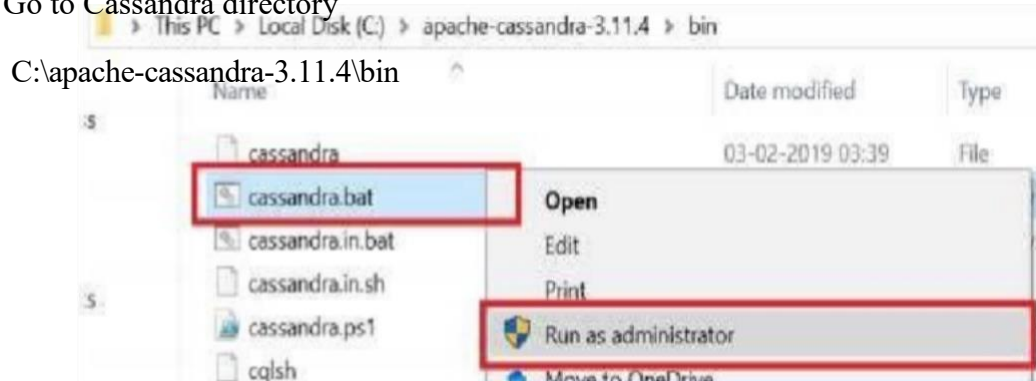
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Practical 1:

Creating Data Model using Cassandra.

Go to Cassandra directory



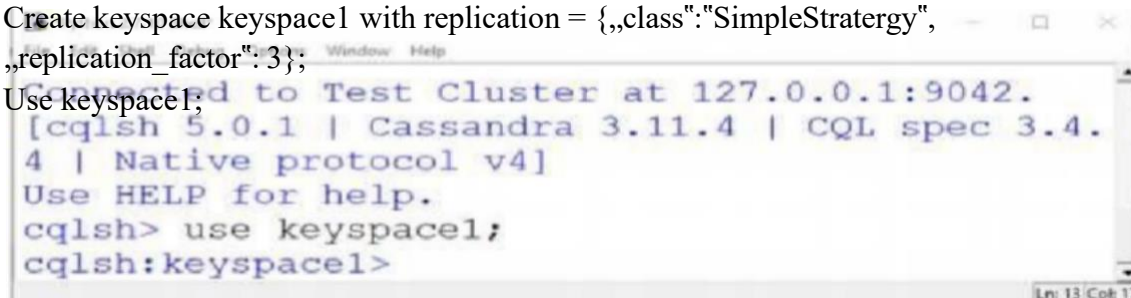
Run Cassandra.bat file

Open C:\apache-cassandra-3.11.4\bin\cqlsh.py with python 2.7 and run

Creating a Keyspace using Cqlsh

Create keyspace keyspace1 with replication = {„class”:„SimpleStrategy”, „replication_factor”: 3};

Use keyspace1;



Create table dept (dept_id int PRIMARY KEY, dept_name text, dept_loc text); Create table emp (emp_id int PRIMARY KEY, emp_name text, dept_id int, email text, phone text);

Insert into dept (dept_id, dept_name, dept_loc) values (1001, 'Accounts', 'Mumbai');

Insert into dept (dept_id, dept_name, dept_loc) values (1002, 'Marketing', 'Delhi');

Insert into dept (dept_id, dept_name, dept_loc) values (1003, 'HR', 'Chennai');

Insert into emp (emp_id, emp_name, dept_id, email, phone) values (1001, 'ABCD', 1001, 'abcd@company.com', '1122334455');

Insert into emp (emp_id, emp_name, dept_id, email, phone) values (1002, 'DEFG', 1001, 'defg@company.com', '2233445566');

Insert into emp (emp_id, emp_name, dept_id, email, phone) values (1003, 'GHIJ', 1002, 'ghij@company.com', '3344556677');

Insert into emp (emp_id, emp_name, dept_id, email, phone) values (1004, 'JKLM', 1002, 'jklm@company.com', '4455667788');

Insert into emp (emp_id, emp_name, dept_id, email, phone) values (1005, 'MNOP', 1003, 'mnop@company.com', '5566778899');

Insert into emp (emp_id, emp_name, dept_id, email, phone) values (1006, 'MNOP', 1003, 'mnop@company.com', '5566778844');

```
cqlsh:keyspace1> select * from emp;
```

emp_id	dept_id	email	emp_name	phone
1006	1003	mnop@company.com	MNOP	5566778844
1004	1002	jklm@company.com	JKLM	4455667788
1005	1003	mnop@company.com	MNOP	5566778899
1001	1001	abcd@company.com	ABCD	1122334455
1003	1002	ghij@company.com	GHIJ	3344556677
1002	1001	defg@company.com	DEFG	2233445566

(6 rows)

```
cqlsh:keyspace1> select * from dept;
```

dept_id	dept_loc	dept_name
1001	Mumbai	Accounts
1003	Chennai	HR
1002	Delhi	Marketing

(3 rows)

update dept set dept_name='Human Resource' where dept_id=1003;

```
cqlsh:keyspace1> select * from dept;
```

dept_id	dept_loc	dept_name
1001	Mumbai	Accounts
1003	Chennai	Human Resource
1002	Delhi	Marketing

(3 rows)

```
cqlsh:keyspace1> delete from emp where emp_id=1006;
```

```
cqlsh:keyspace1> select * from emp;
```

emp_id	dept_id	email	emp_name	phone
1004	1002	jklm@company.com	JKLM	4455667788
1005	1003	mnop@company.com	MNOP	5566778899
1001	1001	abcd@company.com	ABCD	1122334455
1003	1002	ghij@company.com	GHIJ	3344556677
1002	1001	defg@company.com	DEFG	2233445566

(5 rows)

Practical 2

Write Python / R Program to convert from the following formats to HORUS format:

A. Text delimited CSV to HORUS format.

```
# Utility Start CSV to HORUS =====
# Standard Tools import
pandas as pd
# Input Agreement =====
sInputFileName='C:/VKHCG/05-DS/9999-Data/Country_Code.csv'
InputData=pd.read_csv(sInputFileName,encoding="latin-1") print('Input
Data Values =====')
print(InputData)
print('=====')
# Processing Rules =====
ProcessData=InputData
# Remove columns ISO-2-Code and ISO-3-CODE
ProcessData.drop('ISO-2-CODE', axis=1,inplace=True)
ProcessData.drop('ISO-3-Code', axis=1,inplace=True)
# Rename Country and ISO-M49
ProcessData.rename(columns={'Country': 'CountryName'}, inplace=True)
ProcessData.rename(columns={'ISO-M49': 'CountryNumber'}, inplace=True)
# Set new Index
ProcessData.set_index('CountryNumber', inplace=True)
# Sort data by CurrencyNumber
ProcessData.sort_values('CountryName', axis=0, ascending=False, inplace=True)
print('Process Data Values =====') print(ProcessData)
print('=====')
# Output Agreement =====
OutputData=ProcessData
sOutputFileName='C:/VKHCG/05-DS/9999-Data/HORUS-CSV-Country.csv'
OutputData.to_csv(sOutputFileName, index = False) print('CSV to HORUS -
Done')
# Utility done =====
```

OUTPUT :-



CountryNumber	CountryName
1	United States of America
2	China
3	India
4	Germany
5	France
6	United Kingdom
7	Italy
8	Spain
9	Japan
10	South Korea
11	Canada
12	Brazil
13	Mexico
14	Argentina
15	Colombia
16	Venezuela
17	Peru
18	Ecuador
19	Bolivia
20	Paraguay
21	Uruguay
22	Chile
23	Costa Rica
24	Panama
25	Dominican Republic
26	Honduras
27	Nicaragua
28	El Salvador
29	Guatemala
30	Belize
31	Trinidad and Tobago
32	Jamaica
33	Barbados
34	Suriname
35	Guayana Francesa
36	Aruba
37	Curaçao
38	Trinidad y Tobago
39	Jamaica
40	Barbados
41	Suriname
42	Guayana Francesa
43	Aruba
44	Curaçao
45	Trinidad y Tobago
46	Jamaica
47	Barbados
48	Suriname
49	Guayana Francesa
50	Aruba

B. XML to HORUS Format

```
# Utility Start XML to HORUS =====
# Standard Tools import
pandas as pd
import xml.etree.ElementTree as ET def
df2xml(data):
    header = data.columns root =
    ET.Element('root') for row in
    range(data.shape[0]): entry =
    ET.SubElement(root,'entry') for
    index in range(data.shape[1]):
        schild=str(header[index]) child =
        ET.SubElement(entry, schild) if
        str(data[schild][row]) != 'nan':
            child.text = str(data[schild][row])
        else:
            child.text = 'n/a'
    entry.append(child)
    result = ET.tostring(root)
    return result def
xml2df(xml_data):
    root = ET.XML(xml_data)
    all_records = [] for i, child in
    enumerate(root):
        record = {} for
        subchild in child:
            record[subchild.tag] = subchild.text
        all_records.append(record)

    return pd.DataFrame(all_records)
sInputFileName='C:/VKHCG/05-DS/9999-Data/Country_Code.xml' InputData
= open(sInputFileName).read()
print('=====') print('Input
Data Values =====')
print('=====')
print(InputData)
print('=====')
#=====
=
# Processing Rules =====
#=====
=
ProcessDataXML=InputData
# XML to Data Frame
ProcessData=xml2df(ProcessDataXML)
# Remove columns ISO-2-Code and ISO-3-CODE
ProcessData.drop('ISO-2-CODE', axis=1,inplace=True)
ProcessData.drop('ISO-3-Code', axis=1,inplace=True)
```

```

# Rename Country and ISO-M49
ProcessData.rename(columns={'Country': 'CountryName'}, inplace=True)
ProcessData.rename(columns={'ISO-M49': 'CountryNumber'}, inplace=True)
# Set new Index
ProcessData.set_index('CountryNumber', inplace=True)
# Sort data by CountryNumber
ProcessData.sort_values('CountryName', axis=0, ascending=False, inplace=True)
print('=====')
print('Process Data Values =====')
print('=====')
print(ProcessData)
print('=====')
OutputData=ProcessData
sOutputFileName='C:/VKHCG/05-DS/9999-Data/HORUS-XML-Country.csv'
OutputData.to_csv(sOutputFileName, index = False)
print('=====') print('XML
to HORUS - Done')
print('=====')
# Utility done =====

```

Output:

```

===== RESTART: C:\VKHCG\05-DS\9999-Data\XML2HORUS.py =====
Input Data Values =====
Squeezed text (385 lines)
=====
Process Data Values =====
=====
CountryNumber      CountryName
716                Zimbabwe
894                Zambia
887                Yemen
732                Western Sahara
876                Wallis and Futuna Islands
...
16                American Samoa
12                Algeria
8                 Albania
248               Aland Islands
4                 Afghanistan

[247 rows x 1 columns]
=====
XML to HORUS - Done
=====
>>>

```


Practical 3

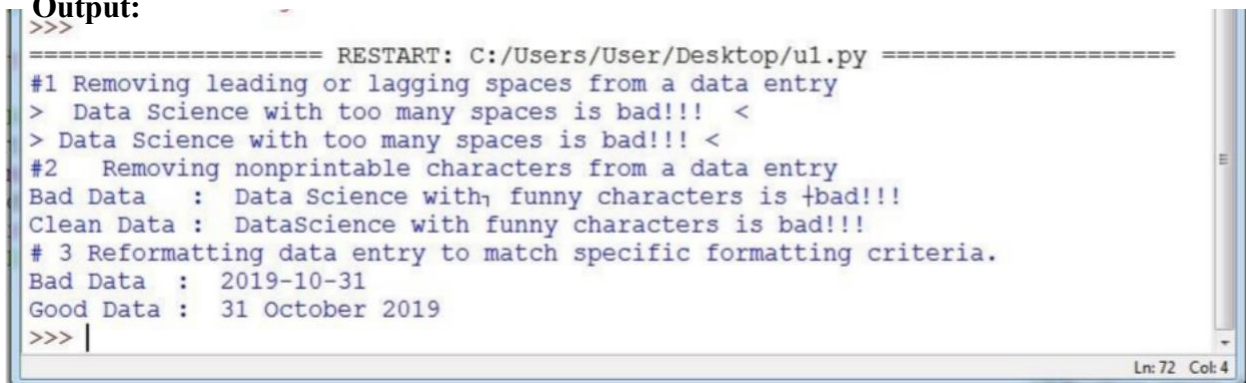
Utilities and Auditing

A. Fixers Utilities:

Fixers enable your solution to take your existing data and fix a specific quality issue.

```
# ----- Program to Demonstrate Fixers utilities -----import
string import datetime as dt
# 1 Removing leading or lagging spaces from a data entry
print('#1 Removing leading or lagging spaces from a data
entry'); baddata = " Data Science with too many spaces is bad!!!"
" print('>',baddata,<') cleandata=baddata.strip()
print('>',cleandata,<')
# 2 Removing nonprintable characters from a data entry
print('#2 Removing nonprintable characters from a data entry')
printable = set(string.printable) baddata = "Data\x00Science
with\x02 funny characters is \x10bad!!!"
cleandata="".join(filter(lambda x: x in string.printable,baddata))
print('Bad Data : ',baddata); print('Clean Data : ',cleandata)
# 3 Reformatting data entry to match specific formatting criteria.
# Convert YYYY/MM/DD to DD Month YYYY
print('# 3 Reformatting data entry to match specific formatting criteria.')
baddate = dt.date(2019, 10, 31) baddata=format(baddate,'%Y-%m-%d')
gooddate = dt.datetime.strptime(baddata,'%Y-%m-%d')
gooddata=format(gooddate,'%d %B %Y')
print('Bad Data : ',baddata) print('Good
Data : ',gooddata)
```

Output:



```
>>>
===== RESTART: C:/Users/User/Desktop/ul.py =====
#1 Removing leading or lagging spaces from a data entry
> Data Science with too many spaces is bad!!! <
> Data Science with too many spaces is bad!!! <
#2 Removing nonprintable characters from a data entry
Bad Data : Data Science with funny characters is bad!!!
Clean Data : DataScience with funny characters is bad!!!
# 3 Reformatting data entry to match specific formatting criteria.
Bad Data : 2019-10-31
Good Data : 31 October 2019
>>> |
```

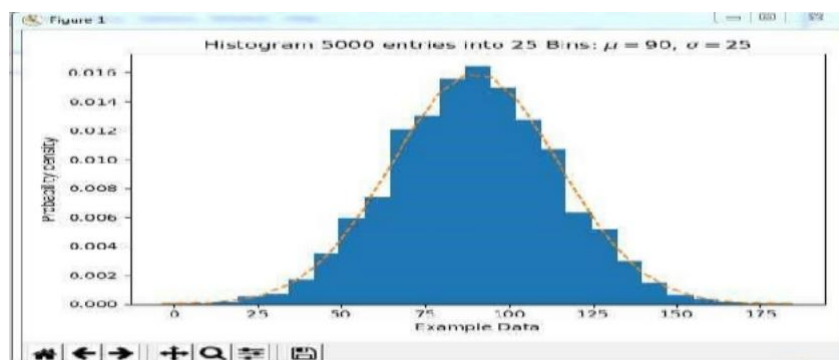
B. Data Binning or Bucketing

Binning is a data preprocessing technique used to reduce the effects of minor observation errors. Statistical data binning is a way to group a number of more or less continuous values into a smaller number of “bins.”

Code :

```
import numpy as np
import matplotlib.mlab as mlab
import matplotlib.pyplot as plt
import scipy.stats as stats
np.random.seed(0)
# example data
mu = 90 # mean of distribution
sigma = 25 # standard deviation of distribution
x = mu + sigma * np.random.randn(5000)
num_bins = 25
fig, ax = plt.subplots() # the histogram of the data
n, bins, patches = ax.hist(x, num_bins, density=1)
# add a 'best fit' line
y = stats.norm.pdf(bins, mu, sigma) # mlab.normpdf(bins, mu, sigma)
ax.plot(bins, y, '--')
ax.set_xlabel('Example Data')
ax.set_ylabel('Probability density')
sTitle=r'Histogram ' + str(len(x)) + ' entries into ' + str(num_bins) + ' Bins:  $\mu =$ ' + str(mu) + '$,  $\sigma =$ ' + str(sigma) + '$'
ax.set_title(sTitle)
fig.tight_layout()
sPathFig='C:/VKHCG/05-DS/4000-UL/0200-DU/DU-Histogram.png'
fig.savefig(sPathFig)
plt.show()
```

Output:



Practical 4

A. Perform the following data processing using R.

Use R-Studio for the following:

```
>library(readr)
```

Warning message:package „readr" was built under R version 3.4.4

Load a table named IP_DATA_ALL.csv.

```
>IP_DATA_ALL <- read_csv("C:/VKHCG/01-  
Vermeulen/00RawData/IP_DATA_ALL.csv")
```

 Parsed with column specification:

```
cols(
```

```
ID = col_double(),
```

```
Country = col_character(),
```

```
`Place Name` = col_character(),
```

```
`Post Code` = col_double(),
```

```
Latitude = col_double(),
```

```
Longitude = col_double(),
```

```
`First IP Number` = col_double(),
```

```
`Last IP Number` = col_double()
```

```
)
```

```
>View(IP_DATA_ALL)
```

```
>spec(IP_DATA_ALL)
```

```
cols(
```

```
ID = col_double(),
```

```
Country = col_character(),
```

```
`Place Name` = col_character(),
```

```
`Post Code` = col_double(),
```

```
Latitude = col_double(),
```

```
Longitude = col_double(),
```

```
`First IP Number` = col_double(),
```

```
Last IP Number` = col_double()
```

```
)
```

This informs you that you have the following eight columns:

- ID of type integer
- Place name of type character
- Post code of type character
- Latitude of type numeric double
- Longitude of type numeric double
- First IP number of type integer
- Last IP number of type integer

```
>library(tibble)
```

```
>set_tidy_names(IP_DATA_ALL, syntactic = TRUE, quiet = FALSE)
```

 New names:

```
Place Name -> Place.Name
```

```
Post Code -> Post.Code
```

```
First IP Number -> First.IP.Number
```

```
Last IP Number -> Last.IP.Number
```

This informs you that four of the field names are not valid and suggests new field names that are valid. You can fix any detected invalid column names by executing

`IP_DATA_ALL_FIX=set_tidy_names(IP_DATA_ALL, syntactic = TRUE, quiet = TRUE)` By using command `View(IP_DATA_ALL_FIX)`, you can check that you have fixed the columns. The new table `IP_DATA_ALL_FIX.csv` will fix the invalid column names with valid names.

```
>apply(IP_DATA_ALL_FIX, typeof)
```

```
ID Country Place.Name Post.Code Latitude
```

```
"double" "character" "character" "double" "double"
```

```
Longitude First.IP.Number Last.IP.Number
```

```
"double" "double" "double"
```

```
>library(data.table)
```

```
>hist_country=data.table(Country=unique(IP_DATA_ALL_FIX[is.na(IP_DATA_ALL_FIX  
['Country']) == 0, ]$Country))
```

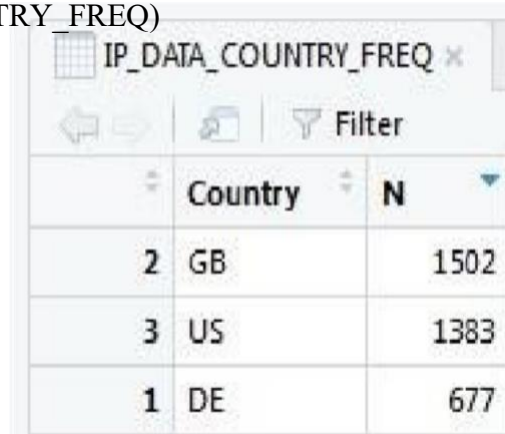
```
>setorder(hist_country,'Country')
```

```
>hist_country_with_id=rowid_to_column(hist_country, var = "RowIDCountry")
```

```
>View(hist_country_fix)
```

```
>IP_DATA_COUNTRY_FREQ=data.table(with(IP_DATA_ALL_FIX, table(Country)))
```

```
>View(IP_DATA_COUNTRY_FREQ)
```



	Country	N
2	GB	1502
3	US	1383
1	DE	677

- The two biggest subset volumes are from the US and GB.

- The US has just over four times the data as GB.

```
hist_latitude =data.table(Latitude=unique(IP_DATA_ALL_FIX  
[is.na(IP_DATA_ALL_with_ID ['Latitude']) == 0, ]$Latitude))
```

```
setkeyv(hist_latitude, 'Latitude') setorder(hist_latitude)
```

```
hist_latitude_with_id=rowid_to_column(hist_latitude, var = "RowID")
```

```
View(hist_latitude_with_id)
```

```
IP_DATA_Latitude_FREQ=data.table(with(IP_DATA_ALL_FIX,table(Latitude)))
```

```
View(IP_DATA_Latitude_FREQ)
```

- The two biggest data volumes are from latitudes 51.5092 and 40.6888.

- The spread appears to be nearly equal between the top-two latitudes.

```
>apply(IP_DATA_ALL_FIX[, 'Latitude'], min, na.rm=TRUE)
```

```
Latitude 40.6888
```

What does this tell you?

Fact: The range of latitude for the Northern Hemisphere is from 0 to 90. So, if you do not have any latitudes farther south than 40.6888, you can improve your retrieve routine.

```
>apply(IP_DATA_ALL_FIX[, 'Country'], min, na.rm=TRUE)
```

```
Country "DE"
```

Minimum business frequency is from DE – Denmark.

```
>sapply(IP_DATA_ALL_FIX['Latitude'], max, na.rm=TRUE) Latitude
```

```
51.5895
```

```
>sapply(IP_DATA_ALL_FIX['Country'], max, na.rm=TRUE)
```

```
Country
```

```
"US"
```

The result is 51.5895. What does this tell you?

Fact: The range in latitude for the Northern Hemisphere is from 0 to 90. So, if you do not have any latitudes more northerly than 51.5895, you can improve your retrieve routine.

```
>sapply(IP_DATA_ALL_FIX ['Latitude'], mean, na.rm=TRUE)
```

```
Latitude
```

```
46.69097
```

```
>sapply(IP_DATA_ALL_FIX ['Latitude'], median, na.rm=TRUE) Latitude
```

```
48.15
```

```
>sapply(IP_DATA_ALL_FIX ['Latitude'], range, na.rm=TRUE)
```

```
Latitude
```

```
[1,] 40.6888 [2,]
```

```
51.5895
```

```
>sapply(IP_DATA_ALL_FIX ['Latitude'], quantile, na.rm=TRUE) Latitude
```

```
0% 40.6888
```

```
25% 40.7588 50%
```

```
48.1500 75%
```

```
51.5092
```

```
100% 51.5895
```

```
>sapply(IP_DATA_ALL_FIX ['Latitude'], sd, na.rm=TRUE)
```

```
Latitude
```

```
4.890387
```

```
>sapply(IP_DATA_ALL_FIX ['Longitude'], sd, na.rm=TRUE)
```

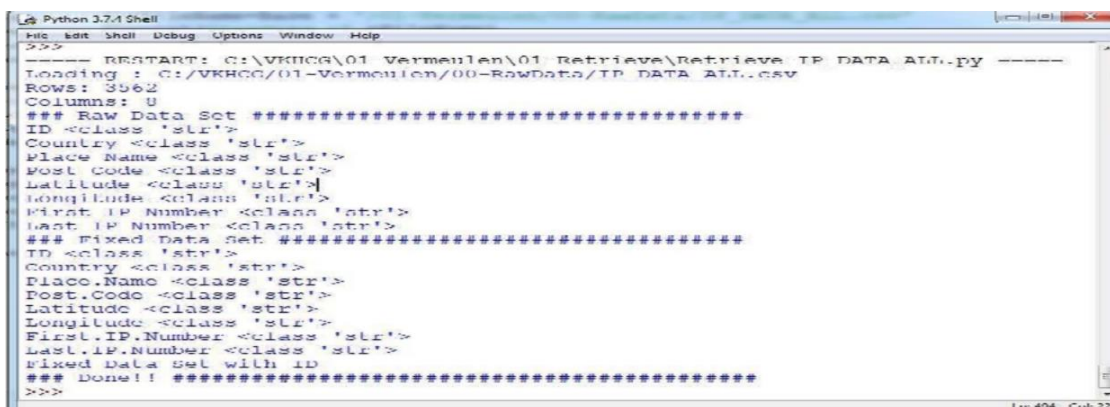
```
Longitude
```

```
38.01702
```

B. Program to retrieve different attributes of data.

```
##### C:\VKHCG\01-Vermeulen\01-Retrieve\Retrive_IP_DATA_ALL.py###
import sys import os
import pandas as pd
#####
Base='C:/VKHCG'
#####
sFileName=Base      +      '/01-Vermeulen/00-RawData/IP_DATA_ALL.csv'
print('Loading :',sFileName)
IP_DATA_ALL=pd.read_csv(sFileName,header=0,low_memory=False,encoding="latin-1")
#####
sFileDir=Base + '/01-Vermeulen/01-Retrieve/01-EDS/02-Python' if
not os.path.exists(sFileDir):
os.makedirs(sFileDir) print('Rows:', IP_DATA_ALL.shape[0])
print('Columns:', IP_DATA_ALL.shape[1]) print('### Raw Data Set
#####') for i in
range(0,len(IP_DATA_ALL.columns)):
print(IP_DATA_ALL.columns[i],type(IP_DATA_ALL.columns[i]))
print('### Fixed Data Set #####')
IP_DATA_ALL_FIX=IP_DATA_ALL
for i in range(0,len(IP_DATA_ALL.columns)):

cNameOld=IP_DATA_ALL_FIX.columns[i] + ''
cNameNew=cNameOld.strip().replace(" ", ".") IP_DATA_ALL_FIX.columns.values[i]
= cNameNew
print(IP_DATA_ALL_FIX.columns[i],type(IP_DATA_ALL_FIX.columns[i]))
#####
#print(IP_DATA_ALL_FIX.head())
##### print('Fixed
Data Set with ID')
IP_DATA_ALL_with_ID=IP_DATA_ALL_FIX
IP_DATA_ALL_with_ID.index.names = ['RowID']
#print(IP_DATA_ALL_with_ID.head()) sFileName2=sFileDir
+ '/Retrieve_IP_DATA.csv'
IP_DATA_ALL_with_ID.to_csv(sFileName2, index = True, encoding="latin-1")
##### print('###
Done!! #####')
#####
```



```
Python 3.7.4 Shell
File Edit Shell Debug Options Window Help
>>>
----- RESTART: C:\VKHCG\01 Vermeulen\01 Retrieve\Retrieve IP DATA ALL.py -----
Loading : C:/VKHCG/01-Vermeulen/00-RawData/IP_DATA_ALL.csv
Rows: 3562
Columns: 8
### Raw Data Set #####
ID <class 'str'>
Country <class 'str'>
Place Name <class 'str'>
Post Code <class 'str'>
Latitude <class 'str'>
Longitude <class 'str'>
First IP Number <class 'str'>
Last IP Number <class 'str'>
### Fixed Data Set #####
ID <class 'str'>
Country <class 'str'>
Place Name <class 'str'>
Post Code <class 'str'>
Latitude <class 'str'>
Longitude <class 'str'>
First IP Number <class 'str'>
Last IP Number <class 'str'>
Fixed Data Set with ID
### Done!! #####
>>>
```

PRACTICAL 05

Assessing Data

Assess Superstep

Data quality refers to the condition of a set of qualitative or quantitative variables. Data quality is a multidimensional measurement of the acceptability of specific data sets. In business, data quality is measured to determine whether data can be used as a basis for reliable intelligence extraction for supporting organizational decisions. Data profiling involves observing in your data sources all the viewpoints that the information offers. The main goal is to determine if individual viewpoints are accurate and complete. The Assess superstep determines what additional processing to apply to the entries that are noncompliant.

Errors

Typically, one of four things can be done with an error to the data.

1. Accept the Error
2. Reject the Error
3. Correct the Error
4. Create a Default Value

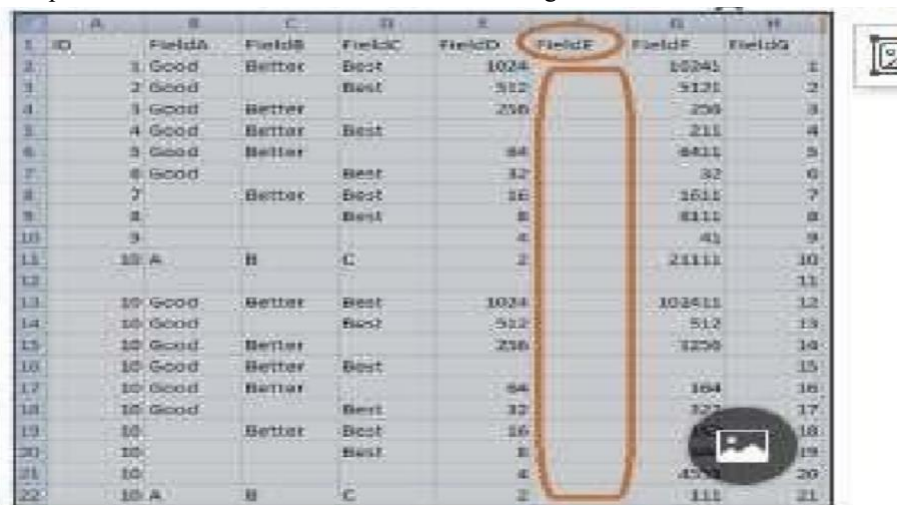
A. Perform error management on the given data using pandas package.

Python pandas package enables several automatic error-management features.

File Location: C:\VKHCG\01-Vermeulen\02-Assess

Missing Values in Pandas:

- i. Drop the Columns Where All Elements Are Missing Values



ID	FieldA	FieldB	FieldC	FieldD	FieldE	FieldF	FieldG
1	1	Good	Better	Best	1024	10241	1
2	2	Good	Better	Best	512	5121	2
3	3	Good	Better	Best	256	2561	3
4	4	Good	Better	Best	128	1281	4
5	5	Good	Better	Best	64	6411	5
6	6	Good	Better	Best	32	32	6
7	7	Better	Best	16	1611	7	7
8	8	Better	Best	8	8111	8	8
9	9	Better	Best	4	41	9	9
10	10	A	B	C	2	21111	10
11	11	A	B	C	2	21111	11
12	12	Good	Better	Best	1024	102411	12
13	13	Good	Better	Best	512	512	13
14	14	Good	Better	Best	256	256	14
15	15	Good	Better	Best	128	128	15
16	16	Good	Better	Best	64	64	16
17	17	Good	Better	Best	32	32	17
18	18	Good	Better	Best	16	16	18
19	19	Good	Better	Best	8	8	19
20	20	Good	Better	Best	4	4	20
21	21	Good	Better	Best	2	2	21
22	22	A	B	C	2	111	21

Code :

```
##### Assess-Good-Bad-01.py#####
# -*- coding: utf-8 -*-
#####
import sys import os import pandas as pd
#####
Base='C:/VKHCG'
#####
print('#####')
print('Working Base :',Base, ' using ', sys.platform)
print('#####')
#####
sInputFileName='Good-or-Bad.csv'
sOutputFileName='Good-or-Bad-01.csv'Company='01-Vermeulen'
#####
Base='C:/VKHCG'
#####
sFileDir=Base + '/' + Company + '/02-Assess/01-EDS/02-Python' if not
os.path.exists(sFileDir):
os.makedirs(sFileDir)
#####
### Import Warehouse
#####
sFileName=Base + '/' + Company + '/00-RawData/' + sInputFileName
print('Loading :',sFileName)
RawData=pd.read_csv(sFileName,header=0)
print('#####')
print('## Raw Data Values')
print('#####') print(RawData)
print('#####')
print('## Data Profile')
print('#####')
print('Rows :',RawData.shape[0]) print('Columns
:',RawData.shape[1])
print('#####')
#####
sFileName=sFileDir + '/' + sInputFileName RawData.to_csv(sFileName,
index = False)
#####
TestData=RawData.dropna(axis=1, how='all')
#####
print('#####')
print('## Test Data Values') print('#####')
print(TestData)
print('#####') print('##
Data Profile')
print('#####')
print('Rows :',TestData.shape[0]) print('Columns
:',TestData.shape[1])
print('#####')
#####
sFileName=sFileDir + '/' + sOutputFileName
TestData.to_csv(sFileName, index = False)
#####
```



```

print('#####') print('###
Done!! #####')
print('#####')
#####
Output:
>>>
===== RESTART: C:\VKHCG\01-Vermeulen\02-Assess\Assess-Good-Bad-01.py
=====
#####
Working Base : C:\VKHCG using win32
#####
Loading : C:\VKHCG\01-Vermeulen\00-RawData\Good-or-Bad.csv
#####
## Raw Data Values
#####
ID FieldA FieldB FieldC FieldD FieldE FieldF FieldG
0 1.0 Good Better Best 1024.0 NaN 10241.0 1
1 2.0 Good NaN Best 512.0 NaN 5121.0 2
2 3.0 Good Better NaN 256.0 NaN 256.0 3
3 4.0 Good Better Best NaN NaN 211.0 4
4 5.0 Good Better NaN 64.0 NaN 6411.0 5
5 6.0 Good NaN Best 32.0 NaN 32.0 6
6 7.0 NaN Better Best 16.0 NaN 1611.0 7
7 8.0 NaN NaN Best 8.0 NaN 8111.0 8
8 9.0 NaN NaN NaN 4.0 NaN 41.0 9
9 10.0 A B C 2.0 NaN 21111.0 10
10 NaN NaN NaN NaN NaN NaN NaN 11
11 10.0 Good Better Best 1024.0 NaN 102411.0 12
12 10.0 Good NaN Best 512.0 NaN 512.0 13
13 10.0 Good Better NaN 256.0 NaN 1256.0 14
14 10.0 Good Better Best NaN NaN NaN 15
15 10.0 Good Better NaN 64.0 NaN 164.0 16
16 10.0 Good NaN Best 32.0 NaN 322.0 17
17 10.0 NaN Better Best 16.0 NaN 163.0 18
18 10.0 NaN NaN Best 8.0 NaN 844.0 19
19 10.0 NaN NaN NaN 4.0 NaN 4555.0 20
20 10.0 A B C 2.0 NaN 111.0 21
All of column E has been deleted, owing to the fact that all values in that column were missing values/errors.
ii. Drop the Columns Where Any of the Elements Is Missing Values
##### Assess-Good-Bad-02.py#####
import sys import os import pandas as pd Base='C:\VKHCG'
sInputFileName='Good-or-Bad.csv' sOutputFileName='Good-or-Bad-02.csv'
Company='01-Vermeulen'
#####
Base='C:\VKHCG'
#####
print('#####')
print('Working Base :',Base, 'using ', sys.platform)
print('#####')
#####sFileDir=Base + '/' + Company
+ '/02-Assess/01-EDS/02-Python' if
not os.path.exists(sFileDir):
os.makedirs(sFileDir)
#####
### Import Warehouse

```

```
#####
sFileName=Base + '/' + Company + '/00-RawData/' + sInputFileName
print('Loading :',sFileName)
RawData=pd.read_csv(sFileName,header=0)
print('#####')
print('## Raw Data Values')
print('#####') print(RawData)
print('#####') print('##
Data Profile')
print('#####')
print('Rows :',RawData.shape[0]) print('Columns
:',RawData.shape[1])
print('#####')
#####
sFileName=sFileDir + '/' + sInputFileName RawData.to_csv(sFileName, index
= False)
#####
TestData=RawData.dropna(axis=1, how='any')
#####
print('#####')
print('## Test Data Values')
print('#####') print(TestData)
print('#####') print('##
Data Profile')
print('#####')
print('Rows :',TestData.shape[0]) print('Columns
:',TestData.shape[1])
print('#####')
#####
sFileName=sFileDir + '/' + sOutputFileName
TestData.to_csv(sFileName, index = False)
#####
print('#####') print('### Done!!
#####') print('#####')
#####
>>>
===== RESTART: C:\VKHCG\01-Vermeulen\02-Assess\Assess-Good-Bad-02.py
=====
#####
Working Base : C:\VKHCG using win32
#####
Loading : C:\VKHCG\01-Vermeulen\00-RawData\Good-or-Bad.csv
#####
## Raw Data Values
#####
ID FieldA FieldB FieldC FieldD FieldE FieldF FieldG
0 1.0 Good Better Best 1024.0 NaN 10241.0 1
1 2.0 Good NaN Best 512.0 NaN 5121.0 2
#####
## Data Profile
#####
Rows : 21
Columns : 8
#####
#####
## Test Data Values
#####
```

```

FieldG
0 1
1 2
#####
## Data Profile
#####
Rows : 21
Columns : 1
#####
#####
### Done!! #####
#####
>>>
iii. Keep Only the Rows That Contain a Maximum of Two Missing Values
##### Assess-Good-Bad-03.py #####
# -*- coding: utf-8 -*-
#####
import sys import os import pandas as pd
#####
sInputFileName='Good-or-Bad.csv' sOutputFileName='Good-or-Bad-03.csv'
Company='01-Vermeulen'
Base='C:/VKHCG'
#####
print('#####') print('Working
Base :',Base, ' using Windows ~~~~')
print('#####')
#####
sFileDir=Base + '/' + Company + '/02-Assess/01-EDS/02-Python' if not
os.path.exists(sFileDir):
os.makedirs(sFileDir)
#####
### Import Warehouse
#####
sFileName=Base + '/' + Company + '/00-RawData/' + sInputFileName
print('Loading :',sFileName)
RawData=pd.read_csv(sFileName,header=0)
print('#####') print('##
Raw Data Values')
print('#####')
print(RawData)
print('#####')
print('## Data Profile')
print('#####')
print('Rows :',RawData.shape[0]) print('Columns
:',RawData.shape[1])
print('#####')
#####
sFileName=sFileDir + '/' + sInputFileName RawData.to_csv(sFileName,
index = False)
#####
TestData=RawData.dropna(thresh=2)
print('#####')
print('## Test Data Values')
print('#####')
print(TestData)
print('#####') print('##
Data Profile')

```

```

print('#####')
print('Rows :',TestData.shape[0]) print('Columns
:',TestData.shape[1])
print('#####')
sFileName=sFileDir + '/' + sOutputFileName
TestData.to_csv(sFileName, index = False)
#####
print('#####') print('###
Done!! #####')
print('#####')
#####

```

A	B	C	D	E	F	G	H
1	1D	1A	1B	1C	1D	1E	1F
2	2D	2A	2B	2C	2D	2E	2F
3	3D	3A	3B	3C	3D	3E	3F
4	4D	4A	4B	4C	4D	4E	4F
5	5D	5A	5B	5C	5D	5E	5F
6	6D	6A	6B	6C	6D	6E	6F
7	7D	7A	7B	7C	7D	7E	7F
8	8D	8A	8B	8C	8D	8E	8F
9	9D	9A	9B	9C	9D	9E	9F
10	10D	10A	10B	10C	10D	10E	10F
11	11D	11A	11B	11C	11D	11E	11F
12	12D	12A	12B	12C	12D	12E	12F
13	13D	13A	13B	13C	13D	13E	13F
14	14D	14A	14B	14C	14D	14E	14F
15	15D	15A	15B	15C	15D	15E	15F
16	16D	16A	16B	16C	16D	16E	16F
17	17D	17A	17B	17C	17D	17E	17F
18	18D	18A	18B	18C	18D	18E	18F
19	19D	19A	19B	19C	19D	19E	19F
20	20D	20A	20B	20C	20D	20E	20F
21	21D	21A	21B	21C	21D	21E	21F

Before After

Row with more than two missing values got deleted.

The next step along the route is to generate a full network routing solution for the company, to resolve the data issues in the retrieve data.

B. Write Python / R program to create the network routing diagram from the given data on routers.

```
##### Assess-Network-Routing-Company.py #####

import sys import os import pandas as pd

#####

pd.options.mode.chained_assignment = None

#####

Base='C:/VKHCG'

#####

print('#####') print('Working Base :!',Base, '
using Windows') print('#####')

##### sInputFileName1='01-Retrieve/01-
EDS/01-R/Retrieve_Country_Code.csv' sInputFileName2='01-Retrieve/01-EDS/02-
Python/Retrieve_Router_Location.csv' sInputFileName3='01-Retrieve/01-EDS/01-R/Retrieve_IP_DATA.csv'
##### sOutputFileName='Assess-Network-
Routing-Company.csv'

Company='01-Vermeulen'

#####

### Import Country Data

#####

sFileName=Base + '/' + Company + '/' + sInputFileName1

print('#####') print('Loading :!',sFileName)

print('#####')

CountryData=pd.read_csv(sFileName,header=0,low_memory=False, encoding="latin-1")

print('Loaded Country:',CountryData.columns.values) print('#####')

#####

## Assess Country Data

#####

print('#####') print('Changed
:',CountryData.columns.values)

CountryData.rename(columns={'Country': 'Country_Name'}, inplace=True)
```

```

CountryData.rename(columns={'ISO-2-CODE': 'Country_Code'}, inplace=True)

CountryData.drop('ISO-M49', axis=1, inplace=True)
CountryData.drop('ISO-3-Code', axis=1, inplace=True)

CountryData.drop('RowID', axis=1, inplace=True) print('To

:',CountryData.columns.values)

print('#####')

#####

#####

### Import Company Data

#####

sFileName=Base + '/' + Company + '/' + sInputFileName2

print('#####') print('Loading :',sFileName)

print('#####')

CompanyData=pd.read_csv(sFileName,header=0,low_memory=False, encoding="latin-

1") print('Loaded Company

:',CompanyData.columns.values)

```

Output:

Go to C:\VKHCG\01-Vermeulen\02-Assess\01-EDS\02-Python folder and open
Assess-Network-Routing-Company.csv

	A	B	C	D	E
1	ny Countr	Company_Place_Name	Company_Latitude	Company_Longitude	Company_Country_Name
2	US	New York	40.7528	-73.9725	United States of America
3	US	New York	40.7214	-74.0052	United States of America
4	US	New York	40.7662	-73.9862	United States of America
5	US	New York	40.7449	-73.9782	United States of America
6	US	New York	40.7605	-73.9933	United States of America
7	US	New York	40.7588	-73.968	United States of America
8	US	New York	40.7637	-73.9727	United States of America
9	US	New York	40.7553	-73.9924	United States of America

Next, Access the the customers location using network router location.

```

#####Assess-Network-Routing-Customer.py#####
import sys import os import pandas as pd
#####
pd.options.mode.chained_assignment = None
#####
Base='C:\VKHCG'
print('#####')
print('Working Base :',Base, ' using ', sys.platform)
print('#####')
#####
sInputFileName=Base+'01-Vermeulen/02-Assess/01-EDS/02-Python/Assess-Network_Routing-Customer.csv'
#####

```

```

sOutputFileName='Assess-Network-Routing-Customer.gml'
Company='01-Vermeulen'
#####
### Import Country Data
#####
sFileName=sInputFileName
print('#####')
print('Loading :',sFileName)
print('#####')
CustomerData=pd.read_csv(sFileName,header=0,low_memory=False, encoding="latin-1")
print('Loaded Country:',CustomerData.columns.values) print('#####')
print(CustomerData.head())
print('#####') print('###
Done!! #####')
print('#####')
#####

```

Output

Assess-Network-Routing-Customer.csv

1	2	3	4	5	
1	lat_Customer_Place_Na	Customer_Latitude	Customer_Longitude	Customer_Country_Name	
2	GW	Gabone	-04.5464	25.9119	Botswana
3	GW	Francistown	-23.3867	27.3367	Botswana
4	GW	Maseru	-19.3833	23.4267	Botswana
5	GW	McLopolele	-04.4167	25.5167	Botswana
6	NE	Nairobi	15.3387	7.1187	Niger
7	MZ	Maputo	-25.3853	32.5492	Mozambique
8	MZ	Tete	-16.1564	31.5867	Mozambique
9	MZ	Quelimane	-17.8708	36.8883	Mozambique
10	MZ	Chimoio	-15.1164	33.4433	Mozambique
11	MZ	Matola	-25.8822	32.4588	Mozambique
12	MZ	Pemba	-12.9808	40.5878	Mozambique
13	MZ	Lichinga	-13.1128	35.3406	Mozambique
14	MZ	Musave	-28.8583	35.3473	Mozambique
15	MZ	Chibuto	-24.8867	33.5308	Mozambique
16	MZ	Pedraza Garcia	-25.4428	31.5653	Mozambique
17	GH	Tema	5.8187	-0.0187	Ghana
18	GH	Kumasi	6.8533	-1.8187	Ghana
19	GH	Takoradi	6.8833	-1.75	Ghana
20	GH	Accra	5.58	-0.2187	Ghana

Practical 6:

Processing Data

Build the time hub, links, and satellites.

Open your Python editor and create a file named Process_Time.py. Save it into directory C:\VKHCG\01-Vermeulen\03-Process.

```
import sys
import os
from datetime import datetime
import timedelta from pytz import timezone,
all_timezones import pandas as pd
import sqlite3 as sq from pandas.io import sql
import uuid
pd.options.mode.chained_assignment = None
if sys.platform == 'linux':
Base=os.path.expanduser('~') + '/VKHCG' else:
Base='C:/VKHCG' print('#####')
print('Working Base:',Base, ' using ', sys.platform)
print('#####')
Company='01-Vermeulen'
InputDir='00-RawData' InputFileName='VehicleData.csv'
sDataBaseDir=Base + '/' + Company + '/03-Process/SQLite' if
not os.path.exists(sDataBaseDir):
os.makedirs(sDataBaseDir)
#####
sDatabaseName=sDataBaseDir + '/Hillman.db' conn1
= sq.connect(sDatabaseName)
#####
sDataVaultDir=Base + '/88-DV' if not os.path.exists(sDataBaseDir):
os.makedirs(sDataBaseDir)
sDatabaseName=sDataVaultDir + '/datavault.db' conn2
= sq.connect(sDatabaseName)
base = datetime(2018,1,1,0,0)
numUnits=10*365*24
date_list = [base - timedelta(hours=x) for x in range(0, numUnits)]
t=0 for i in date_list:
now_utc=i.replace(tzinfo=timezone('UTC'))
sDateTime=now_utc.strftime("%Y-%m-%d %H:%M:%S")
print(sDateTime) sDateTimeKey=sDateTime.replace(' ','-')
.replace(':', '-') t+=1
IDNumber=str(uuid.uuid4())
TimeLine=[('ZoneBaseKey', ['UTC']),
('IDNumber', [IDNumber]),
('nDateTimeValue', [now_utc]),
('DateTimeValue', [sDateTime]),
('DateTimeKey', [sDateTimeKey])] if
t==1:
TimeFrame = pd.DataFrame.from_items(TimeLine) else:
TimeRow = pd.DataFrame.from_items(TimeLine)
TimeFrame = TimeFrame.append(TimeRow)
#####
TimeHub=TimeFrame[['IDNumber','ZoneBaseKey','DateTimeKey','DateTimeValue']]
```



```

TimeHubIndex=TimeHub.set_index(['IDNumber'],inplace=False)
TimeFrame.set_index(['IDNumber'],inplace=True)
sTable = 'Process-Time'
print('Storing :',sDatabaseName,' Table:',sTable)
TimeHubIndex.to_sql(sTable, conn1, if_exists="replace") sTable
= 'Hub-Time'
print('Storing :',sDatabaseName,' Table:',sTable)
TimeHubIndex.to_sql(sTable, conn2, if_exists="replace")
active_timezones=all_timezones z=0 for zone in
active_timezones:
t=0 for j in
range(TimeFrame.shape[0]):
now_date=TimeFrame['nDateTimeValue'][j]
DateTimeKey=TimeFrame['DateTimeKey'][j]
now_utc=now_date.replace(tzinfo=timezone('UTC'))
sDateTime=now_utc.strftime("%Y-%m-%d %H:%M:%S")
now_zone = now_utc.astimezone(timezone(zone))
sZoneDateTime=now_zone.strftime("%Y-%m-%d %H:%M:%S")
print(sZoneDateTime) t+=1 z+=1
IDZoneNumber=str(uuid.uuid4())
TimeZoneLine=[('ZoneBaseKey', ['UTC']),
('IDZoneNumber', [IDZoneNumber]), ('DateTimeKey',
[DateTimeKey]),
('UTCDateTimeValue', [sDateTime]),
('Zone', [zone]),
('DateTimeValue', [sZoneDateTime])] if
t==1:
TimeZoneFrame = pd.DataFrame.from_items(TimeZoneLine) else:
TimeZoneRow = pd.DataFrame.from_items(TimeZoneLine)
TimeZoneFrame = TimeZoneFrame.append(TimeZoneRow)
TimeZoneFrameIndex=TimeZoneFrame.set_index(['IDZoneNumber'],inplace=False)
sZone=zone.replace('/', '-').replace(' ', '') sTable = 'Process-Time-'+sZone
print('Storing :',sDatabaseName,' Table:',sTable)
TimeZoneFrameIndex.to_sql(sTable, conn1, if_exists="replace") sTable
= 'Satellite-Time-'+sZone
print('Storing :',sDatabaseName,' Table:',sTable)
TimeZoneFrameIndex.to_sql(sTable, conn2, if_exists="replace")
print('#####') print('Vacuum Databases')
sSQL="VACUUM;"
sql.execute(sSQL,conn1) sql.execute(sSQL,conn2)
print('#####')
print('### Done!! #####') You
have built your first hub and satellites for time in the data vault.
The data vault has been built in directory ..\VKHCG\88-DV\datavault.db. You can access it with your SQLite tools
Golden Nominal
A golden nominal record is a single person"s record, with distinctive references for use by all systems. This
gives the system a single view of the person. I use first name, other names, last name, and birth date as my
golden nominal. The data we have in the assess directory requires a birth date to become a golden nominal.
The proram will generate a golden nominal using our sample data set.
Open your Python editor and create a file called Process-People.py in the ..
C:\VKHCG\04-Clark\03-Process directory.
#####
import sys import os import sqlite3 as sq import pandas as pd from pandas.io
import sql from datetime import datetime, timedelta from pytz import timezone,
all_timezones from random import randint import uuid if sys.platform ==
'linux':
Base=os.path.expanduser('~') + '/VKHCG' else:

```

```

Base='C:/VKHCG'
print('#####')
print('Working Base :',Base, ' using ', sys.platform)
print('#####') Company='04-Clark'
sInputFileName='02-Assess/01-EDS/02-Python/Assess_People.csv'
sDataBaseDir=Base + '/' + Company + '/03-Process/SQLite' if not
os.path.exists(sDataBaseDir):
os.makedirs(sDataBaseDir)
sDatabaseName=sDataBaseDir + '/clark.db'
conn1 = sq.connect(sDatabaseName)
sDataVaultDir=Base + '/88-DV' if not
os.path.exists(sDataBaseDir):
os.makedirs(sDataBaseDir)
sDatabaseName=sDataVaultDir + '/datavault.db' conn2
= sq.connect(sDatabaseName)
### Import Female Data sFileName=Base + '/' +
Company + '/' + sInputFileName
print('#####')
print('Loading :',sFileName)
print('#####') print(sFileName)
RawData=pd.read_csv(sFileName,header=0,low_memory=False,encoding="latin-1")
RawData.drop_duplicates(subset=None, keep='first', inplace=True) start_date =
datetime(1900,1,1,0,0,0)
start_date_utc=start_date.replace(tzinfo=timezone('UTC'))
HoursBirth=100*365*24
RawData['BirthDateUTC']=RawData.apply(lambda row:
(start_date_utc + timedelta(hours=randint(0, HoursBirth))),axis=1)
zonemax=len(all_timezones)-1
RawData['TimeZone']=RawData.apply(lambda row:
(all_timezones[randint(0, zonemax)]),axis=1)
RawData['BirthDateISO']=RawData.apply(lambda row:
row["BirthDateUTC"].astimezone(timezone(row["TimeZone"])),axis=1)
RawData['BirthDateKey']=RawData.apply(lambda row:
row["BirthDateUTC"].strftime("%Y-%m-%d %H:%M:%S"),axis=1)
RawData['BirthDate']=RawData.apply(lambda row:
row["BirthDateISO"].strftime("%Y-%m-%d %H:%M:%S"),axis=1)
RawData['PersonID']=RawData.apply(lambda row:
str(uuid.uuid4()),axis=1)
Data=RawData.copy()
Data.drop('BirthDateUTC', axis=1,inplace=True)
Data.drop('BirthDateISO', axis=1,inplace=True) indexed_data
= Data.set_index(['PersonID'])
print('#####')
print('#####') sTable='Process_Person'
print('Storing :',sDatabaseName,' Table:',sTable)
indexed_data.to_sql(sTable, conn1, if_exists="replace")
print('#####')
PersonHubRaw=Data[['PersonID','FirstName','SecondName','LastName','BirthDateKey']]
PersonHubRaw['PersonHubID']=RawData.apply(lambda row: str(uuid.uuid4()),axis=1)

```

```

PersonHub=PersonHubRaw.drop_duplicates(subset=None, \
keep='first',\ inplace=False) indexed_PersonHub =
PersonHub.set_index(['PersonHubID']) sTable = 'Hub-Person'
print('Storing :',sDatabaseName,' Table:',sTable)
indexed_PersonHub.to_sql(sTable, conn2, if_exists="replace")
PersonSatelliteGenderRaw=Data[['PersonID','FirstName','SecondName','LastName'\
,'BirthDateKey','Gender']]
PersonSatelliteGenderRaw['PersonSatelliteID']=RawData.apply(lambda row:
str(uuid.uuid4()))
,axis=1)
PersonSatelliteGender=PersonSatelliteGenderRaw.drop_duplicates(subset=None, \
keep='first', \ inplace=False) indexed_PersonSatelliteGender =
PersonSatelliteGender.set_index(['PersonSatelliteID']) sTable = 'Satellite-Person-Gender'
print('Storing :',sDatabaseName,' Table:',sTable)
indexed_PersonSatelliteGender.to_sql(sTable, conn2, if_exists="replace")
#####
PersonSatelliteBirthdayRaw=Data[['PersonID','FirstName','SecondName','LastName','\
'BirthDateKey','TimeZone','BirthDate']]
PersonSatelliteBirthdayRaw['PersonSatelliteID']=RawData.apply(lambda row:
str(uuid.uuid4()))
,axis=1)
PersonSatelliteBirthday=PersonSatelliteBirthdayRaw.drop_duplicates(subset=None, \ keep='first',\ inplace=False)
indexed_PersonSatelliteBirthday = PersonSatelliteBirthday.set_index(['PersonSatelliteID']) sTable
= 'Satellite-Person-Names'
print('Storing :',sDatabaseName,' Table:',sTable)
indexed_PersonSatelliteBirthday.to_sql(sTable, conn2, if_exists="replace")
sFileDir=Base + '/' + Company + '/03-Process/01-EDS/02-Python' if not
os.path.exists(sFileDir):
os.makedirs(sFileDir) sOutputFileName =
sTable + '.csv' sFileName=sFileDir + '/' +
sOutputFileName
print('#####')
print('Storing :', sFileName)
print('#####') RawData.to_csv(sFileName,
index = False) print('#####')
print('#####') print('Vacuum
Databases')
sSQL="VACUUM;"
sql.execute(sSQL,conn1) sql.execute(sSQL,conn2)
print('#####')
print('### Done!! #####')
Output :
It will apply golden nominal rules by assuming nobody born before January 1, 1900, dropping to two ISO
complex date time structures, as the code does not translate into SQLite"s data types and saves your new
golden nominal to a CSV file.
Load the person into the data vault
===== RESTART: C:\VKHCG\04-Clark\03-Process\Process-People.py =====
Working Base : C:\VKHCG using win32
Loading : C:\VKHCG\04-Clark\02-Assess\01-EDS\02-Python\Assess_People.csv
C:\VKHCG\04-Clark\02-Assess\01-EDS\02-Python\Assess_People.csv
Storing : C:\VKHCG\88-DV\datavault.db Table: Process_Person
Storing : C:\VKHCG\88-DV\datavault.db Table: Satellite-Person-Gender
Storing : C:\VKHCG\88-DV\datavault.db Table: Satellite-Person-Names
Storing : C:\VKHCG\04-Clark\03-Process\01-EDS\02-Python\Satellite-Person-Names.csv
Vacuum Databases
#####
### Done!! #####

```

Vehicles

The international classification of vehicles is a complex process. There are standards, but these are not universally applied or similar between groups or countries.

Let's load the vehicle data for Hillman Ltd into the data vault, as we will need it later. Create a new file named Process-Vehicle-Logistics.py in the Python editor in directory ..\VKHCG\03-Hillman\03-Process.

```
# -*- coding: utf-8 -*- import sys import os
import pandas as pd import sqlite3 as sq
from pandas.io import sql import uuid
pd.options.mode.chained_assignment =
None if sys.platform == 'linux':
Base=os.path.expanduser('~') + '/VKHCG'
else:
Base='C:/VKHCG'
print('#####')
print('Working Base :',Base, ' using ', sys.platform)
print('#####')
Company='03-Hillman' InputDir='00-RawData'
InputFileName='VehicleData.csv' sDataBaseDir=Base + '/'
+ Company + '/03-Process/SQLite' if not
os.path.exists(sDataBaseDir):
os.makedirs(sDataBaseDir)
sDatabaseName=sDataBaseDir + '/Hillman.db'
conn1 = sq.connect(sDatabaseName)
sDataVaultDir=Base + '/88-DV' if not
os.path.exists(sDataVaultDir):
os.makedirs(sDataVaultDir)
sDatabaseName=sDataVaultDir + '/datavault.db' conn2 =
sq.connect(sDatabaseName) sFileName=Base + '/' + Company + '/' +
InputDir + '/' + InputFileName print('#####') print('Loading
:',sFileName)
VehicleRaw=pd.read_csv(sFileName,header=0,low_memory=False,encoding="latin-1") sTable='Process_Vehicles'
print('Storing :',sDatabaseName,' Table:',sTable) VehicleRaw.to_sql(sTable,
conn1, if_exists="replace")
VehicleRawKey=VehicleRaw[['Make','Model']].copy()
VehicleKey=VehicleRawKey.drop_duplicates()
VehicleKey['ObjectKey']=VehicleKey.apply(lambda row:
str('(' + str(row['Make']).strip().replace(' ', '-').replace('/', '-').lower() +
')-( ' + str(row['Model']).strip().replace(' ', '-').replace(' ', '-').lower()
+')')
,axis=1)
VehicleKey['ObjectType']=VehicleKey.apply(lambda row:
'vehicle',axis=1)
VehicleKey['ObjectUUID']=VehicleKey.apply(lambda row:
str(uuid.uuid4()),axis=1)
### Vehicle Hub
#
VehicleHub=VehicleKey[['ObjectType','ObjectKey','ObjectUUID']].copy()
VehicleHub.index.name='ObjectHubID' sTable = 'Hub-Object-Vehicle'
print('Storing :',sDatabaseName,' Table:',sTable)
VehicleHub.to_sql(sTable, conn2, if_exists="replace")
### Vehicle Satellite
#
VehicleSatellite=VehicleKey[['ObjectType','ObjectKey','ObjectUUID','Make','Model']].copy()
VehicleSatellite.index.name='ObjectSatelliteID' sTable = 'Satellite-Object-Make-Model'
print('Storing :',sDatabaseName,' Table:',sTable)
VehicleSatellite.to_sql(sTable, conn2, if_exists="replace")
### Vehicle Dimension sView='Dim-Object'
```

```

print('Storing :',sDatabaseName,' View:',sView) sSQL="CREATE
VIEW IF NOT EXISTS [" + sView + "]" AS" sSQL=sSQL+ "
SELECT DISTINCT" sSQL=sSQL+ " H.ObjectType,"
sSQL=sSQL+ " H.ObjectKey AS VehicleKey," sSQL=sSQL+ "
TRIM(S.Make) AS VehicleMake," sSQL=sSQL+ "
TRIM(S.Model) AS VehicleModel"
sSQL=sSQL+ " FROM"
sSQL=sSQL+ " [Hub-Object-Vehicle] AS H"
sSQL=sSQL+ " JOIN" sSQL=sSQL+ " [Satellite-
Object-Make-Model] AS S"
sSQL=sSQL+ " ON"
sSQL=sSQL+ " H.ObjectType=S.ObjectType"
sSQL=sSQL+ " AND"
sSQL=sSQL+ " H.ObjectUUID=S.ObjectUUID;"
sql.execute(sSQL,conn2) print('#####')
print('Loading :',sDatabaseName,' Table:',sView)
sSQL=" SELECT DISTINCT" sSQL=sSQL+ "
VehicleMake," sSQL=sSQL+ " VehicleModel"
sSQL=sSQL+ " FROM"
sSQL=sSQL+ " [" + sView + "]"
sSQL=sSQL+ " ORDER BY"
sSQL=sSQL+ " VehicleMake"
sSQL=sSQL+ " AND" sSQL=sSQL+
" VehicleMake;"
DimObjectData=pd.read_sql_query(sSQL, conn2)
DimObjectData.index.name='ObjectDimID'
DimObjectData.sort_values(['VehicleMake','VehicleModel'],inplace=True, ascending=True)
print('#####') print(DimObjectData)
#####
print('#####') print('Vacuum Databases')
sSQL="VACUUM;"
sql.execute(sSQL,conn1) sql.execute(sSQL,conn2)
print('#####')
#####
conn1.close() conn2.close()
#####
#print('### Done!! #####')
#####

```



Human-Environment Interaction

In the Python editor, open a new file named Process_Location.py in directory ..\VKHCG\01- Vermeulen\03-Process.

```

import sys
import os
import pandas as pd

```

```

import sqlite3 as sq from
pandas.io import sql
import uuid
Base='C:/VKHCG' print('#####')
print('Working Base :',Base, 'using ', sys.platform)
print('#####')
Company='01-Vermeulen'
InputAssessGraphName='Assess_All_Animals.gml'
EDSAssessDir='02-Assess/01-EDS'
InputAssessDir=EDSAssessDir + '/02-Python'
sFileAssessDir=Base + '/' + Company + '/' + InputAssessDir if
not os.path.exists(sFileAssessDir):
os.makedirs(sFileAssessDir) sDataBaseDir=Base + '/' +
Company + '/03-Process/SQLite' if not
os.path.exists(sDataBaseDir):
os.makedirs(sDataBaseDir)
sDatabaseName=sDataBaseDir + '/Vermeulen.db'
conn1 = sq.connect(sDatabaseName)
sDataVaultDir=Base + '/88-DV' if not
os.path.exists(sDataVaultDir):
os.makedirs(sDataVaultDir)
sDatabaseName=sDataVaultDir + '/datavault.db'
conn2 = sq.connect(sDatabaseName) t=0
tMax=360*180 for Longitude in range(-
180,180,10): for Latitude in range(-90,90,10):
t+=1
IDNumber=str(uuid.uuid4())
LocationName='L'+format(round(Longitude,3)*1000, '+07d') +\ '-' +format(round(Latitude,3)*1000, '+07d')
print('Create:',t, 'of ',tMax,':',LocationName)
LocationLine=[('ObjectBaseKey', ['GPS']),
('IDNumber', [IDNumber]),
('LocationNumber', [str(t)]),
('LocationName', [LocationName]),
('Longitude', [Longitude]),
('Latitude', [Latitude])] if
t==1:
LocationFrame = pd.DataFrame.from_items(LocationLine) else:
LocationRow = pd.DataFrame.from_items(LocationLine)
LocationFrame = LocationFrame.append(LocationRow)
LocationHubIndex=LocationFrame.set_index(['IDNumber'],inplace=False)
sTable = 'Process-Location'
print('Storing :',sDatabaseName,' Table:',sTable)
LocationHubIndex.to_sql(sTable, conn1, if_exists="replace") sTable
= 'Hub-Location'
print('Storing :',sDatabaseName,' Table:',sTable)
LocationHubIndex.to_sql(sTable, conn2, if_exists="replace")
print('#####')
print('Vacuum Databases')
sSQL="VACUUM;"
sql.execute(sSQL,conn1)
sql.execute(sSQL,conn2)
print('#####') print('###
Done!! #####

```

```

Python 2.7.4 Shell
File Edit Shell Debug Options Window Help
Create: 645 of 64800 : L+170000--050000
Create: 646 of 64800 : L+170000--060000
Create: 647 of 64800 : L+170000--070000
Create: 648 of 64800 : L+170000--080000
Storing : C:/VKHCG/04-IV/datavault.db Table: Process-Location
Storing : C:/VKHCG/04-IV/datavault.db Table: Hub-Location
#####
Vacuum Databases
#####
### Done!! #####
>>>

```

Forecasting

Forecasting is the ability to project a possible future, by looking at historical data. The datavault enables these types of investigations, owing to the complete history it collects as it processes the source's systems data. A data scientist supply answers to such questions as the following:

- What should we buy?
- What should we sell?
- Where will our next business come from?

People want to know what you calculate to determine what is about to happen.

Open a new file in your Python editor and save it as Process-Shares-Data.py in directory C: \VKHCG\04-Clark\03-Process. I will guide you through this process. You will require a library called quandl type pip install quandl in cmd

```

#####
import sys import os import sqlite3 as sq import quandl import pandas as pd
#####
Base='C:/VKHCG'
print('#####')
print('Working Base :',Base, ' using ', sys.platform)
print('#####')
#####
Company='04-Clark'
sInputFileName='00-RawData/VKHCG_Shares.csv'
sOutputFileName='Shares.csv'
#####
sDataBaseDir=Base + '/' + Company + '/03-Process/SQLite' if
not os.path.exists(sDataBaseDir): os.makedirs(sDataBaseDir)
#####
sFileDir1=Base + '/' + Company + '/01-Retrieve/01-EDS/02-Python' if not
os.path.exists(sFileDir1):
os.makedirs(sFileDir1)
#####
sFileDir2=Base + '/' + Company + '/02-Assess/01-EDS/02-Python' if not
os.path.exists(sFileDir2):
os.makedirs(sFileDir2)
#####
sFileDir3=Base + '/' + Company + '/03-Process/01-EDS/02-Python' if not
os.path.exists(sFileDir3):
os.makedirs(sFileDir3)
#####
sDatabaseName=sDataBaseDir + '/clark.db' conn
= sq.connect(sDatabaseName)

```



```
#####
### Import Share Names Data
#####
sFileName=Base + '/' + Company + '/' + sInputFileName
print('#####')
print('Loading :',sFileName)
print('#####')
RawData=pd.read_csv(sFileName,header=0,low_memory=False, encoding="latin-1")
RawData.drop_duplicates(subset=None, keep='first', inplace=True) print('Rows
:',RawData.shape[0]) print('Columns:',RawData.shape[1]) print('#####')
#####
sFileName=sFileDir1 + '/Retrieve_' + sOutputFileName
print('#####')
print('Storing :', sFileName) print('#####')
RawData.to_csv(sFileName, index = False)
print('#####')
#####
sFileName=sFileDir2 + '/Assess_' + sOutputFileName
print('#####')
print('Storing :', sFileName) print('#####')
RawData.to_csv(sFileName, index = False)
print('#####')
#####
sFileName=sFileDir3 + '/Process_' + sOutputFileName
print('#####')
print('Storing :', sFileName) print('#####')
RawData.to_csv(sFileName, index = False)
print('#####')
#####
### Import Shares Data Details nShares=RawData.shape[0]
#nShares=6 for sShare in
range(nShares):
sShareName=str(RawData['Shares'][sShare])
ShareData = quandl.get(sShareName)
UnitsOwn=RawData['Units'][sShare]
ShareData['UnitsOwn']=ShareData.apply(lambda row:(UnitsOwn),axis=1)
ShareData['ShareCode']=ShareData.apply(lambda
row:(sShareName),axis=1) print('#####') print('Share
:',sShareName) print('Rows :',ShareData.shape[0])
print('Columns:',ShareData.shape[1]) print('#####')
#####
print('#####') sTable=str(RawData['sTable'][sShare])
print('Storing :',sDatabaseName,' Table:',sTable) ShareData.to_sql(sTable, conn,
if_exists="replace") print('#####')
#####
sOutputFileName = sTable.replace("/", "-") + '.csv' sFileName=sFileDir1 +
'/Retrieve_' + sOutputFileName print('#####')
print('Storing :', sFileName)
print('#####') ShareData.to_csv(sFileName,
index = False) print('#####')
#####
sOutputFileName = sTable.replace("/", "-") + '.csv' sFileName=sFileDir2 +
'/Assess_' + sOutputFileName print('#####')
print('Storing :', sFileName)
print('#####') ShareData.to_csv(sFileName,
index = False) print('#####')
```



```
#####
sOutputFileName = sTable.replace("/", "-") + '.csv' sFileName=sFileDir3 +
'/Process_' + sOutputFileName print('#####')
print('Storing :', sFileName)
print('#####') ShareData.to_csv(sFileName,
index = False) print('#####')
print('### Done!! #####')
#####
```

Output:

===== RESTART: C:\VKHCG\04-Clark\03-Process\Process-Shares-Data.py =====

Working Base : C:\VKHCG using win32

Loading : C:\VKHCG\04-Clark\00-RawData\VKHCG_Shares.csv

Rows : 10

Columns: 3

Storing : C:\VKHCG\04-Clark\01-Retrieve\01-EDS\02-Python\Retrieve_Shares.csv

Storing : C:\VKHCG\04-Clark\02-Assess\01-EDS\02-Python\Assess_Shares.csv

Storing : C:\VKHCG\04-Clark\03-Process\01-EDS\02-Python\Process_Shares.csv

Share : WIKI/GOOGL

Rows : 3424

Columns: 14

Storing : C:\VKHCG\04-Clark\03-Process\SQLite\clark.db Table: WIKI_Google Storing :

C:\VKHCG\04-Clark\01-Retrieve\01-EDS\02-Python\Retrieve_WIKI_Google.csv Storing :

C:\VKHCG\04-Clark\02-Assess\01-EDS\02-Python\Assess_WIKI_Google.

Practical 7

Transforming Data

Transform Superstep

C: \VKHCG\01-Vermeulen\04-Transform.

```
import sys import os from
datetime import datetime
from pytz import timezone
import pandas as pd import
sqlite3 as sq import uuid
pd.options.mode.chained_assignment = None
#####
Base='C:/VKHCG'
print('#####') print('Working
Base :',Base, ' using ', sys.platform)
print('#####')
#####
Company='01-Vermeulen'
InputDir='00-RawData'
InputFileName='VehicleData.csv'
#####
sDataBaseDir=Base + '/' + Company + '/04-Transform/SQLite' if
not os.path.exists(sDataBaseDir):
os.makedirs(sDataBaseDir)
#####
sDatabaseName=sDataBaseDir + '/Vermeulen.db' conn1
= sq.connect(sDatabaseName)
#####
sDataVaultDir=Base + '/88-DV' if not os.path.exists(sDataVaultDir):
os.makedirs(sDataVaultDir)
#####
sDatabaseName=sDataVaultDir + '/datavault.db' conn2
= sq.connect(sDatabaseName)
#####
sDataWarehouseDir=Base + '/99-DW' if not
os.path.exists(sDataWarehouseDir):
os.makedirs(sDataWarehouseDir)
#####
sDatabaseName=sDataWarehouseDir + '/datawarehouse.db' conn3
= sq.connect(sDatabaseName)
#####
print('\n#####')
print('Time Category') print('UTC
Time')
BirthDateUTC = datetime(1960,12,20,10,15,0)
BirthDateZoneUTC=BirthDateUTC.replace(tzinfo=timezone('UTC'))
BirthDateZoneStr=BirthDateZoneUTC.strftime("%Y-%m-%d %H:%M:%S")
BirthDateZoneUTCStr=BirthDateZoneUTC.strftime("%Y-%m-%d %H:%M:%S (%Z) (%z)")
print(BirthDateZoneUTCStr)
print('#####')
print('Birth Date in Reykjavik :) BirthZone
= 'Atlantic/Reykjavik'
BirthDate = BirthDateZoneUTC.astimezone(timezone(BirthZone))
```

```

BirthDateStr=BirthDate.strftime("%Y-%m-%d %H:%M:%S (%Z) (%z)")
BirthDateLocal=BirthDate.strftime("%Y-%m-%d %H:%M:%S")
print(BirthDateStr)

print('#####')
#####
IDZoneNumber=str(uuid.uuid4())
sDateTimeKey=BirthDateZoneStr.replace(' ','-').replace(':','-')
TimeLine=[('ZoneBaseKey', ['UTC']),
('IDNumber', [IDZoneNumber]),
('DateTimeKey', [sDateTimeKey]),
('UTCDateTimeValue', [BirthDateZoneUTC]),
('Zone', [BirthZone]),
('DateTimeValue', [BirthDateStr])]
TimeFrame = pd.DataFrame.from_items(TimeLine)
#####
TimeHub=TimeFrame[['IDNumber','ZoneBaseKey','DateTimeKey','DateTimeValue']]
TimeHubIndex=TimeHub.set_index(['IDNumber'],inplace=False)
#####
sTable = 'Hub-Time-Gunnarsson'
print("\n#####") print('Storing
:',sDatabaseName,'\n Table:',sTable)
print("\n#####")
TimeHubIndex.to_sql(sTable, conn2, if_exists="replace") sTable = 'Dim-Time-
Gunnarsson'
TimeHubIndex.to_sql(sTable, conn3, if_exists="replace")
#####
TimeSatellite=TimeFrame[['IDNumber','DateTimeKey','Zone','DateTimeValue']]
TimeSatelliteIndex=TimeSatellite.set_index(['IDNumber'],inplace=False)
#####
BirthZoneFix=BirthZone.replace(' ','-').replace('/','-') sTable =
'Satellite-Time-' + BirthZoneFix + '-Gunnarsson'
print("\n#####")
print('Storing :',sDatabaseName,'\n Table:',sTable)
print("\n#####")
TimeSatelliteIndex.to_sql(sTable, conn2, if_exists="replace")
sTable = 'Dim-Time-' + BirthZoneFix + '-Gunnarsson'
TimeSatelliteIndex.to_sql(sTable, conn3, if_exists="replace")
#####
print("\n#####")
print('Person Category') FirstName =
'Guðmundur' LastName = 'Gunnarsson'
print('Name:',FirstName,LastName) print('Birth
Date:',BirthDateLocal) print('Birth
Zone:',BirthZone) print('UTC Birth
Date:',BirthDateZoneStr)
print('#####')
)
#####
IDPersonNumber=str(uuid.uuid4())
PersonLine=[('IDNumber', [IDPersonNumber]),
('FirstName', [FirstName]),
('LastName', [LastName]),
('Zone', ['UTC']),
('DateTimeValue', [BirthDateZoneStr])]
PersonFrame = pd.DataFrame.from_items(PersonLine)
#####

```

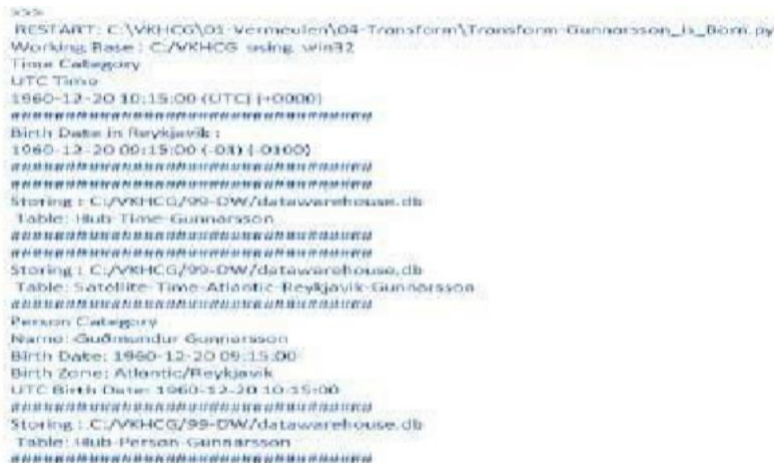
```

TimeHub=PersonFrame
TimeHubIndex=TimeHub.set_index(['IDNumber'],inplace=False)
#####
sTable = 'Hub-Person-Gunnarsson'
print("\n#####") print('Storing
:',sDatabaseName,'\n Table:',sTable)
print("\n#####")

TimeHubIndex.to_sql(sTable, conn2, if_exists="replace") sTable
= 'Dim-Person-Gunnarsson'
TimeHubIndex.to_sql(sTable, conn3, if_exists="replace")
#####

```

Output :



```

>>>
RESTART: C:\VKHCG\01-Vermeulen\04-Transform\Transform-Gunnarsson_Is_Born.py
Working Base: C:\VKHCG using win32
Time Category
UTC Time
1960-12-20 10:15:00 (UTC) [+0000]
Birth Date in Reykjavik:
1960-12-20 09:15:00 (-03) [-0100]
Storing: C:\VKHCG\99-DW\datawarehouse.db
Table: Hub Time-Gunnarsson
Storing: C:\VKHCG\99-DW\datawarehouse.db
Table: Satellite Time-Atlantic-Reykjavik-Gunnarsson
Person Category
Name: Guðmundur Gunnarsson
Birth Date: 1960-12-20 09:15:00
Birth Zone: Atlantic/Reykjavik
UTC Birth Date: 1960-12-20 10:15:00
Storing: C:\VKHCG\99-DW\datawarehouse.db
Table: Hub Person-Gunnarsson

```

You must build three items: **dimension Person**, **dimension Time**, and **factPersonBornAtTime**. Open your Python editor and create a file named Transform-Gunnarsson-Sun-Model.py in directory C:\VKHCG\01-Vermeulen\04-Transform.

```

#####
import sys import os from datetime import datetime from pytz import timezone import
pandas as pd import sqlite3 as sq import uuid
pd.options.mode.chained_assignment = None
#####
if sys.platform == 'linux':
Base=os.path.expanduser('~') + '/VKHCG' else:
Base='C:/VKHCG'
print('#####') print('Working
Base:',Base, ' using ', sys.platform)
print('#####')
#####
Company='01-Vermeulen'
#####
sDataBaseDir=Base + '/' + Company + '/04-Transform/SQLite' if
not os.path.exists(sDataBaseDir):
os.makedirs(sDataBaseDir)
#####
sDatabaseName=sDataBaseDir + '/Vermeulen.db' conn1
= sq.connect(sDatabaseName)
#####
sDataWarehousetDir=Base + '/99-DW' if not
os.path.exists(sDataWarehousetDir): os.makedirs(sDataWarehousetDir)

```

```
#####
sDatabaseName=sDataWarehouseDir + '/datawarehouse.db' conn2
= sq.connect(sDatabaseName)
#####
print("\n#####")
print('Time Dimension') BirthZone
= 'Atlantic/Reykjavik'
BirthDateUTC = datetime(1960,12,20,10,15,0)
BirthDateZoneUTC=BirthDateUTC.replace(tzinfo=timezone('UTC'))
BirthDateZoneStr=BirthDateZoneUTC.strftime("%Y-%m-%d %H:%M:%S")
BirthDateZoneUTCStr=BirthDateZoneUTC.strftime("%Y-%m-%d %H:%M:%S (%Z) (%z)")
BirthDate = BirthDateZoneUTC.astimezone(timezone(BirthZone))
BirthDateStr=BirthDate.strftime("%Y-%m-%d %H:%M:%S (%Z) (%z)")
BirthDateLocal=BirthDate.strftime("%Y-%m-%d %H:%M:%S")
#####
IDTimeNumber=str(uuid.uuid4())
TimeLine=[('TimeID', [IDTimeNumber]),
('UTCDate', [BirthDateZoneStr]),
('LocalTime', [BirthDateLocal]),
('TimeZone', [BirthZone])]
TimeFrame = pd.DataFrame.from_items(TimeLine)
#####
DimTime=TimeFrame
DimTimeIndex=DimTime.set_index(['TimeID'],inplace=False)
#####
sTable = 'Dim-Time' print("\n#####")
print('Storing :',sDatabaseName,'\n Table:',sTable)
print("\n#####")
DimTimeIndex.to_sql(sTable, conn1, if_exists="replace")
DimTimeIndex.to_sql(sTable, conn2, if_exists="replace")
#####
print("\n#####") print('Dimension
Person')
print("\n#####")
FirstName = 'Guðmundur'
LastName = 'Gunnarsson'
#####
IDPersonNumber=str(uuid.uuid4())
PersonLine=[('PersonID', [IDPersonNumber]),
('FirstName', [FirstName]),
('LastName', [LastName]),
('Zone', ['UTC']),
('DateTimeValue', [BirthDateZoneStr])]
PersonFrame = pd.DataFrame.from_items(PersonLine)
#####
DimPerson=PersonFrame
DimPersonIndex=DimPerson.set_index(['PersonID'],inplace=False)
#####
sTable = 'Dim-Person' print("\n#####")
print('Storing :',sDatabaseName,'\n Table:',sTable)
print("\n#####")
DimPersonIndex.to_sql(sTable, conn1, if_exists="replace")
DimPersonIndex.to_sql(sTable, conn2, if_exists="replace")
#####
print("\n#####") print('Fact
- Person - time')
```

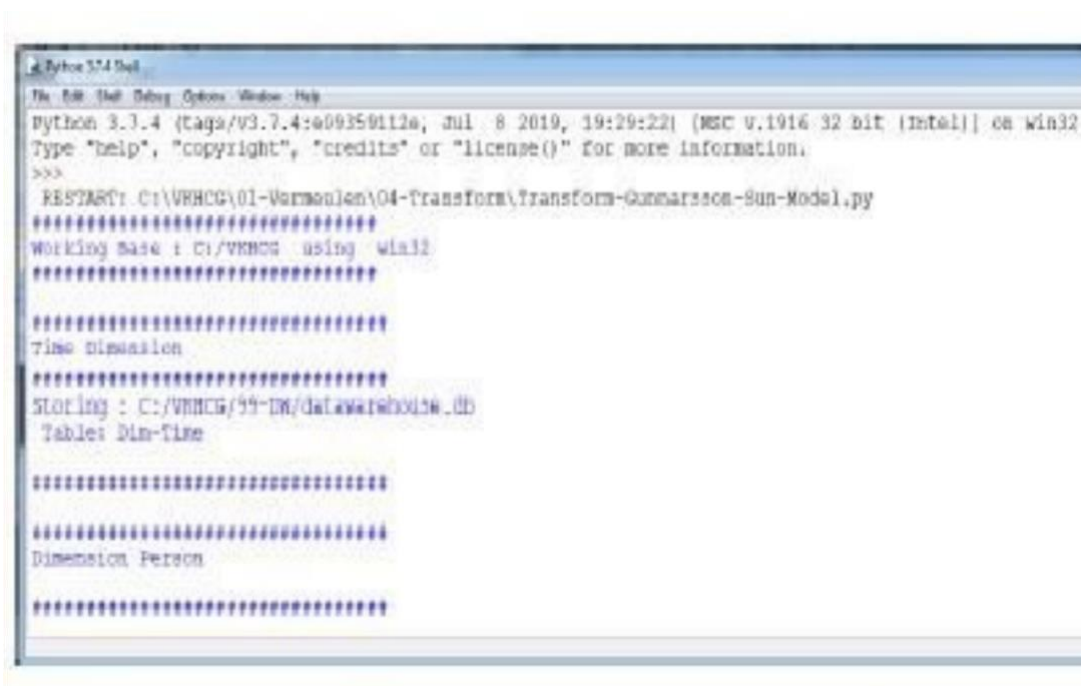
```

print("\n#####")
IDFactNumber=str(uuid.uuid4())

PersonTimeLine=[('IDNumber', [IDFactNumber]),
('IDPersonNumber', [IDPersonNumber]),
('IDTimeNumber', [IDTimeNumber])]
PersonTimeFrame = pd.DataFrame.from_items(PersonTimeLine)
#####
FctPersonTime=PersonTimeFrame
FctPersonTimeIndex=FctPersonTime.set_index(['IDNumber'],inplace=False)
#####
sTable = 'Fact-Person-Time' print("\n#####")
print('Storing :',sDatabaseName,'\n Table:',sTable)
print("\n#####")
FctPersonTimeIndex.to_sql(sTable, conn1, if_exists="replace")
FctPersonTimeIndex.to_sql(sTable, conn2, if_exists="replace")
#####

```

Output:



```

Python 3.7.4 Shell
File Edit Shell Debug Options Window Help
Python 3.7.4 (tags/v3.7.4:009350112e; Jul 8 2019, 19:29:22) [MSC v.1916 32 bit (Intel)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
RESTART: C:\VKHCG\01-Vermeulen\04-Transform\Transform-Gunnarsen-Sun-Model.py
#####
Working base : C:\VKHCG using win32
#####
#####
Time dimension
#####
Storing : C:\VKHCG\01-Vermeulen\datawarehouse.db
Table: Dim-Time
#####
#####
Dimension Person
#####

```

Building a Data Warehouse

Open the Transform-Sun-Models.py file from directory C:\VKHCG\01-Vermeulen\04Transform.

```

#####
import sys import os from datetime import datetime from pytz import timezone import
pandas as pd import sqlite3 as sq import uuid
pd.options.mode.chained_assignment = None
#####
if sys.platform == 'linux':
Base=os.path.expanduser('~') + '/VKHCG' else:
Base='C:/VKHCG'
print('#####') print('Working
Base :',Base, ' using ', sys.platform)
print('#####')

```

```

#####
Company='01-Vermeulen'
#####
sDataBaseDir=Base + '/' + Company + '/04-Transform/SQLite' if
not os.path.exists(sDataBaseDir):
os.makedirs(sDataBaseDir)
#####

sDatabaseName=sDataBaseDir + '/Vermeulen.db' conn1
= sq.connect(sDatabaseName)
#####
##### sDataVaultDir=Base + '/88-DV'
if not os.path.exists(sDataVaultDir):
os.makedirs(sDataVaultDir)
#####
sDatabaseName=sDataVaultDir + '/datavault.db' conn2
= sq.connect(sDatabaseName)
#####
sDataWarehouseDir=Base + '/99-DW' if not
os.path.exists(sDataWarehouseDir):
os.makedirs(sDataWarehouseDir)
#####
sDatabaseName=sDataWarehouseDir + '/datawarehouse.db' conn3
= sq.connect(sDatabaseName)
#####
sSQL=" SELECT DateTimeValue FROM [Hub-Time];"
DateDataRow=pd.read_sql_query(sSQL, conn2)
DateData=DateDataRow.head(1000)
print(DateData)
#####
print('\n#####') print('Time
Dimension')
print('\n#####')
t=0 mt=DateData.shape[0] for i in range(mt):
BirthZone = ('Atlantic/Reykjavik','Europe/London','UCT')
for j in range(len(BirthZone)): t+=1 print(t,mt*3)
BirthDateUTC = datetime.strptime(DateData['DateTimeValue'][i],"%Y-%m-%d %H:%M:%S")
BirthDateZoneUTC=BirthDateUTC.replace(tzinfo=timezone('UTC'))
BirthDateZoneStr=BirthDateZoneUTC.strftime("%Y-%m-%d %H:%M:%S")
BirthDateZoneUTCStr=BirthDateZoneUTC.strftime("%Y-%m-%d %H:%M:%S (%Z) (%z)")
BirthDate = BirthDateZoneUTC.astimezone(timezone(BirthZone[j]))
BirthDateStr=BirthDate.strftime("%Y-%m-%d %H:%M:%S (%Z) (%z)")
BirthDateLocal=BirthDate.strftime("%Y-%m-%d %H:%M:%S")
#####
IDTimeNumber=str(uuid.uuid4())
TimeLine=[('TimeID', [str(IDTimeNumber)]),
('UTCDate', [str(BirthDateZoneStr)]),
('LocalTime', [str(BirthDateLocal)]),
('TimeZone', [str(BirthZone)])] if
t==1:
TimeFrame = pd.DataFrame.from_items(TimeLine) else:
TimeRow = pd.DataFrame.from_items(TimeLine)
TimeFrame=TimeFrame.append(TimeRow)
#####
DimTime=TimeFrame
DimTimeIndex=DimTime.set_index(['TimeID'],inplace=False)

```

```
#####
sTable = 'Dim-Time' print("\n#####")
print('Storing :',sDatabaseName,'\n Table:',sTable)
print("\n#####")
DimTimeIndex.to_sql(sTable, conn1, if_exists="replace")
DimTimeIndex.to_sql(sTable, conn3, if_exists="replace")

#####
sSQL=" SELECT " + \
" FirstName," + \
" SecondName," + \
" LastName," + \
" BirthDateKey " + \
" FROM [Hub-Person];"
PersonDataRaw=pd.read_sql_query(sSQL, conn2)
PersonData=PersonDataRaw.head(1000)
#####
print("\n#####") print('Dimension
Person')
print("\n#####")
t=0 mt=DateData.shape[0] for i in range(mt):
t+=1
print(t,mt)
FirstName = str(PersonData["FirstName"])
SecondName = str(PersonData["SecondName"]) if
len(SecondName) > 0:
SecondName=""
LastName = str(PersonData["LastName"])
BirthDateKey = str(PersonData["BirthDateKey"])
#####
IDPersonNumber=str(uuid.uuid4())
PersonLine=[('PersonID', [str(IDPersonNumber)]),
('FirstName', [FirstName]),
('SecondName', [SecondName]),
('LastName', [LastName]),
('Zone', [str('UTC')]),
('BirthDate', [BirthDateKey])] if
t==1:
PersonFrame = pd.DataFrame.from_items(PersonLine) else:
PersonRow = pd.DataFrame.from_items(PersonLine)
PersonFrame = PersonFrame.append(PersonRow)
#####
DimPerson=PersonFrame print(DimPerson)
DimPersonIndex=DimPerson.set_index(['PersonID'],inplace=False)
#####
sTable = 'Dim-Person' print("\n#####")
print('Storing :',sDatabaseName,'\n Table:',sTable)
print("\n#####")
DimPersonIndex.to_sql(sTable, conn1, if_exists="replace")
DimPersonIndex.to_sql(sTable, conn3, if_exists="replace")
#####
```

Output:

You have successfully performed data vault to data warehouse transformation.

Simple Linear Regression

Linear regression is used if there is a relationship or significant association between the variables. This can be checked by scatterplots. If no linear association appears between the

variables, fitting a linear regression model to the data will not provide a useful model. A linear regression line has equations in the following form:

$$Y = a + bX,$$

Where, X = explanatory variable and

Y = dependent variable b

= slope of the line

a = intercept (the value of y when x = 0)

```
#####
```

```
# -*- coding: utf-8 -*-
```

```
#####
```

```
import sys import os import pandas as pd
```

```
import sqlite3 as sq import
```

```
matplotlib.pyplot as plt
```

```
import numpy as np
```

```
from sklearn import datasets, linear_model
```

```
from sklearn.metrics import mean_squared_error, r2_score
```

```
#####
```

```
Base='C:/VKHCG'
```

```
print('#####') print('Working
```

```
Base:',Base, ' using ', sys.platform)
```

```
print('#####')
```

```
#####
```

```
#####
```

```
Company='01-Vermeulen'
```

```
#####
```

```
sDataBaseDir=Base + '/' + Company + '/04-Transform/SQLite' if
```

```
not os.path.exists(sDataBaseDir):
```

```
os.makedirs(sDataBaseDir)
```

```
#####
```

```
sDatabaseName=sDataBaseDir + '/Vermeulen.db' conn1
```

```
= sq.connect(sDatabaseName)
```

```
#####
```

```
sDataVaultDir=Base + '/88-DV' if not os.path.exists(sDataVaultDir):
```

```
os.makedirs(sDataVaultDir)
```

```
#####
```

```
sDatabaseName=sDataVaultDir + '/datavault.db' conn2
```

```
= sq.connect(sDatabaseName)
```

```
#####
```

```
sDataWarehouseDir=Base + '/99-DW' if
```

```
not os.path.exists(sDataWarehouseDir):
```

```
os.makedirs(sDataWarehouseDir)
```

```
#####
```

```
sDatabaseName=sDataWarehouseDir + '/datawarehouse.db' conn3
```

```
= sq.connect(sDatabaseName)
```

```
#####
```

```
t=0 tMax=((300-100)/10)*((300-30)/5) for heightSelect in range(100,300,10):
```

```
for weightSelect in range(30,300,5): height = round(heightSelect/100,3)
```

```
weight = int(weightSelect) bmi = weight/(height*height) if bmi <= 18.5:
```

```
BMI_Result=1 elif bmi > 18.5 and bmi < 25: BMI_Result=2 elif bmi > 25
```

```
and bmi < 30: BMI_Result=3 elif bmi > 30:
```

```
BMI_Result=4 else:
```

```
BMI_Result=0
```

```
PersonLine=[('PersonID', [str(t)]),
```

```
('Height', [height]),
```

```

('Weight', [weight]),
('bmi', [bmi]),
('Indicator', [BMI_Result])
t+=1
print('Row:',t,'of',tMax) if
t==1:

PersonFrame = pd.DataFrame.from_items(PersonLine) else:
PersonRow = pd.DataFrame.from_items(PersonLine)
PersonFrame = PersonFrame.append(PersonRow)
#####
DimPerson=PersonFrame
DimPersonIndex=DimPerson.set_index(['PersonID'],inplace=False)
#####
sTable = 'Transform-BMI' print("\n#####")
print('Storing :',sDatabaseName,'\n Table:',sTable)
print("\n#####")
DimPersonIndex.to_sql(sTable, conn1, if_exists="replace")
#####
#####
sTable = 'Person-Satellite-BMI'
print("\n#####") print('Storing
:',sDatabaseName,'\n Table:',sTable)
print("\n#####")
DimPersonIndex.to_sql(sTable, conn2, if_exists="replace")
#####
#####
sTable = 'Dim-BMI' print("\n#####")
print('Storing :',sDatabaseName,'\n Table:',sTable)
print("\n#####")
DimPersonIndex.to_sql(sTable, conn3, if_exists="replace")
#####
fig = plt.figure()
PlotPerson=DimPerson[DimPerson['Indicator']==1]
x=PlotPerson['Height']
y=PlotPerson['Weight'] plt.plot(x,
y, ".")
PlotPerson=DimPerson[DimPerson['Indicator']==2]
x=PlotPerson['Height']
y=PlotPerson['Weight'] plt.plot(x,
y, "o")
PlotPerson=DimPerson[DimPerson['Indicator']==3]
x=PlotPerson['Height'] y=PlotPerson['Weight']
plt.plot(x, y, "+")
PlotPerson=DimPerson[DimPerson['Indicator']==4]
x=PlotPerson['Height']
y=PlotPerson['Weight']
plt.plot(x, y, "^")
plt.axis('tight') plt.title("BMI
Curve")
plt.xlabel("Height(meters)")
plt.ylabel("Weight(kg)")

plt.plot()
# Load the diabetes dataset diabetes =
datasets.load_diabetes() # Use only one feature

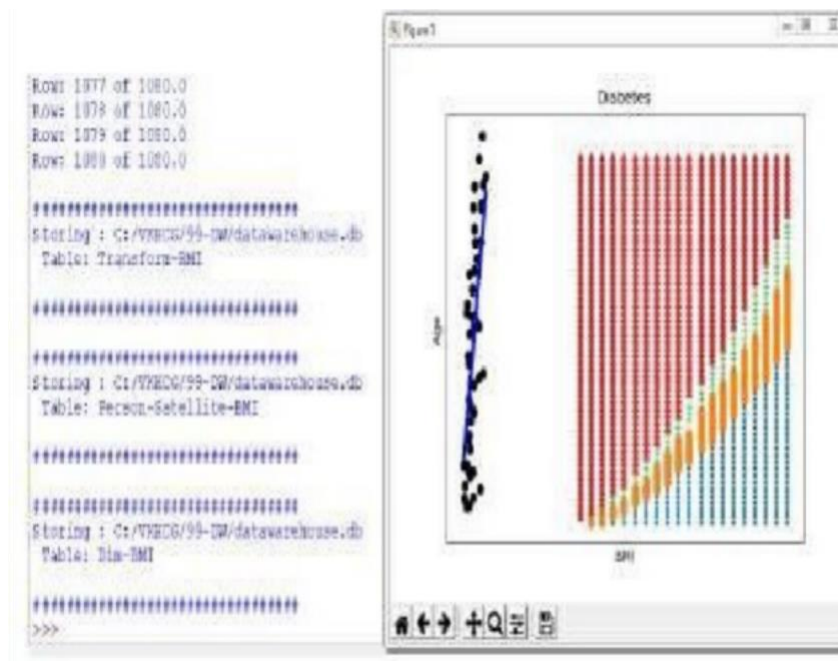
```

```

diabetes_X = diabetes.data[:, np.newaxis, 2]
diabetes_X_train = diabetes_X[:-30]
diabetes_X_test = diabetes_X[-50:]
diabetes_y_train = diabetes.target[:-30]
diabetes_y_test = diabetes.target[-50:] regr =
linear_model.LinearRegression()
regr.fit(diabetes_X_train, diabetes_y_train)
diabetes_y_pred = regr.predict(diabetes_X_test)
print('Coefficients: \n', regr.coef_) print("Mean
squared error: %.2f"
% mean_squared_error(diabetes_y_test, diabetes_y_pred))
print('Variance score: %.2f % r2_score(diabetes_y_test, diabetes_y_pred))
plt.scatter(diabetes_X_test, diabetes_y_test, color='black')
plt.plot(diabetes_X_test, diabetes_y_pred, color='blue', linewidth=3)
plt.xticks(()) plt.yticks(()) plt.axis('tight') plt.title("Diabetes")
plt.xlabel("BMI") plt.ylabel("Age") plt.show()

```

Output:



Practical 8:

Organizing Data

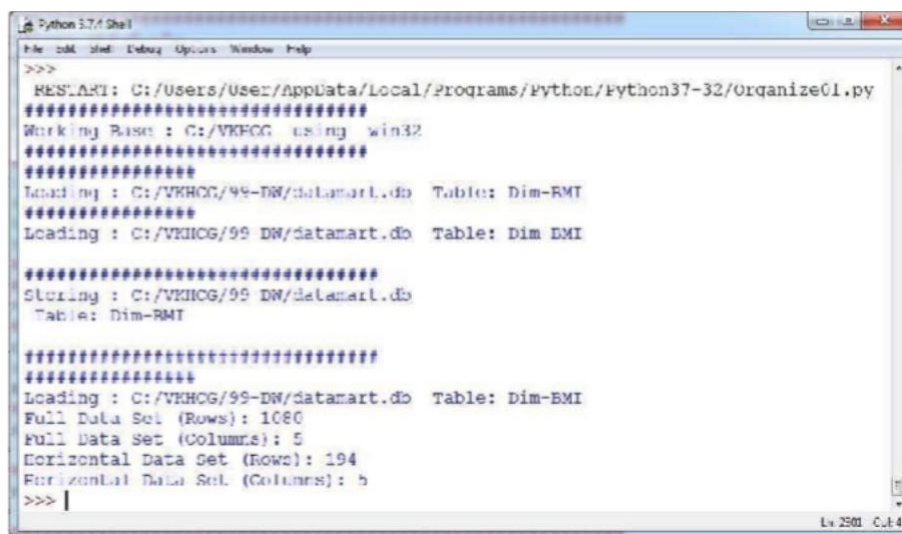
C:\VKHCG\01- Vermeulen\05-Organise\ Organize

Horizontal.py

```
#####
import sys import os import pandas as pd import sqlite3 as sq
#####
Base='C:/VKHCG'
print('#####') print('Working
Base:',Base, ' using ', sys.platform)
print('#####')
#####
#####
Company='01-Vermeulen'
#####
sDataWarehouseDir=Base + '/99-DW' if not
os.path.exists(sDataWarehouseDir): os.makedirs(sDataWarehouseDir)
#####
sDatabaseName=sDataWarehouseDir + '/datawarehouse.db' conn1
= sq.connect(sDatabaseName)
#####
sDatabaseName=sDataWarehouseDir + '/datamart.db' conn2
= sq.connect(sDatabaseName)
#####
print('#####') sTable = 'Dim-BMI'
print('Loading:',sDatabaseName,' Table:',sTable)
sSQL="SELECT * FROM [Dim-BMI];"
PersonFrame0=pd.read_sql_query(sSQL, conn1)
print('#####') sTable = 'Dim-BMI'
print('Loading:',sDatabaseName,' Table:',sTable)
sSQL="SELECT PersonID,\
Height,\
Weight,\ bmi,\
Indicator\
FROM [Dim-BMI]\
WHERE \
Height > 1.5 \ and
Indicator = 1\
ORDER BY \
Height,\
Weight;"
PersonFrame1=pd.read_sql_query(sSQL, conn1)
#####
DimPerson=PersonFrame1
DimPersonIndex=DimPerson.set_index(['PersonID'],inplace=False)
```

```
#####
sTable = 'Dim-BMI' print('\n#####')
print('Storing :',sDatabaseName,'\n Table:',sTable)
print('\n#####')
#DimPersonIndex.to_sql(sTable, conn2, if_exists="replace")
#####
print('#####')

sTable = 'Dim-BMI'
print('Loading :',sDatabaseName,' Table:',sTable) sSQL="SELECT
* FROM [Dim-BMI];"
PersonFrame2=pd.read_sql_query(sSQL, conn2) print('Full
Data Set (Rows):', PersonFrame0.shape[0]) print('Full Data
Set (Columns):', PersonFrame0.shape[1]) print('Horizontal
Data Set (Rows):', PersonFrame2.shape[0])
print('Horizontal Data Set (Columns):', PersonFrame2.shape[1])
```



```
Python 3.7.4 Shell
File Edit Shell Debug Options Window Help
>>>
RESTART: C:/Users/User/AppData/Local/Programs/Python/Python37-32/Organize01.py
#####
Working Base : C:/VKHCG using win32
#####
Loading : C:/VKHCG/99-DW/datamart.db Table: Dim-BMT
#####
Loading : C:/VKHCG/99-DW/datamart.db Table: Dim-BMI
#####
Storing : C:/VKHCG/99-DW/datamart.db
Table: Dim-BMT
#####
Loading : C:/VKHCG/99-DW/datamart.db Table: Dim-BMI
Full Data Set (Rows): 1080
Full Data Set (Columns): 5
Horizontal Data Set (Rows): 194
Horizontal Data Set (Columns): 5
>>> |
```

Vertical Style

C:\VKHCG\01-Vermeulen\05-Organise\ Organize-Vertical.py

```
import sys import os import pandas as pd import sqlite3 as sq
#####
Base='C:/VKHCG'
print('#####') print('Working
Base :',Base, ' using ', sys.platform)
print('#####')
#####
#####
Company='01-Vermeulen'
#####
sDataWarehouseDir=Base + '/99-DW' if not
os.path.exists(sDataWarehouseDir):
os.makedirs(sDataWarehouseDir)
#####
sDatabaseName=sDataWarehouseDir + '/datawarehouse.db' conn1
= sq.connect(sDatabaseName)
#####
```

```

sDatabaseName=sDataWarehouseDir + '/datamart.db' conn2
= sq.connect(sDatabaseName)
#####
print('#####')
sTable = 'Dim-BMI' print('Loading
:',sDatabaseName,' Table:',sTable)
sSQL="SELECT * FROM [Dim-BMI];"
PersonFrame0=pd.read_sql_query(sSQL, conn1)
#####

print('#####')
sTable = 'Dim-BMI' print('Loading
:',sDatabaseName,' Table:',sTable)
print('#####')
sSQL="SELECT \
Height,\
Weight,\
Indicator\
FROM [Dim-BMI];"
PersonFrame1=pd.read_sql_query(sSQL, conn1)
#####
DimPerson=PersonFrame1
DimPersonIndex=DimPerson.set_index(['Indicator'],inplace=False)
#####
sTable = 'Dim-BMI-Vertical'
print('\n#####') print('Storing
:',sDatabaseName,'\n Table:',sTable)
print('\n#####')
DimPersonIndex.to_sql(sTable, conn2, if_exists="replace")
#####
print('#####') sTable = 'Dim-BMI-Vertical'
print('Loading :',sDatabaseName,' Table:',sTable) sSQL="SELECT
* FROM [Dim-BMI-Vertical];"
PersonFrame2=pd.read_sql_query(sSQL, conn2)
#####
print('#####') print('Full Data
Set (Rows):', PersonFrame0.shape[0]) print('Full Data Set
(Columns):', PersonFrame0.shape[1])
print('#####')
print('Horizontal Data Set (Rows):', PersonFrame2.shape[0])
print('Horizontal Data Set (Columns):', PersonFrame2.shape[1])
print('#####')
#####

```

OUTPUT:

```
===== RESTART: C:\VKHCG\01-Vermeulen\05-Organise\Organize-Vertical.py =====
#####
Working Base : C:/VKHCG using win32
#####
Loading : C:/VKHCG/99-DW/datamart.db Table: Dim-BMI
#####
Loading : C:/VKHCG/99-DW/datamart.db Table: Dim-BMI
#####
Storing : C:/VKHCG/99-DW/datamart.db
Table: Dim-BMI-Vertical
#####
#####
Loading : C:/VKHCG/99-DW/datamart.db Table: Dim-BMI-Vertical
#####
Full Data Set (Rows): 1080
Full Data Set (Columns): 5
#####
Horizontal Data Set (Rows): 1080
Horizontal Data Set (Columns): 3
#####
```

Island Style

C:\VKHCG\01-Vermeulen\05-Organise\ Organize-Island.py

```
import sys import os import pandas as pd import sqlite3 as sq
#####
Base='C:/VKHCG'
print('#####') print('Working
Base :',Base, ' using ', sys.platform)
print('#####')
#####
#####

Company='01-Vermeulen'
#####
sDataWarehouseDir=Base + '/99-DW' if not
os.path.exists(sDataWarehouseDir):
os.makedirs(sDataWarehouseDir)
#####
sDatabaseName=sDataWarehouseDir + '/datawarehouse.db' conn1
= sq.connect(sDatabaseName)
#####
sDatabaseName=sDataWarehouseDir + '/datamart.db' conn2
= sq.connect(sDatabaseName)
#####
print('#####') sTable = 'Dim-BMI'
print('Loading :',sDatabaseName,' Table:',sTable) sSQL="SELECT
* FROM [Dim-BMI];"
PersonFrame0=pd.read_sql_query(sSQL, conn1)
#####
print('#####') sTable = 'Dim-BMI'
print('Loading :',sDatabaseName,' Table:',sTable)
sSQL="SELECT \
Height,\
Weight,\
Indicator\
FROM [Dim-BMI]\
WHERE Indicator > 2\
```



```

ORDER BY \
Height,\
Weight;"
PersonFrame1=pd.read_sql_query(sSQL, conn1)
#####
DimPerson=PersonFrame1
DimPersonIndex=DimPerson.set_index(['Indicator'],inplace=False)
#####
sTable = 'Dim-BMI-Vertical'
print('\n#####') print('Storing
:',sDatabaseName,'\n Table:',sTable)
print('\n#####')
DimPersonIndex.to_sql(sTable, conn2, if_exists="replace")
#####
print('#####')
sTable = 'Dim-BMI-Vertical' print('Loading
:',sDatabaseName,' Table:',sTable)
print('#####')
sSQL="SELECT * FROM [Dim-BMI-Vertical];"
PersonFrame2=pd.read_sql_query(sSQL, conn2)
#####
print('#####') print('Full Data
Set (Rows):', PersonFrame0.shape[0]) print('Full Data Set
(Columns):', PersonFrame0.shape[1])
print('#####')
print('Horizontal Data Set (Rows):', PersonFrame2.shape[0])
print('Horizontal Data Set (Columns):', PersonFrame2.shape[1])
print('#####')
##### Output:

```

```

Python 3.7.4 Shell
File Edit Shell Debug Options Window Help
>>>
===== RESTART: C:/VKHCG/01-Vermeulen/05-Organise/Organize-Island.py =====
Working Base : C:/VKHCG using win32
#####
Loading : C:/VKHCG/99-DW/datamart.db Table: Dim-BMI
#####
Loading : C:/VKHCG/99-DW/datamart.db Table: Dim-BMI

#####
Storing : C:/VKHCG/99-DW/datamart.db
Table: Dim-BMI-Vertical

#####
Loading : C:/VKHCG/99-DW/datamart.db Table: Dim-BMI-Vertical
#####
Full Data Set (Rows): 1080
Full Data Set (Columns): 5
#####
Horizontal Data Set (Rows): 771
Horizontal Data Set (Columns): 3
#####
>>>

```

Secure Vault Style

C:\VKHCG\01-Vermeulen\05-Organise\ Organize-SecureVault.py

```

import sys import
os import pandas
as pd import
sqlite3 as sq

```



```
#####
Base='C:/VKHCG'
print('#####') print('Working
Base:',Base, ' using ', sys.platform)
print('#####')
#####
#####
Company='01-Vermeulen'
#####
sDataWarehouseDir=Base + '/99-DW' if not
os.path.exists(sDataWarehouseDir):
os.makedirs(sDataWarehouseDir)
#####
sDatabaseName=sDataWarehouseDir + '/datawarehouse.db' conn1
= sq.connect(sDatabaseName)
#####
sDatabaseName=sDataWarehouseDir + '/datamart.db' conn2
= sq.connect(sDatabaseName)
#####
print('#####') sTable = 'Dim-BMI' print('Loading
:',sDatabaseName,' Table:',sTable) sSQL="SELECT * FROM [Dim-BMI];"
PersonFrame0=pd.read_sql_query(sSQL, conn1)
#####
print('#####') sTable = 'Dim-BMI' print('Loading
:',sDatabaseName,' Table:',sTable)
sSQL="SELECT \
Height,\
Weight,\
Indicator,\
CASE Indicator\
WHEN 1 THEN 'Pip'\
WHEN 2 THEN 'Norman'\
WHEN 3 THEN 'Grant'\ ELSE
'Sam'\

END AS Name\
FROM [Dim-BMI]\
WHERE Indicator > 2\
ORDER BY \
Height,\
Weight;"
PersonFrame1=pd.read_sql_query(sSQL, conn1)
#####
DimPerson=PersonFrame1
DimPersonIndex=DimPerson.set_index(['Indicator'],inplace=False)
#####
sTable = 'Dim-BMI-Secure' print("\n#####")
print('Storing :',sDatabaseName,"\n Table:",sTable)
print("\n#####")
DimPersonIndex.to_sql(sTable, conn2, if_exists="replace")
```

```
#####
print('#####')
sTable = 'Dim-BMI-Secure' print('Loading
:',sDatabaseName,' Table:',sTable)
print('#####')
sSQL="SELECT * FROM [Dim-BMI-Secure] WHERE Name = 'Sam';"
PersonFrame2=pd.read_sql_query(sSQL, conn2)
#####
print('#####') print('Full Data
Set (Rows):', PersonFrame0.shape[0]) print('Full Data Set
(Columns):', PersonFrame0.shape[1])
print('#####')
print('Horizontal Data Set (Rows):', PersonFrame2.shape[0])
print('Horizontal Data Set (Columns):', PersonFrame2.shape[1])
print('Only Sam Data') print(PersonFrame2.head())
print('#####')
##### Output:
```

```
Python 3.7.4 Shell
File Edit Shell Debug Options Window Help
>>>
==== RESTART: C:\VKHCG\01-Vermeulen\05-Organise\Organize-Secure-Vault.py ====
#####
Working Base : C:/VKHCG using win32
#####
Loading : C:/VKHCG/99-DW/datamart.db Table: Dim-BMI
#####
Loading : C:/VKHCG/99-DW/datamart.db Table: Dim-BMI
#####
Storing : C:/VKHCG/99-DW/datamart.db
Table: Dim-BMI-Secure
#####
Loading : C:/VKHCG/99-DW/datamart.db Table: Dim-BMI-Secure
#####
Full Data Set (Rows): 1080
Full Data Set (Columns): 5
#####
Horizontal Data Set (Rows): 692
Horizontal Data Set (Columns): 4
Only Sam Data
Indicator Height Weight Name
0 4 1.0 35 Sam
1 4 1.0 40 Sam
2 4 1.0 45 Sam
3 4 1.0 50 Sam
4 4 1.0 55 Sam
#####
```

Association Rule Mining

C:\VKHCG\01-Vermeulen\05-Organise\ Organize-Association-

Rule.py import

sys import os

import pandas as pd

from

mlxtend.frequent_p

attners import

apriori from

mlxtend.frequent_p

attners import

association_rules

#####

Base='C:/VKHCG'

print('#####') print('Working

```

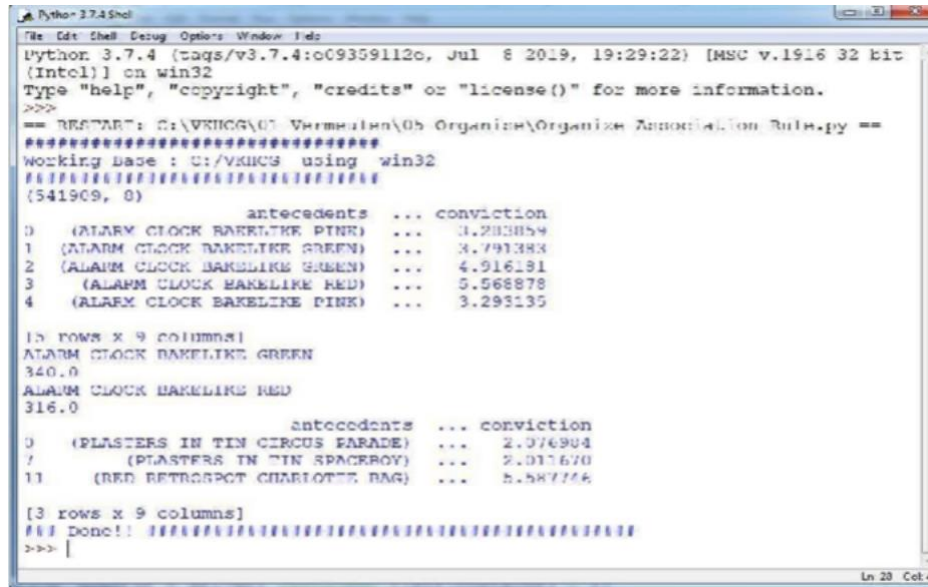
Base :,Base, ' using ', sys.platform)
print('#####')
#####
Company='01-Vermeulen'
InputFileName='Online-Retail-Billboard.xlsx'
EDSAssessDir='02-Assess/01-EDS'
InputAssessDir=EDSAssessDir + '/02-Python'
#####
sFileAssessDir=Base + '/' + Company + '/' + InputAssessDir if
not os.path.exists(sFileAssessDir):
os.makedirs(sFileAssessDir)
#####
sFileName=Base+'/' + Company + '/00-RawData/' + InputFileName
##### df
= pd.read_excel(sFileName)
print(df.shape)
#####
df['Description'] = df['Description'].str.strip()
df.dropna(axis=0, subset=['InvoiceNo'], inplace=True)
df['InvoiceNo'] = df['InvoiceNo'].astype('str') df =
df[~df['InvoiceNo'].str.contains('C')] basket =
(df[df['Country'] == "France"] .groupby(['InvoiceNo',
'Description']))['Quantity']
.sum().unstack().reset_index().fillna(0)
.set_index('InvoiceNo'))
#####
def encode_units(x): if x <= 0: return 0 if x >= 1: return 1
#####
basket_sets = basket.applymap(encode_units) basket_sets.drop('POSTAGE',
inplace=True, axis=1)
frequent_itemsets = apriori(basket_sets, min_support=0.07, use_colnames=True)
rules = association_rules(frequent_itemsets, metric="lift", min_threshold=1)
print(rules.head()) rules[ (rules['lift'] >= 6) & (rules['confidence'] >= 0.8) ]
#####
sProduct1='ALARM CLOCK BAKELIKE GREEN'
print(sProduct1) print(basket[sProduct1].sum())
sProduct2='ALARM CLOCK BAKELIKE RED' print(sProduct2)
print(basket[sProduct2].sum())
##### basket2
= (df[df['Country'] == "Germany"]
.groupby(['InvoiceNo', 'Description'])['Quantity']
.sum().unstack().reset_index().fillna(0)
.set_index('InvoiceNo')) basket_sets2 = basket2.applymap(encode_units)
basket_sets2.drop('POSTAGE', inplace=True, axis=1) frequent_itemsets2 =
apriori(basket_sets2, min_support=0.05, use_colnames=True) rules2 =
association_rules(frequent_itemsets2, metric="lift", min_threshold=1)

print(rules2[ (rules2['lift'] >= 4) &
(rules2['confidence'] >= 0.5)])
##### print('###
Done!! #####')

```

#####

Output:



```
Python 2.7.4 Shell
File Edit Shell Debug Options Window Help
Python 3.7.4 (tags/v3.7.4:0c9359112c, Jul 8 2019, 19:29:22) [MSC v.1916 32 bit
Type "help", "copyright", "credits" or "license()" for more information.
>>>
== RESTART: C:\VKHCG\01-Vermeulen\05-Organise\Organise-NetworkRouting-Company.py ==
#####
Working base : C:/VKHCG using win32
#####
(541909, 0)

antecedents ... conviction
0 (ALARM CLOCK BAKELIKE PINK) ... 3.283854
1 (ALARM CLOCK BAKELIKE GREEN) ... 3.791383
2 (ALARM CLOCK BAKELIKE GREEN) ... 4.916181
3 (ALARM CLOCK BAKELIKE RED) ... 5.568878
4 (ALARM CLOCK BAKELIKE PINK) ... 3.293135

[5 rows x 3 columns]
ALARM CLOCK BAKELIKE GREEN
340.0
ALARM CLOCK BAKELIKE RED
316.0

antecedents ... conviction
0 (PLASTERS IN TIN CIRCUS FARADE) ... 2.076904
1 (PLASTERS IN TIN SPACEROY) ... 2.013670
11 (RED RETROSPOT CHARLOTTE BAG) ... 2.587766

[3 rows x 3 columns]
### Done!! #####
>>> |
```

Create a Network Routing Diagram

I will guide you through a possible solution for the requirement, by constructing an island-style Organise superstep that uses a graph data model to reduce the records and the columns on the data set.

C:\VKHCG\01-Vermeulen\05-Organise\ Organise-NetworkRouting-Company.py

```
import sys import os import
pandas as pd import
networkx as nx import
matplotlib.pyplot as plt

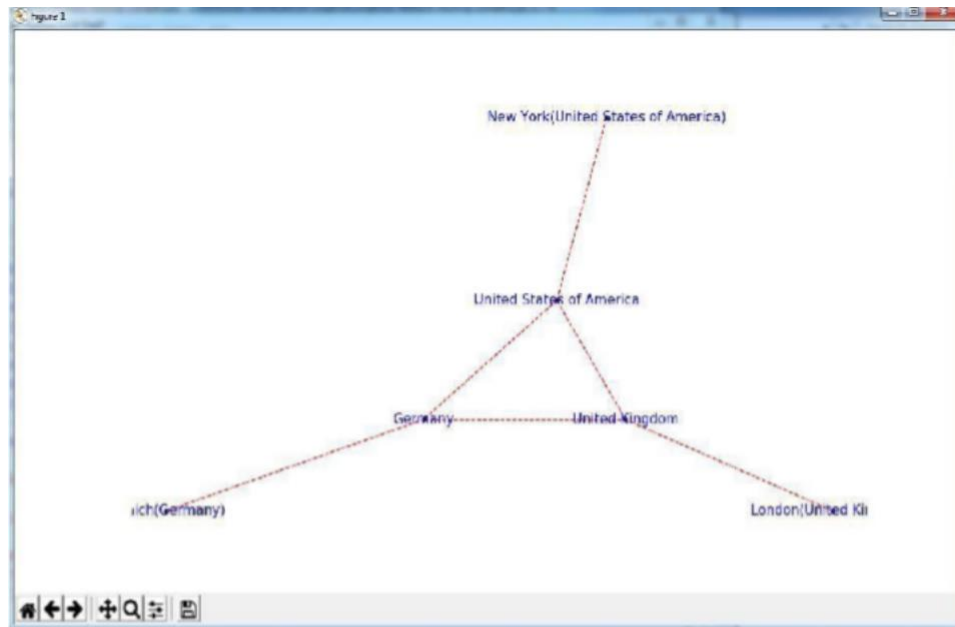
#####
pd.options.mode.chained_assignment = None
#####
Base='C:/VKHCG'
#####
print('#####') print('Working
Base :',Base, ' using ', sys.platform)
print('#####')
#####
sInputFileName='02-Assess/01-EDS/02-Python/Assess-Network-Routing-Company.csv'
#####
sOutputFileName1='05-Organise/01-EDS/02-Python/Organise-Network-
RoutingCompany.gml'
sOutputFileName2='05-Organise/01-EDS/02-Python/Organise-Network-Routing-
Company.png'
Company='01-Vermeulen'
#####
#####
### Import Country Data
#####
```

```

sFileName=Base + '/' + Company + '/' + sInputFileName
print('#####') print('Loading
:',sFileName)

print('#####')
CompanyData=pd.read_csv(sFileName,header=0,low_memory=False, encoding="latin-1")
print('#####') print(CompanyData.head())
print(CompanyData.shape) G=nx.Graph() for i in range(CompanyData.shape[0]): for j in
range(CompanyData.shape[0]):
Node0=CompanyData['Company_Country_Name'][i]
Node1=CompanyData['Company_Country_Name'][j] if
Node0 != Node1:
G.add_edge(Node0,Node1) for i in
range(CompanyData.shape[0]):
Node0=CompanyData['Company_Country_Name'][i]
Node1=CompanyData['Company_Place_Name'][i] + '('+
CompanyData['Company_Country_Name'][i] + ')' if
Node0 != Node1:
G.add_edge(Node0,Node1) print('Nodes:',
G.number_of_nodes()) print('Edges:',
G.number_of_edges())
sFileName=Base + '/' + Company + '/' + sOutputFileName1
print('#####') print('Storing
:',sFileName)
print('#####')
nx.write_gml(G, sFileName)
sFileName=Base + '/' + Company + '/' + sOutputFileName2
print('#####') print('Storing
Graph Image:',sFileName)
print('#####')
plt.figure(figsize=(15, 15)) pos=nx.spectral_layout(G,dim=2)
nx.draw_networkx_nodes(G,pos, node_color='k', node_size=10, alpha=0.8)
nx.draw_networkx_edges(G, pos,edge_color='r', arrows=False, style='dashed')
nx.draw_networkx_labels(G,pos,font_size=12,font_family='sans-serif,font_color='b')
plt.axis('off')
plt.savefig(sFileName,dpi=600)
plt.show()
print('#####') print('###
Done!! #####')
print('#####')

```



Picking Content for Billboards C:\VKHCG\02-Krennwallner\05-Organise\ Organisebillboards.py

```
#####
import sys import os import pandas as pd import networkx as nx import
matplotlib.pyplot as plt import numpy as np
#####
pd.options.mode.chained_assignment = None
#####
Base='C:/VKHCG'
#####
print('#####') print('Working
Base :',Base, ' using ', sys.platform)
print('#####')
#####
sInputFileName='02-Assess/01-EDS/02-Python/Assess-DE-Billboard-Visitor.csv'
#####
sOutputFileName1='05-Organise/01-EDS/02-Python/Organise-Billboards.gml'
sOutputFileName2='05-Organise/01-EDS/02-Python/Organise-Billboards.png'
Company='02-Krennwallner'
#####
#####
### Import Company Data
#####
sFileName=Base + '/' + Company + '/' + sInputFileName
print('#####') print('Loading
:',sFileName)
print('#####')
BillboardDataRow=pd.read_csv(sFileName,header=0,low_memory=False, encoding="latin-
1")
print('#####')
#####
print(BillboardDataRow.head()) print(BillboardDataRow.shape)
BillboardData=BillboardDataRow
```

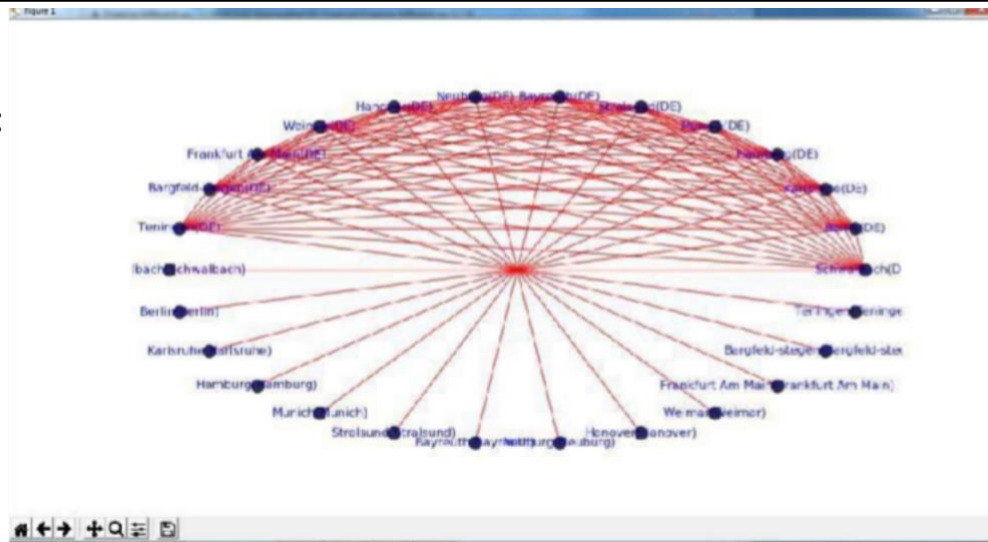
```

sSample=list(np.random.choice(BillboardData.shape[0],20))
#####
G=nx.Graph() for
i in sSample: for
j in sSample:
Node0=BillboardData['BillboardPlaceName'][i] + '('+ BillboardData['BillboardCountry'][i] +
')'
Node1=BillboardData['BillboardPlaceName'][j] + '('+ BillboardData['BillboardCountry'][i] +
')'
if Node0 != Node1:
G.add_edge(Node0,Node1) for
i in sSample:
Node0=BillboardData['BillboardPlaceName'][i] + '('+ BillboardData['VisitorPlaceName'][i] +
')'
Node1=BillboardData['BillboardPlaceName'][i] + '('+ BillboardData['VisitorCountry'][i] + ')'
if Node0 != Node1:
G.add_edge(Node0,Node1) print('Nodes:',
G.number_of_nodes())
print('Edges:', G.number_of_edges())
#####
sFileName=Base + '/02-Krennwallner/' + sOutputFileName1
print('#####') print('Storing
:',sFileName)

print('#####') nx.write_gml(G,
sFileName)
#####
sFileName=Base + '/02-Krennwallner/' + sOutputFileName2
print('#####')
print('Storing Graph Image:',sFileName)
print('#####')
plt.figure(figsize=(15, 15)) pos=nx.circular_layout(G,dim=2)
nx.draw_networkx_nodes(G,pos, node_color='k', node_size=150, alpha=0.8)
nx.draw_networkx_edges(G, pos,edge_color='r', arrows=False, style='solid')
nx.draw_networkx_labels(G,pos,font_size=12,font_family='sans-serif,font_color='b')
plt.axis('off')
plt.savefig(sFileName,dpi=600)
plt.show()
#####
print('#####') print('###
Done!! #####')
print('#####')
#####

```


OUTPUT:



Create a Delivery Route C:\VKHCG\03-Hillman\05-Organise\Organise-Routes.py

```
import sys
import os
import pandas as pd

#####
Base='C:/VKHCG'
#####
print('#####') print('Working
Base : ',Base, ' using ', sys.platform)
print('#####')
#####
sInputFileName='02-Assess/01-EDS/02-Python/Assess_Shipping_Routes.txt'
#####
sOutputFileName='05-Organise/01-EDS/02-Python/Organise-Routes.csv'
Company='03-Hillman'
#####
#####
### Import Routes Data
#####
sFileName=Base + '/' + Company + '/' + sInputFileName
print('#####')

print('Loading : ',sFileName)
print('#####')
RouteDataRow=pd.read_csv(sFileName,header=0,low_memory=False, sep='|',
encoding="latin-1")
print('#####')
#####
RouteStart=RouteDataRow[RouteDataRow['StartAt']=='WH-KA13']
#####
RouteDistance=RouteStart[RouteStart['Cost']=='DistanceMiles']
RouteDistance=RouteDistance.sort_values(by=['Measure'], ascending=False)
#####
RouteMax=RouteStart["Measure"].max()
RouteMaxCost=round((((RouteMax/1000)*1.5*2)),2)
print('#####') print('Maximum
(£) per day:')
```

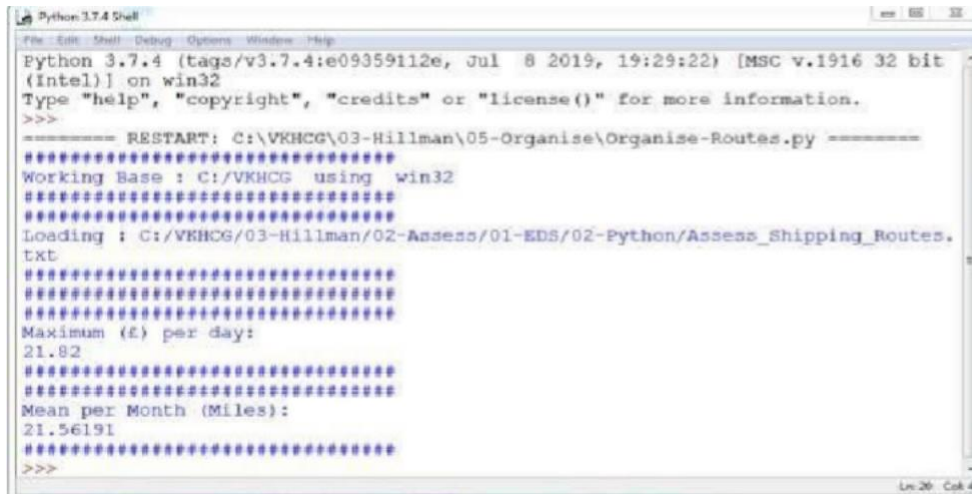


```

print(RouteMaxCost)
print('#####')
#####
RouteMean=RouteStart["Measure"].mean()
RouteMeanMonth=round((((RouteMean/1000)*2*30)),6)
print('#####')
print('Mean per Month (Miles):') print(RouteMeanMonth)
print('#####')

```

Output:



```

Python 3.7.4 Shell
File Edit Shell Debug Options Window Help
Python 3.7.4 (tags/v3.7.4:e09359112e, Jul 8 2019, 19:29:22) [MSC v.1916 32 bit
(Intel)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: C:\VKHCG\03-Hillman\05-Organise\Organise-Routes.py =====
#####
Working Base : C:/VKHCG using win32
#####
Loading : C:/VKHCG/03-Hillman/02-Assess/01-EDS/02-Python/Assess_Shipping_Routes.
txt
#####
Maximum (£) per day:
21.82
#####
Mean per Month (Miles):
21.56191
#####
>>>

```

Clark Ltd

Our financial services company has been tasked to investigate the options to convert 1 million pounds sterling into extra income. Mr. Clark Junior suggests using the simple variance in the daily rate between the British pound sterling and the US dollar, to generate extra income from trading. Your chief financial officer wants to know if this is feasible?

Simple Forex Trading Planner

Your challenge is to take 1 million US dollars or just over six hundred thousand pounds sterling and, by simply converting it between pounds sterling and US dollars, achieve a profit. Are you up to this challenge?

The Program will help you how to model this problem and achieve a positive outcome. The forex data has been collected on a daily basis by Clark's accounting department, from previous overseas transactions.

C:\VKHCG\04-Clark\05-Organise\Organise-Forex.py

```

import sys import os import pandas as pd import sqlite3 as sq import re
#####
Base='C:/VKHCG'
#####

print('#####') print('Working
Base:',Base, ' using ', sys.platform)
print('#####')
#####
sInputFileName='03-Process/01-EDS/02-Python/Process_ExchangeRates.csv'

```

```
#####
sOutputFileName='05-Organise/01-EDS/02-Python/Organise-Forex.csv'
Company='04-Clark'
#####
sDatabaseName=Base + '/' + Company + '/05-Organise/SQLite/clark.db' conn
= sq.connect(sDatabaseName)
#conn = sq.connect(':memory:')
#####
#####
### Import Forex Data
#####
sFileName=Base + '/' + Company + '/' + sInputFileName
print('#####') print('Loading
:',sFileName)
print('#####')
ForexDataRow=pd.read_csv(sFileName,header=0,low_memory=False,encoding="latin-1")
print('#####')
#####
ForexDataRow.index.names = ['RowID']
sTable='Forex_All'
print('Storing :',sDatabaseName,' Table:',sTable) ForexDataRow.to_sql(sTable,
conn, if_exists="replace")
#####
sSQL="SELECT 1 as Bag\
, CAST(min(Date) AS VARCHAR(10)) as Date \
,CAST(1000000.0000000 as NUMERIC(12,4)) as Money \
,'USD' as Currency \
FROM Forex_All \
;"
sSQL=re.sub("\s\s+", " ", sSQL) nMoney=pd.read_sql_query(sSQL,
conn)
#####
nMoney.index.names = ['RowID'] sTable='MoneyData'
print('Storing :',sDatabaseName,' Table:',sTable)
nMoney.to_sql(sTable, conn, if_exists="replace")
#####
sTable='TransactionData' print('Storing :',sDatabaseName,' Table:',sTable)
nMoney.to_sql(sTable, conn, if_exists="replace")
#####
ForexDay=pd.read_sql_query("SELECT Date FROM Forex_All GROUP BY Date;", conn)
#####
t=0 for i in range(ForexDay.shape[0]):
sDay1=ForexDay['Date'][i]
sDay=str(sDay1) sSQL='\
SELECT M.Bag as Bag, \ F.Date
as Date, \
round(M.Money * F.Rate,6) AS Money, \
F.CodeIn AS PCurrency, \
F.CodeOut AS Currency \
FROM MoneyData AS M \
```

```

JOIN \ ( \
SELECT CodeIn, CodeOut, Date, Rate FROM Forex_All WHERE
CodeIn = "USD" AND CodeOut = "GBP" \
UNION \
SELECT CodeOut AS CodeIn, CodeIn AS CodeOut, Date, (1/Rate) AS Rate FROM \
Forex_All WHERE CodeIn = "USD" AND CodeOut = "GBP" \
) AS F \
ON \
M.Currency=F.CodeIn \
AND \
F.Date="" +sDay + "";' sSQL=re.sub("\s\s+",
" ", sSQL)
ForexDayRate=pd.read_sql_query(sSQL, conn) for
j in range(ForexDayRate.shape[0]):
sBag=str(ForexDayRate['Bag'][j])
nMoney=str(round(ForexDayRate['Money'][j],2))
sCodeIn=ForexDayRate['PCurrency'][j] sCodeOut=ForexDayRate['Currency'][j]
sSQL='UPDATE MoneyData SET Date= "" + sDay + "", ' sSQL= sSQL + '
Money = ' + nMoney + ', Currency="" + sCodeOut + ""' sSQL= sSQL + '
WHERE Bag=' + sBag + ' AND Currency="" + sCodeIn + "";'
sSQL=re.sub("\s\s+", " ", sSQL) cur = conn.cursor() cur.execute(sSQL)
conn.commit() t+=1
print('Trade:', t, sDay, sCodeOut, nMoney) sSQL='
\
INSERT INTO TransactionData ( \
RowID, \
Bag, \
Date, \
Money, \
Currency \
) \
SELECT ' + str(t) + ' AS RowID, \
Bag, \
Date, \
Money, \
Currency \
FROM MoneyData ;'
sSQL=re.sub("\s\s+", " ", sSQL)
cur = conn.cursor()
cur.execute(sSQL) conn.commit()
#####
sSQL="SELECT RowID, Bag, Date, Money, Currency FROM TransactionData ORDER BY
RowID;"
sSQL=re.sub("\s\s+", " ", sSQL)
TransactionData=pd.read_sql_query(sSQL, conn)
OutputFile=Base + '/' + Company + '/' + sOutputFileName
TransactionData.to_csv(OutputFile, index = False)
#####

```

Practical 9

Generating Data

Report Superstep

The Report superstep is the step in the ecosystem that enhances the data science findings with the art of storytelling and data visualization. You can perform the best data science, but if you cannot execute a respectable and trustworthy Report step by turning your data science into actionable business insights, you have achieved no advantage for your business.

Vermeulen PLC

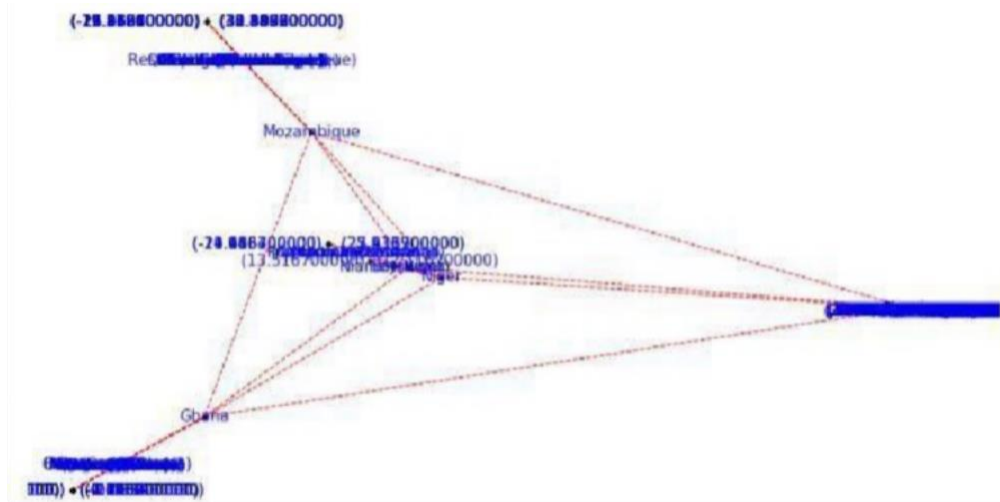
Vermeulen requires a map of all their customers" data links. Can you provide a report to deliver this? I will guide you through an example that delivers this requirement.

C:\VKHCG\01-Vermeulen\06-Report\Report-Network-RoutingCustomer.py

```
#####
import sys import os import pandas as pd import networkx as nx import matplotlib.pyplot
as plt
#####
pd.options.mode.chained_assignment = None
##### if
sys.platform == 'linux':
Base=os.path.expanduser('~') + 'VKHCG' else:
Base='C:/VKHCG'
#####
print('#####') print('Working
Base:',Base, ' using ', sys.platform)
print('#####')
#####
sInputFileName='02-Assess/01-EDS/02-Python/Assess-Network-Routing-Customer.csv'
#####
sOutputFileName1='06-Report/01-EDS/02-Python/Report-Network-Routing-Customer.gml'
sOutputFileName2='06-Report/01-EDS/02-Python/Report-Network-Routing-Customer.png'
Company='01-Vermeulen'
#####
#####
### Import Country Data
#####
sFileName=Base + '/' + Company + '/' + sInputFileName
print('#####') print('Loading
:',sFileName)
print('#####')
CustomerDataRow=pd.read_csv(sFileName,header=0,low_memory=False,encoding="latin-
1")
CustomerData=CustomerDataRow.head(100) print('Loaded
Country:',CustomerData.columns.values)
print('#####')
```

```
#####
print(CustomerData.head())
print(CustomerData.shape)
#####
G=nx.Graph() for i in
range(CustomerData.shape[0]): for j
in range(CustomerData.shape[0]):
Node0=CustomerData['Customer_Country_Name'][i]
Node1=CustomerData['Customer_Country_Name'][j] if
Node0 != Node1:

G.add_edge(Node0,Node1)
for i in range(CustomerData.shape[0]):
Node0=CustomerData['Customer_Country_Name'][i]
Node1=CustomerData['Customer_Place_Name'][i] + '('+
CustomerData['Customer_Country_Name'][i] + ')'
Node2=('(' + "{:.9f}".format(CustomerData['Customer_Latitude'][i]) + ')\
(' + "{:.9f}".format(CustomerData['Customer_Longitude'][i]) + ')') if
Node0 != Node1:
G.add_edge(Node0,Node1) if
Node1 != Node2:
G.add_edge(Node1,Node2) print('Nodes:',
G.number_of_nodes())
print('Edges:', G.number_of_edges())
#####
sFileName=Base + '/' + Company + '/' + sOutputFileName1
print('#####') print('Storing
:',sFileName)
print('#####') nx.write_gml(G,
sFileName)
#####
sFileName=Base + '/' + Company + '/' + sOutputFileName2
print('#####') print('Storing
Graph Image:',sFileName)
print('#####')
plt.figure(figsize=(25, 25)) pos=nx.spectral_layout(G,dim=2)
nx.draw_networkx_nodes(G,pos, node_color='k', node_size=10, alpha=0.8)
nx.draw_networkx_edges(G, pos,edge_color='r', arrows=False, style='dashed')
nx.draw_networkx_labels(G,pos,font_size=12,font_family='sans-serif,font_color='b')
plt.axis('off')
plt.savefig(sFileName,dpi=600)
plt.show()
print('#####') print('###
Done!! #####')
print('#####')
```



Krennwallner AG

The Krennwallner marketing department wants to deploy the locations of the billboards onto the company web server. Can you prepare three versions of the locations" web pages?

- Locations clustered into bubbles when you zoom out
- Locations as pins
- Locations as heat map

Picking Content for Billboards

C:\VKHCG\02-Krennwallner\06-Report\Report_Billboard.py

```
import sys import os import pandas as pd
```

```
from folium.plugins import FastMarkerCluster, HeatMap
from folium import Marker, Map import webbrowser
#####
Base='C:/VKHCG'
print('#####') print('Working
Base :,Base, ' using ', sys.platform)
print('#####')
#####
sFileName=Base+'/02-Krennwallner/01-Retrieve/01-
EDS/02Python/Retrieve_DE_Billboard_Locations.csv'
df = pd.read_csv(sFileName,header=0,low_memory=False, encoding="latin-1")
df.fillna(value=0, inplace=True) print(df.shape)
#####
t=0 for i in range(df.shape[0]):
try:
sLongitude=df["Longitude"][i] sLongitude=float(sLongitude)
except Exception: sLongitude=float(0.0) try:
sLatitude=df["Latitude"][i]
sLatitude=float(sLatitude)
except Exception:
sLatitude=float(0.0) try:
```

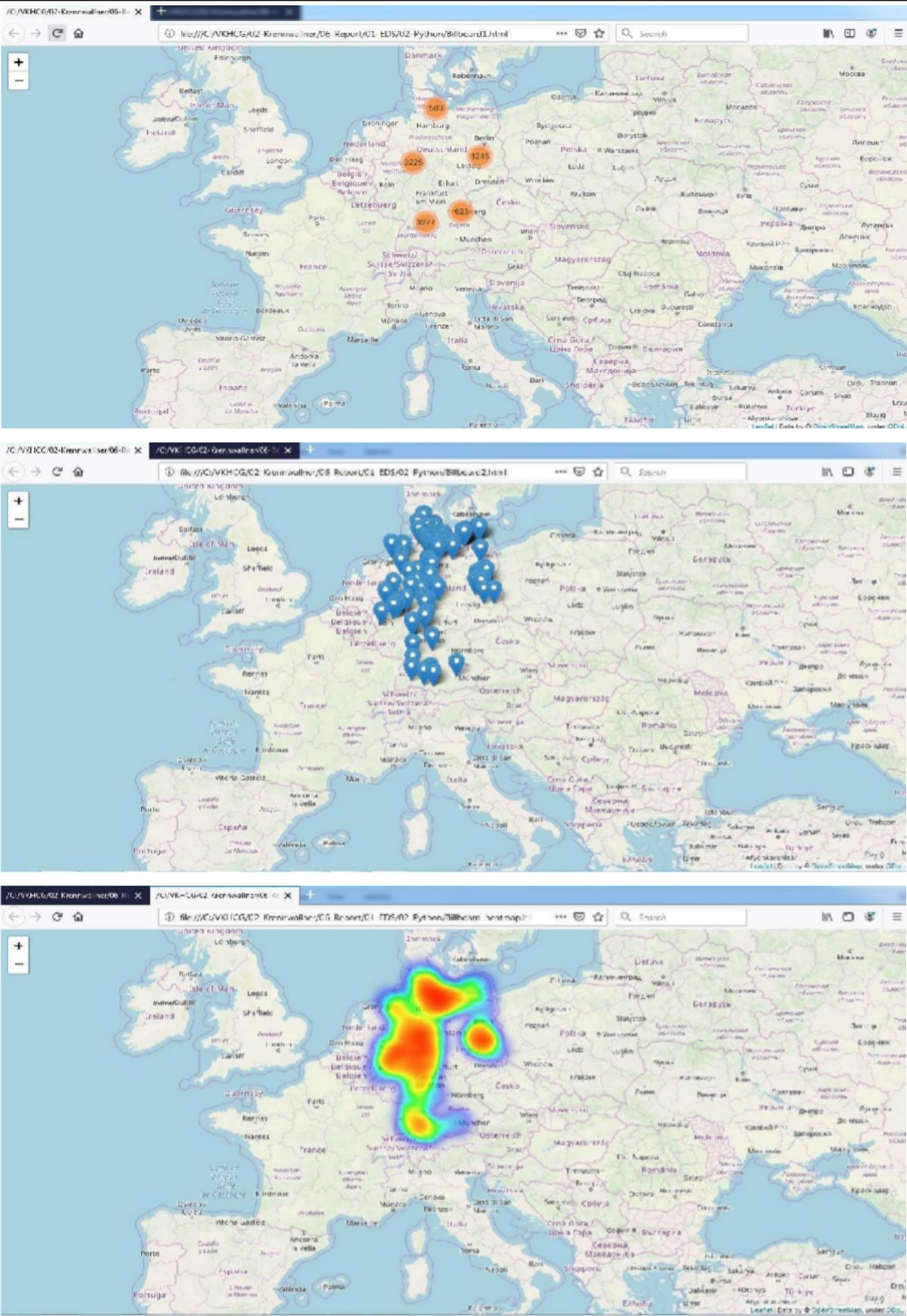
```

sDescription=df["Place_Name"][i] + '(' + df["Country"][i]+')' except
Exception:
sDescription='VKHCG' if sLongitude != 0.0
and sLatitude != 0.0:
DataClusterList=list([sLatitude, sLongitude])
DataPointList=list([sLatitude, sLongitude, sDescription])
t+=1 if t==1:
DataCluster=[DataClusterList]
DataPoint=[DataPointList] else:
DataCluster.append(DataClusterList)
DataPoint.append(DataPointList)
data=DataCluster pins=pd.DataFrame(DataPoint)
pins.columns = [ 'Latitude','Longitude','Description']
#####
stops_map1 = Map(location=[48.1459806, 11.4985484], zoom_start=5) marker_cluster
= FastMarkerCluster(data).add_to(stops_map1)
sFileNameHtml=Base+'/02-Krennwallner/06-Report/01-EDS/02-Python/Billboard1.html'
stops_map1.save(sFileNameHtml)
webbrowser.open('file://' + os.path.realpath(sFileNameHtml))
#####
stops_map2 = Map(location=[48.1459806, 11.4985484], zoom_start=5) for
name, row in pins.iloc[:100].iterrows():
Marker([row["Latitude"],row["Longitude"]],
popup=row["Description"]).add_to(stops_map2)
sFileNameHtml=Base+'/02-Krennwallner/06-Report/01-EDS/02-Python/Billboard2.html'
stops_map2.save(sFileNameHtml)
webbrowser.open('file://' + os.path.realpath(sFileNameHtml))
#####
stops_heatmap = Map(location=[48.1459806, 11.4985484], zoom_start=5)
stops_heatmap.add_child(HeatMap([[row["Latitude"], row["Longitude"]] for name, row in
pins.iloc[:100].iterrows()])))

sFileNameHtml=Base+'/02-Krennwallner/06-Report/01-
EDS/02Python/Billboard_heatmap.html'
stops_heatmap.save(sFileNameHtml)
webbrowser.open('file://' + os.path.realpath(sFileNameHtml))
##### print('###
Done!! #####')
#####

```


OUTPUT:



Hillman Ltd

Dr. Hillman Sr. has just installed a camera system that enables the company to capture video and, therefore, indirectly, images of all containers that enter or leave the warehouse. Can you convert the number on the side of the containers into digits?

Reading the Containers

C:\VKHCG\03-Hillman\06-Report\ Report_Reading_Container.py

```
from time import time
import numpy as np
import matplotlib.pyplot as plt
from matplotlib import offsetbox
from sklearn import (manifold, datasets, decomposition, ensemble, discriminant_analysis,
random_projection)
digits = datasets.load_digits(n_class=6)
X = digits.data
y = digits.target
n_samples, n_features = X.shape
n_neighbors = 30
def plot_embedding(X, title=None):
    x_min, x_max = np.min(X, 0), np.max(X, 0)
    X = (X - x_min) / (x_max - x_min)
    plt.figure(figsize=(10, 10))
    ax = plt.subplot(111)
    for i in range(X.shape[0]):
        plt.text(X[i, 0], X[i, 1], str(digits.target[i]), color=plt.cm.Set1(y[i] / 10.), fontdict={'weight': 'bold', 'size': 9})
    if hasattr(offsetbox, 'AnnotationBbox'):
        # only print thumbnails with matplotlib > 1.0
        shown_images = np.array([[1., 1.]])
        for i in range(digits.data.shape[0]):
            dist = np.sum((X[i] - shown_images) ** 2, 1)
            if np.min(dist) < 4e-3:
                # don't show points that are too close
                continue
            shown_images = np.r_[shown_images, [X[i]]]
            imagebox = offsetbox.AnnotationBbox(offsetbox.OffsetImage(digits.images[i], cmap=plt.cm.gray_r), X[i])
            ax.add_artist(imagebox)
        plt.xticks([], plt.yticks([]))
        if title is not None:
            plt.title(title)
    n_img_per_row = 20
    img = np.zeros((10 * n_img_per_row, 10 * n_img_per_row))
    for i in range(n_img_per_row):
        ix = 10 * i + 1
        for j in range(n_img_per_row):
            iy = 10 * j + 1
            img[ix:ix + 8, iy:iy + 8] = X[i * n_img_per_row + j].reshape((8, 8))
    plt.figure(figsize=(10, 10))
    plt.imshow(img, cmap=plt.cm.binary)
    plt.xticks([], plt.yticks([]))
    plt.title('A selection from the 64-dimensional digits dataset')
    print("Computing random projection")
    rp = random_projection.SparseRandomProjection(n_components=2, random_state=42)
    X_projected = rp.fit_transform(X)
    plot_embedding(X_projected, "Random Projection of the digits")
    print("Computing PCA projection")
```

```

t0 = time()
X_pca = decomposition.TruncatedSVD(n_components=2).fit_transform(X)
plot_embedding(X_pca,"Principal Components projection of the digits (time %.2fs)"
%(time() - t0))
print("Computing Linear Discriminant Analysis projection")
X2 = X.copy()
X2.flat[:,X.shape[1] + 1] += 0.01 # Make X invertible t0 = time() X_lda =
discriminant_analysis.LinearDiscriminantAnalysis(n_components=2).fit_transform(X2, y)
plot_embedding(X_lda,"Linear Discriminant projection of the digits (time %.2fs)" %(time() -
t0))
print("Computing Isomap embedding") t0
= time()
X_iso = manifold.Isomap(n_neighbors, n_components=2).fit_transform(X) print("Done.")
plot_embedding(X_iso,"Isomap projection of the digits (time %.2fs)" %(time() - t0))
print("Computing LLE embedding")
clf = manifold.LocallyLinearEmbedding(n_neighbors, n_components=2,method='standard')
t0 = time()
X_lle = clf.fit_transform(X)
print("Done. Reconstruction error: %g" % clf.reconstruction_error_)
plot_embedding(X_lle,"Locally Linear Embedding of the digits (time %.2fs)" %(time() - t0))
print("Computing modified LLE embedding")
clf = manifold.LocallyLinearEmbedding(n_neighbors, n_components=2,
method='modified') t0 = time()
X_mlle = clf.fit_transform(X)
print("Done. Reconstruction error: %g" % clf.reconstruction_error_)
plot_embedding(X_mlle,"Modified Locally Linear Embedding of the digits (time %.2fs)"
%(time() - t0))
print("Computing Hessian LLE embedding")
clf = manifold.LocallyLinearEmbedding(n_neighbors, n_components=2,method='hessian') t0
= time()
X_hlle = clf.fit_transform(X)
print("Done. Reconstruction error: %g" % clf.reconstruction_error_)
plot_embedding(X_hlle,"Hessian Locally Linear Embedding of the digits (time %.2fs)"
%(time() - t0))
print("Computing LTSA embedding")
clf = manifold.LocallyLinearEmbedding(n_neighbors, n_components=2,method='ltsa') t0
= time()
X_ltsa = clf.fit_transform(X) print("Done. Reconstruction error:
%g" % clf.reconstruction_error_) plot_embedding(X_ltsa,"Local
Tangent Space Alignment of the digits (time %.2fs)" %(time()
- t0))
print("Computing MDS embedding")
clf = manifold.MDS(n_components=2, n_init=1, max_iter=100) t0
= time()
X_mds = clf.fit_transform(X) print("Done.
Stress: %f" % clf.stress_)
plot_embedding(X_mds,"MDS embedding of the digits (time %.2fs)" %(time() - t0))
print("Computing Totally Random Trees embedding")

```

```

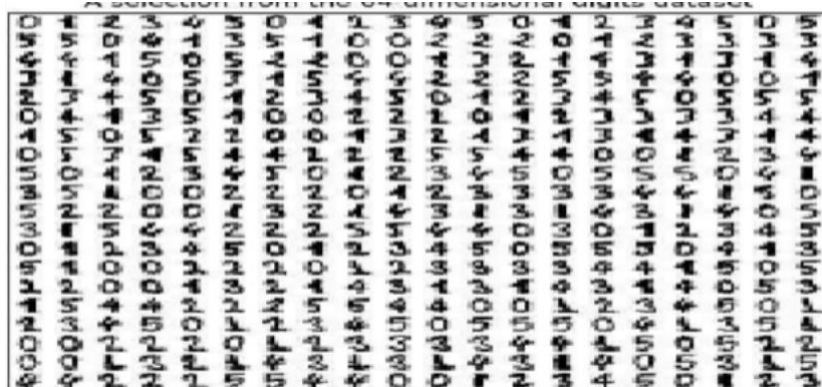
hasher = ensemble.RandomTreesEmbedding(n_estimators=200, random_state=0,
max_depth=5) t0 = time()
X_transformed = hasher.fit_transform(X)
pca = decomposition.TruncatedSVD(n_components=2) X_reduced
= pca.fit_transform(X_transformed)
plot_embedding(X_reduced,"Random forest embedding of the digits (time %.2fs)" %(time() -
t0))
print("Computing Spectral embedding")
embedder = manifold.SpectralEmbedding(n_components=2, random_state=0,
eigen_solver="arpack")
t0 = time()
X_se = embedder.fit_transform(X)
plot_embedding(X_se,"Spectral embedding of the digits (time %.2fs)" %(time() - t0))
print("Computing t-SNE embedding")
tsne = manifold.TSNE(n_components=2, init='pca', random_state=0) t0
= time()
X_tsne = tsne.fit_transform(X)
plot_embedding(X_tsne,"t-SNE embedding of the digits (time %.2fs)" %(time() - t0))
plt.show()

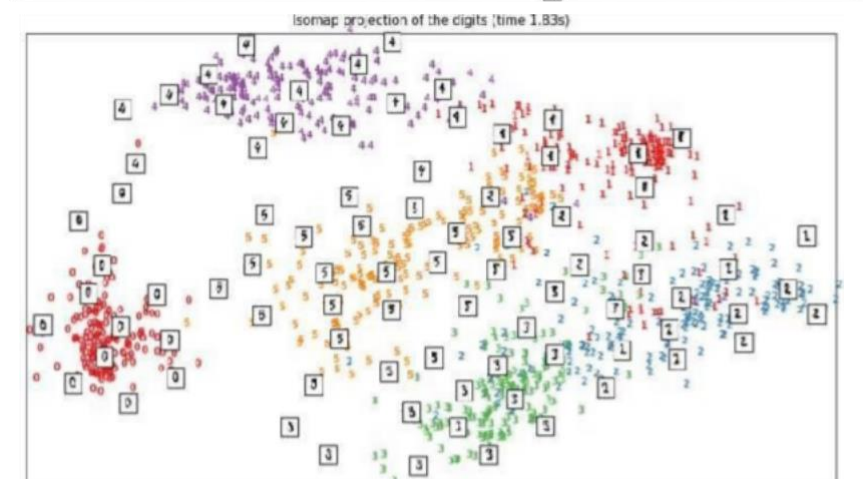
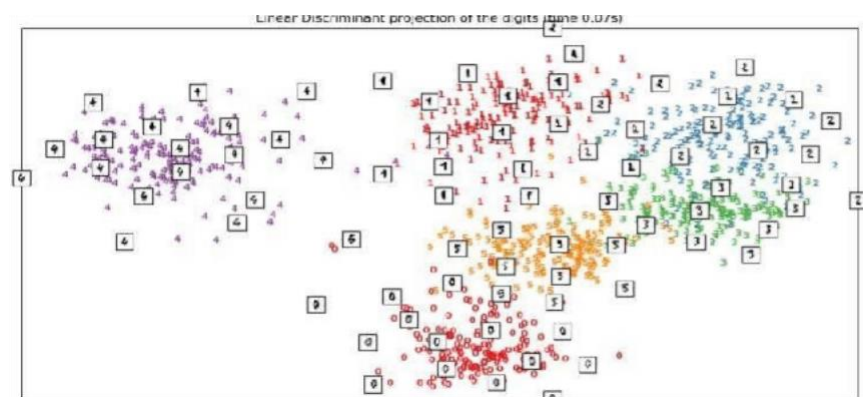
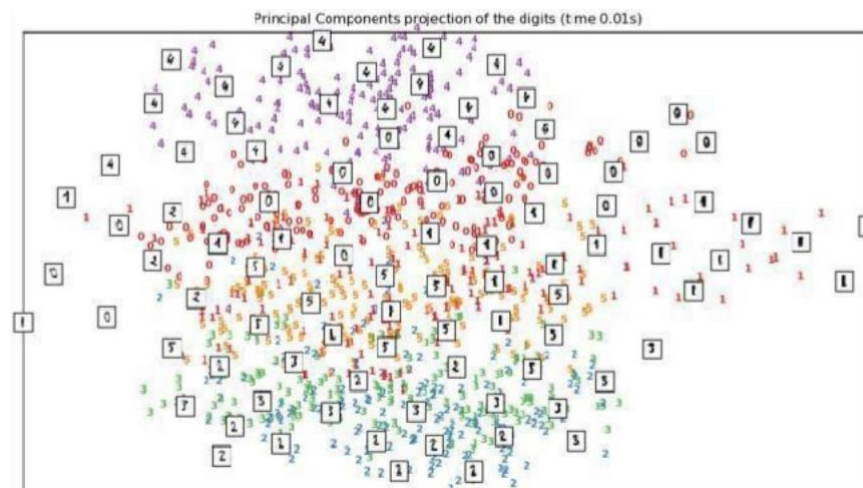
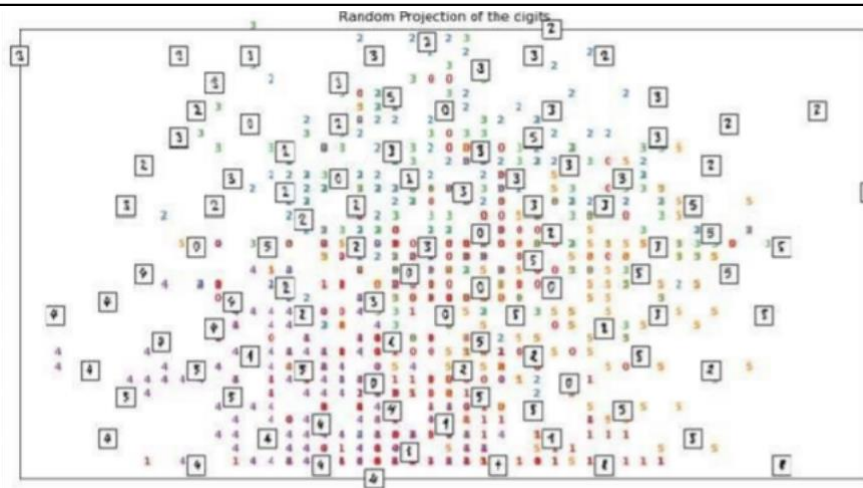
```

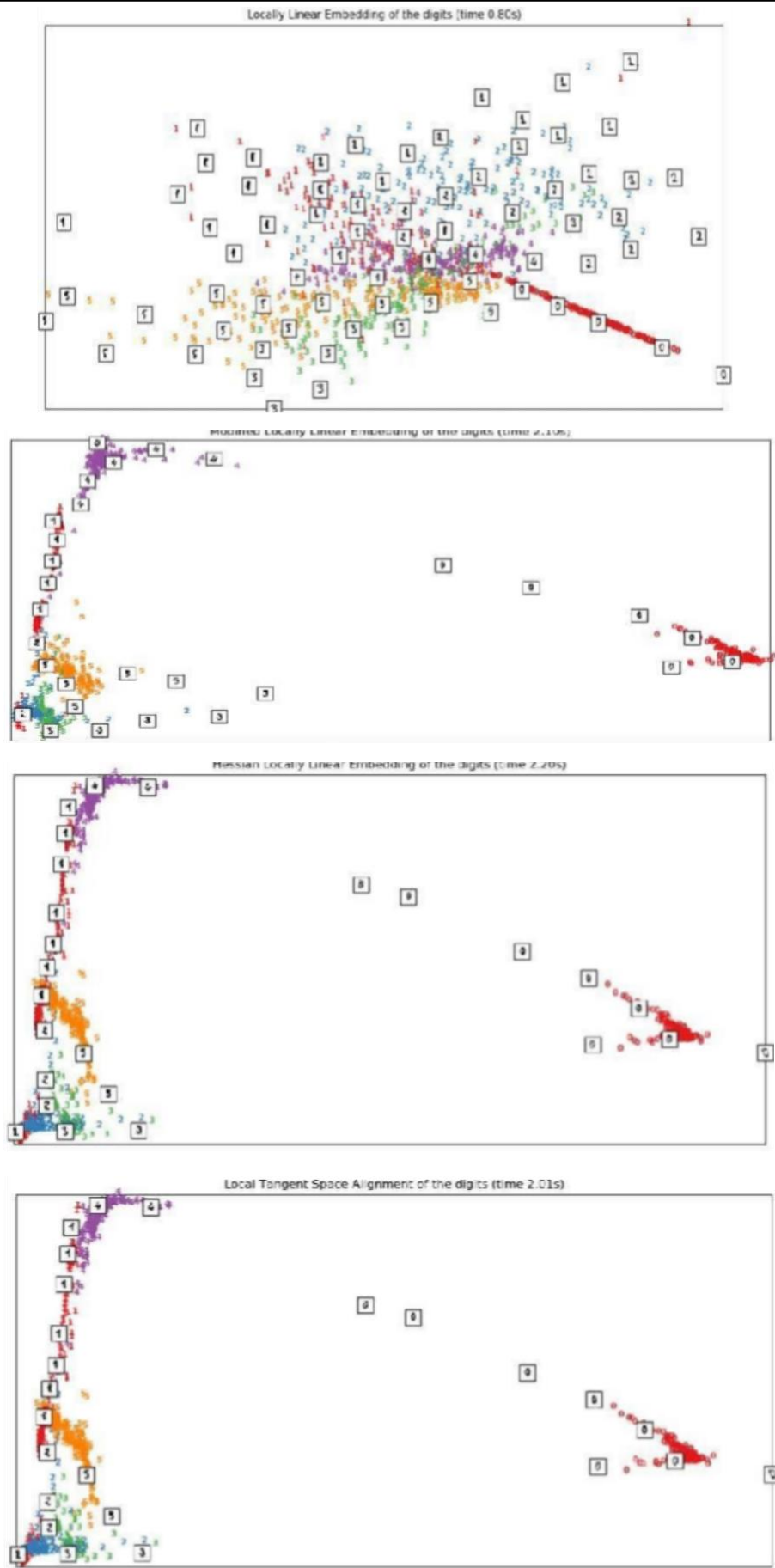
```

Python 3.7.4 Shell
File Edit Shell Debug Options Window Help
Python 3.7.4 (tags/v3.7.4:e09359112e, Jul 8 2019, 19:29:22) [MSC v.1916 32 bit
(Intel)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: C:/VKHCG/03-Hillman/06-Report/Report_Reading_Container.py =====
1. Computing random projection
2. Computing PCA projection
3. Computing Linear Discriminant Analysis projection
4. Computing Isomap embedding
Done.
5. Computing LLE embedding
Done. Reconstruction error: 1.63544e-06
6. Computing modified LLE embedding
Done. Reconstruction error: 0.360655
7. Computing Hessian LLE embedding
Done. Reconstruction error: 0.212804
8. Computing LTSA embedding
Done. Reconstruction error: 0.212804
9. Computing MDS embedding
Done. Stress: 136501329.149015
10. Computing Totally Random Trees embedding
11. Computing Spectral embedding
12. Computing t-SNE embedding
Ln: 3 Col: 4

```







Clark Ltd

The financial company in VKHCG is the Clark accounting firm that VKHCG owns with a 60% stake. The accountants are the financial advisers to the group and handle everything to do with the complex work of international accounting.

Financials

The VKHCG companies did well last year, and the teams at Clark must prepare a balance sheet for each company in the group. The companies require a balance sheet for each company, to be produced using the template (Balance-Sheet-Template.xlsx) that can be found in the example directory (..\VKHCG\04-Clark\00-RawData).

The Program will guide you through a process that will enable you to merge the data science with preformatted Microsoft Excel template, to produce a balance sheet for each of the VKHCG companies.

C:\VKHCG\04-Clark\06-Report\Report-Balance-Sheet.py

```
import sys import os import pandas as pd import sqlite3 as sq import re
from openpyxl import load_workbook
#####
Base='C:\VKHCG'
#####
print('#####') print('Working
Base:',Base, ' using ', sys.platform)
print('#####')
#####
sInputTemplateName='00-RawData/Balance-Sheet-Template.xlsx'
##### sOutputFileName='06-
Report/01-EDS/02-Python/Report-Balance-Sheet'

Company='04-Clark'
#####
sDatabaseName=Base + '/' + Company + '/06-Report/SQLite/clark.db' conn =
sq.connect(sDatabaseName)
#conn = sq.connect(':memory:')
#####
### Import Balance Sheet Data
##### for
y in range(1,13):
sInputFileName='00-RawData/BalanceSheets' + str(y).zfill(2) + '.csv'
sFileName=Base + '/' + Company + '/' + sInputFileName
print('#####') print('Loading
:',sFileName) print('#####')
ForexDataRow=pd.read_csv(sFileName,header=0,low_memory=False,encoding="latin-1")
print('#####')
#####
ForexDataRow.index.names = ['RowID']
sTable='BalanceSheets' print('Storing
:',sDatabaseName,' Table:',sTable) if y == 1:
print('Load Data')
ForexDataRow.to_sql(sTable, conn, if_exists="replace") else:
print('Append Data')
ForexDataRow.to_sql(sTable, conn, if_exists="append")
##### sSQL="SELECT
\
Year, \
Quarter, \
Country, \
Company, \
CAST(Year AS INT) || 'Q' || CAST(Quarter AS INT) AS sDate, \
Company || '(' || Country || ')' AS sCompanyName , \
```

```

CAST(Year AS INT) || 'Q' || CAST(Quarter AS INT) || '-' || \
Company || '-' || Country AS sCompanyFile \
FROM BalanceSheets \
GROUP BY \
Year, \
Quarter, \
Country, \
Company \
HAVING Year is not null \
;"
sSQL=re.sub("\s\s+", " ", sSQL) sDatesRaw=pd.read_sql_query(sSQL,
conn)
print(sDatesRaw.shape) sDates=sDatesRaw.head(5)
#####
## Loop Dates
#####
for i in range(sDates.shape[0]): sFileName=Base + '/' + Company + '/' +
sInputTemplateName wb = load_workbook(sFileName)
ws=wb.get_sheet_by_name("Balance-Sheet") sYear=sDates['sDate'][i]
sCompany=sDates['sCompanyName'][i]
sCompanyFile=sDates['sCompanyFile'][i] sCompanyFile=re.sub("\s+", "",
sCompanyFile)\

ws['D3'] = sYear ws['D5']
= sCompany
sFields =
pd.DataFrame( [
['Cash','D16', 1],
['Accounts_Receivable','D17', 1],
['Doubtful_Accounts','D18', 1],
['Inventory','D19', 1],
['Temporary_Investment','D20', 1],
['Prepaid_Expenses','D21', 1],
['Long_Term_Investments','D24', 1],
['Land','D25', 1], ['Buildings','D26',
1],
['Depreciation_Buildings','D27', -1],
['Plant_Equipment','D28', 1],
['Depreciation_Plant_Equipment','D29', -1],
['Furniture_Fixtures','D30', 1],
['Depreciation_Furniture_Fixtures','D31', -1],
['Accounts_Payable','H16', 1],
['Short_Term_Notes','H17', 1],
['Current_Long_Term_Notes','H18', 1],
['Interest_Payable','H19', 1],
['Taxes_Payable','H20', 1],
['Accrued_Payroll','H21', 1],
['Mortgage','H24', 1],
['Other_Long_Term_Liabilities','H25', 1],
['Capital_Stock','H30', 1]
]

```



```

)
nYear=str(int(sDates['Year'][i])) nQuarter=str(int(sDates['Quarter'][i]))
sCountry=str(sDates['Country'][i])
sCompany=str(sDates['Company'][i])
sFileName=Base + '/' + Company + '/' + sOutputFileName + \
'-' + sCompanyFile + '.xlsx'
print(sFileName) for j in
range(sFields.shape[0]):
sSumField=sFields[0][j]
sCellField=sFields[1][j] nSumSign=sFields[2][j]
sSQL="SELECT \
Year, \
Quarter, \
Country, \
Company, \
SUM(" + sSumField + ") AS nSumTotal \
FROM BalanceSheets \
GROUP BY \
Year, \
Quarter, \
Country, \
Company \ HAVING
\

Year=" + nYear + " \
AND \
Quarter=" + nQuarter + " \
AND \
Country=" + sCountry + " \
AND \
Company=" + sCompany + " \
;"
sSQL=re.sub("\s\s+", " ", sSQL)
sSumRaw=pd.read_sql_query(sSQL, conn) ws[sCellField]
= sSumRaw["nSumTotal"][0] * nSumSign print('Set
cell',sCellField,' to ', sSumField,'Total')
wb.save(sFileName)

```

Output:

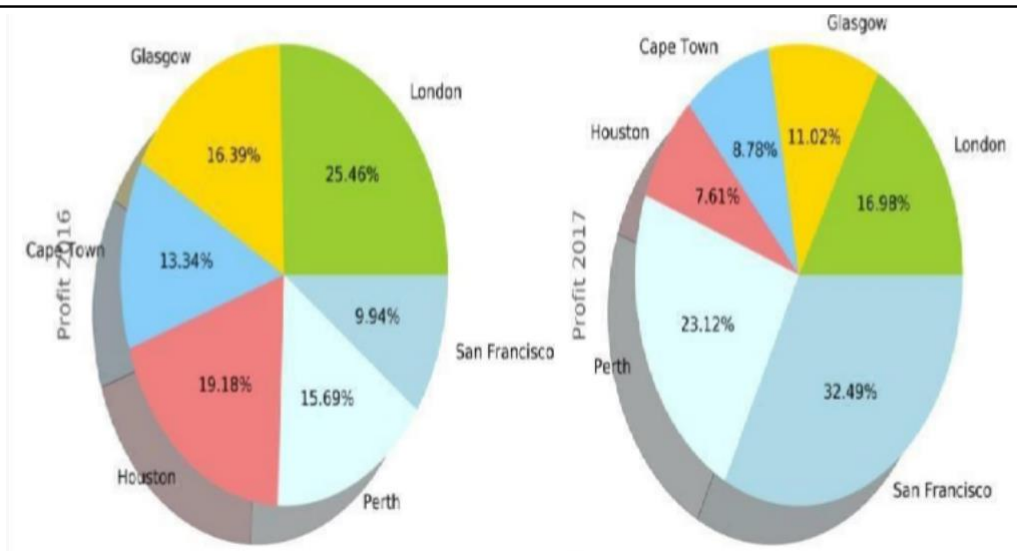
Graphics

This section will now guide you through a number of visualizations that particularly useful in presenting data to my customers.

Pie Graph

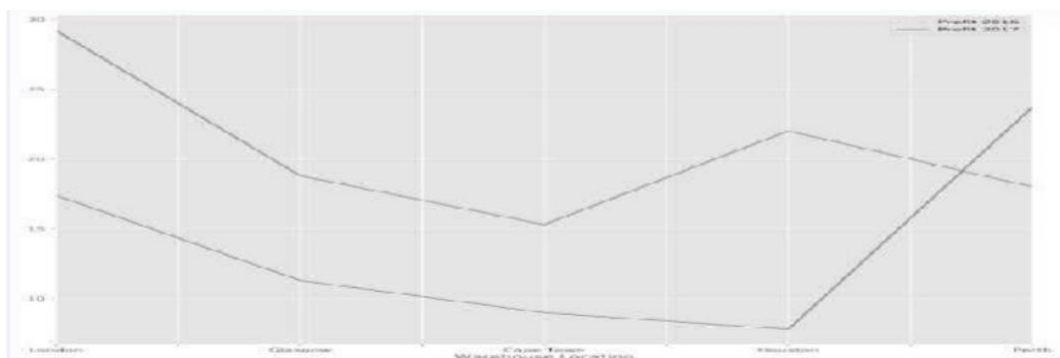
Double Pie

C:\VKHCG\01-Vermeulen\06-Report\Report_Graph_A.py



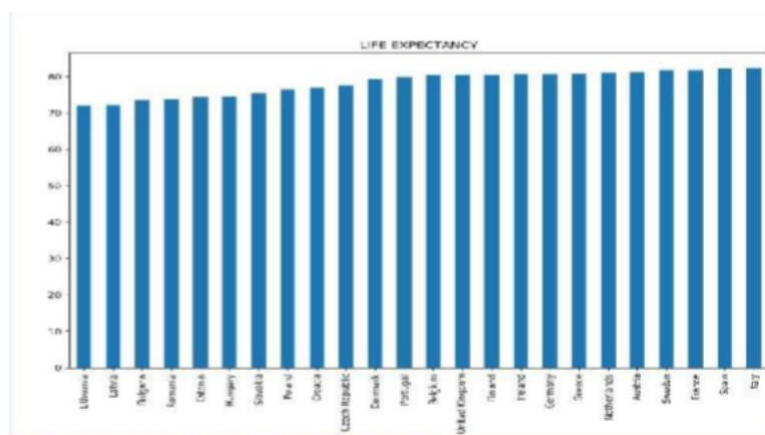
Line Graph

C:/VKHCG/01-Vermeulen/06-Report/Report_Graph_A.py



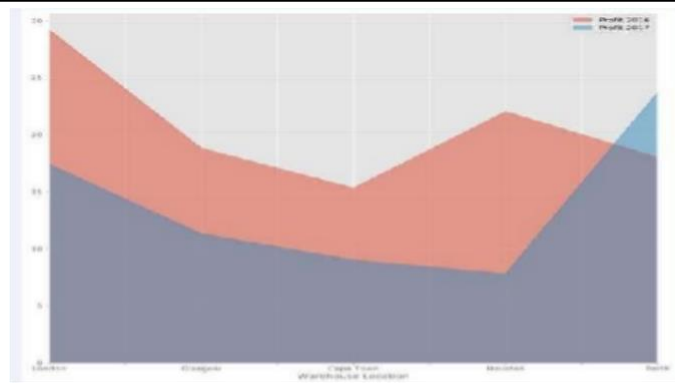
Bar Graph / Horizontal Bar Graph

C:/VKHCG/01-Vermeulen/06-Report/Report_Graph_A.py



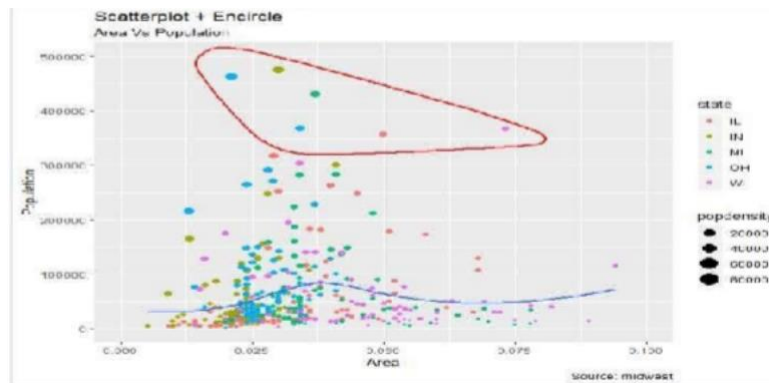
Area Graph

C:/VKHCG/01-Vermeulen/06-Report/Report_Graph_A.py



SCATTER GRAPH

C:/ VKHCG/03-HILLMAN/06-REPORT/REPORT-SCATTERPLOT-WITH-ENCIRCLING.R



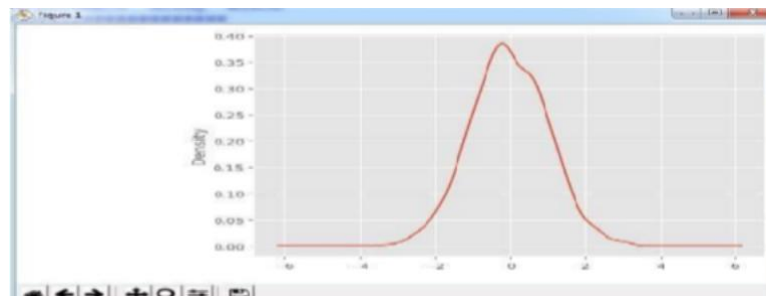
Hexbin:

Program : C:\VKHCG\01-Vermeulen\06-Report\Report_Graph_A.py



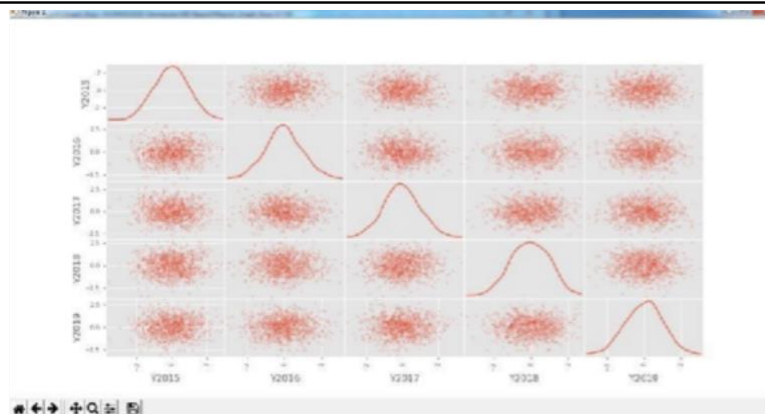
Kernel Density Estimation (KDE) Graph

C:\VKHCG\01-Vermeulen\06-Report\Report_Graph_B.py



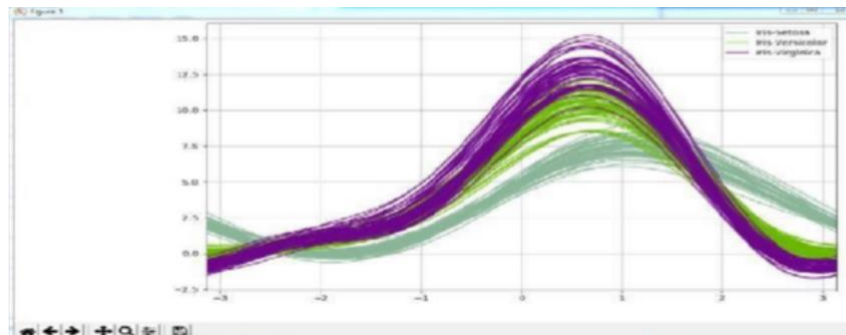
Scatter Matrix Graph

C:\VKHCG\01-Vermeulen\06-Report\Report_Graph_B.py



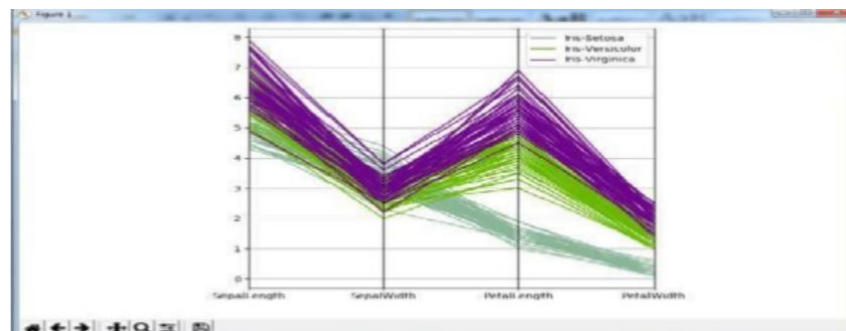
Andrews' Curves

C:\VKHCG\01-Vermeulen\06-Report\Report_Graph_C.py



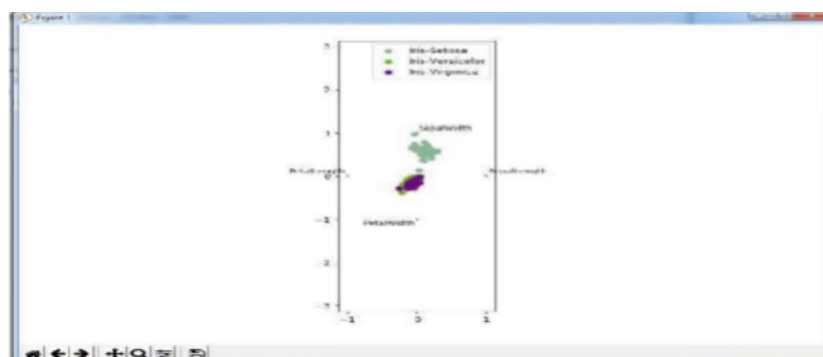
Parallel Coordinates

C:\VKHCG\01-Vermeulen\06-Report\Report_Graph_C.py



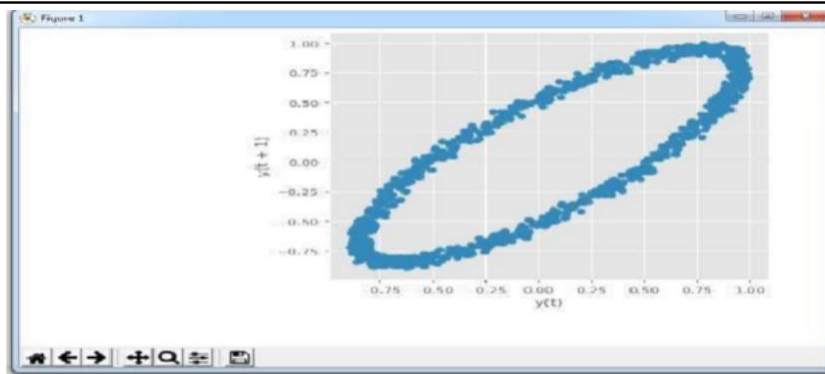
RADVIZ Method

C:\VKHCG\01-Vermeulen\06-Report\Report_Graph_C.py



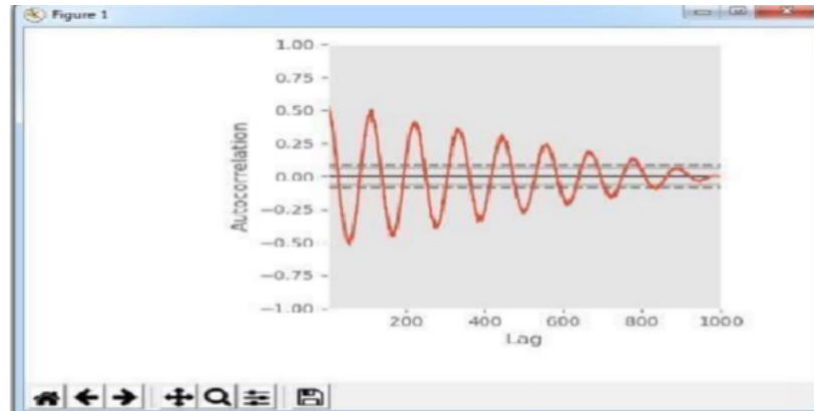
Lag Plot

C:\VKHCG\01-Vermeulen\06-Report\Report_Graph_D.py



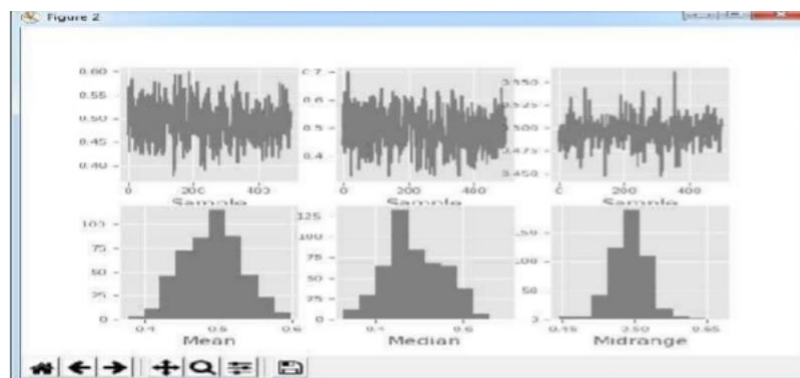
Autocorrelation Plot

C:\VKHCG\01-Vermeulen\06-Report\Report_Graph_D.py



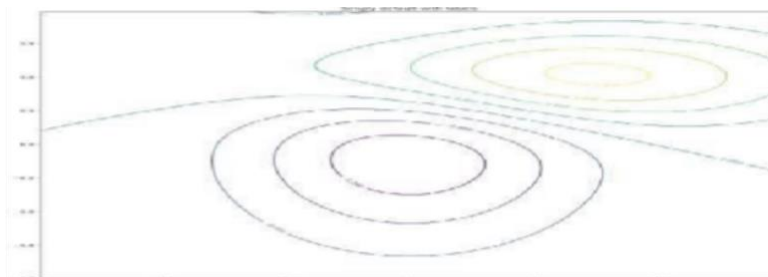
Bootstrap Plot

C:\VKHCG\01-Vermeulen\06-Report\Report_Graph_D.py

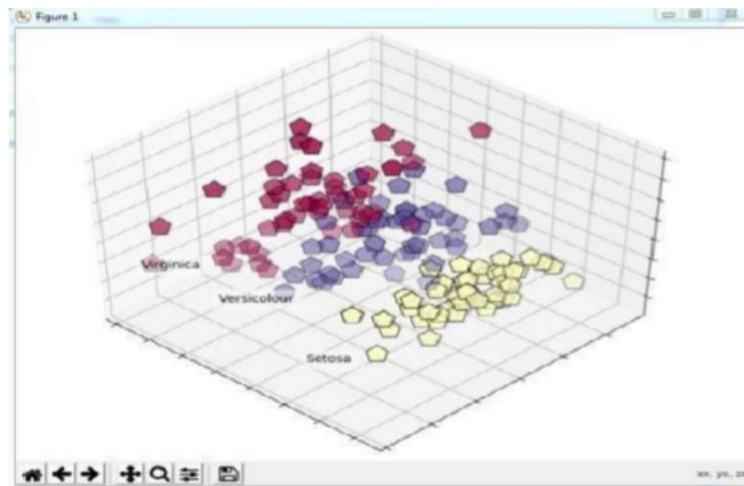


Contour Graphs

C:\VKHCG\01-Vermeulen\06-Report\Report_Graph_G.py



3D Graphs



Practical 10

Data Visualization with Power BI

Case Study : Sales Data

Step 1: Connect to an Excel workbook

1. Launch Power BI Desktop.
2. From the Home ribbon, select **Get Data**. Excel is one of the **Most Common** data connections, so you can select it directly from the **Get Data** menu.

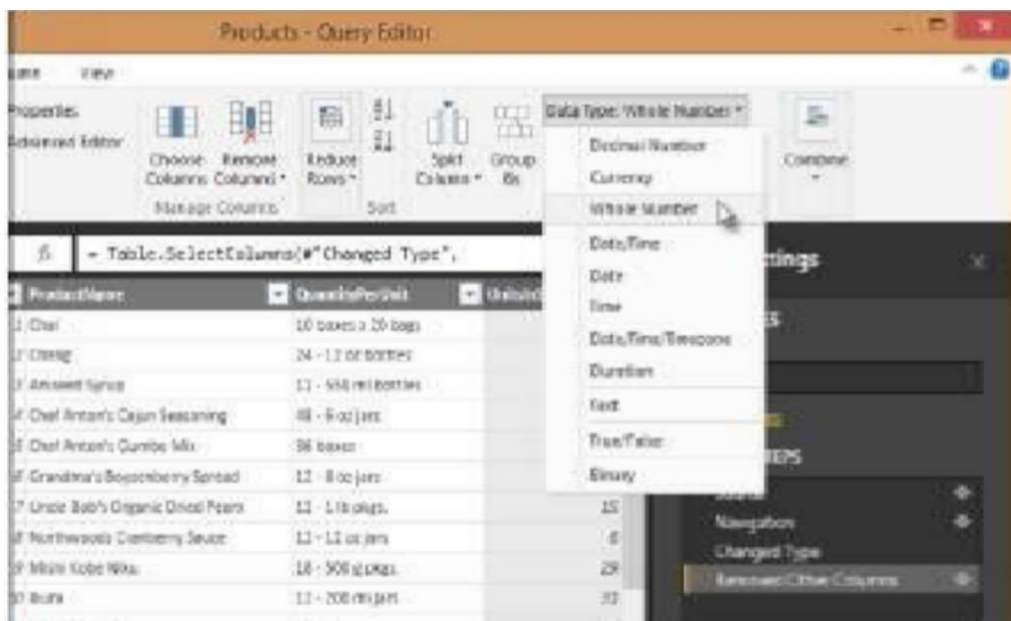
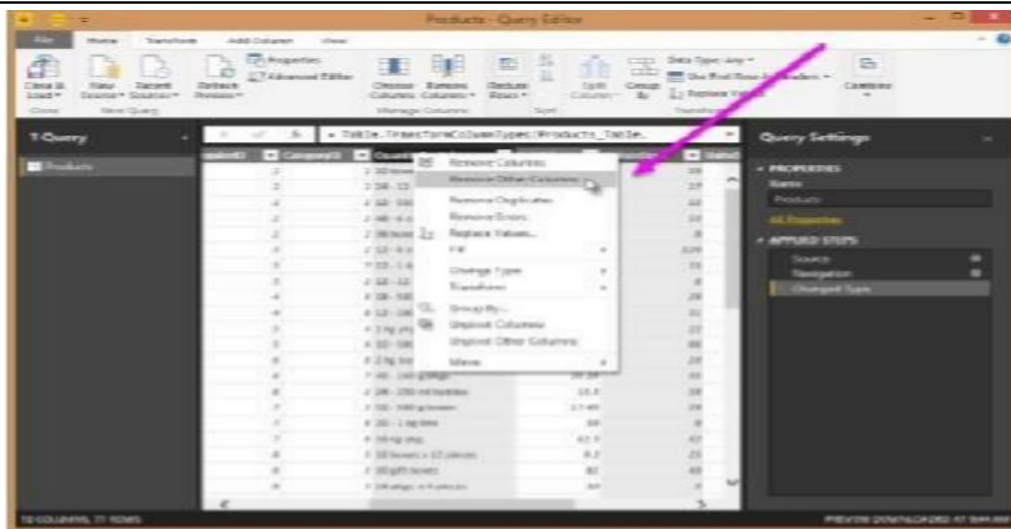


3. If you select the Get Data button directly, you can also select **File > Excel** and select **Connect**.
4. In the **Open File** dialog box, select the **Products.xlsx** file.



You can also open the Query Editor by selecting **Edit Queries** from the Home ribbon in Power BI Desktop. The following steps are performed in Query Editor.

1. In Query Editor, select the **ProductID**, **ProductName**, **QuantityPerUnit**, and **UnitsInStock** columns
(use **Ctrl+Click** to select more than one column, or **Shift+Click** to select columns that are beside each other)
2. Select **Remove Columns - Remove Other Columns** from the ribbon, or right-click on a column header and click **Remove Other Columns**.



Step 3: Change the data type of the UnitsInStock column

For the Excel workbook, products in stock will always be a whole number, so in this step you confirm the UnitsInStock column's datatype is Whole Number.

1. Select the UnitsInStock column.
2. Select the Data Type drop-down button in the Home ribbon.
3. If not already a Whole Number, select Whole Number for data type from the drop down (the Data Type: button also displays the data type for the current selection).

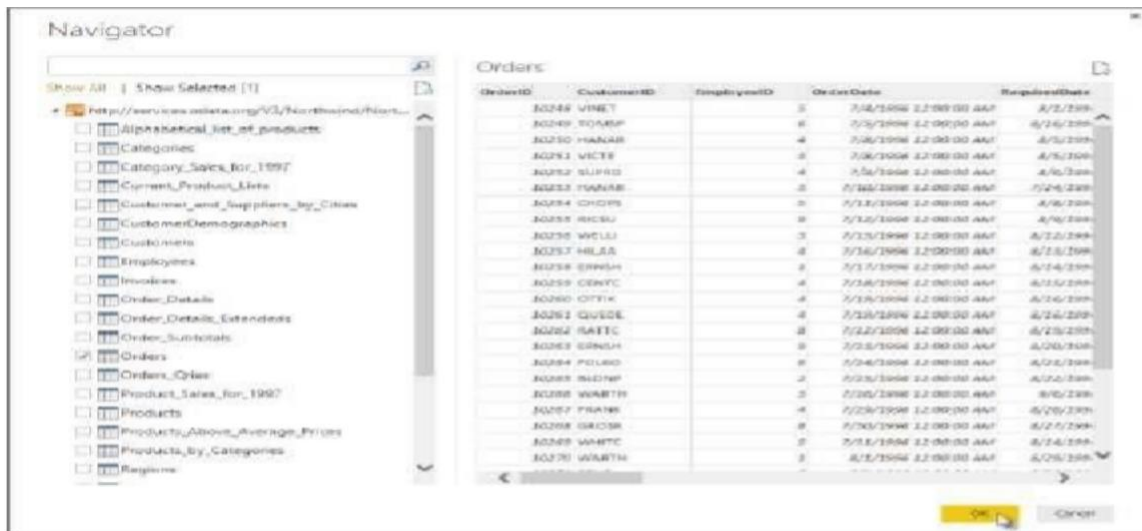
Task 2: Import order data from an OData feed

You import data into Power BI Desktop from the sample Northwind OData feed at the following URL, which you can copy (and then paste) in the steps below:

<http://services.odata.org/V3/Northwind/Northwind.svc/>

Step 1: Connect to an OData feed

1. From the **Home** ribbon tab in Query Editor, select **Get Data**.
2. Browse to the **OData Feed** data source.
3. In the **OData Feed** dialog box, paste the **URL** for the Northwind OData feed.
4. Select **OK**.



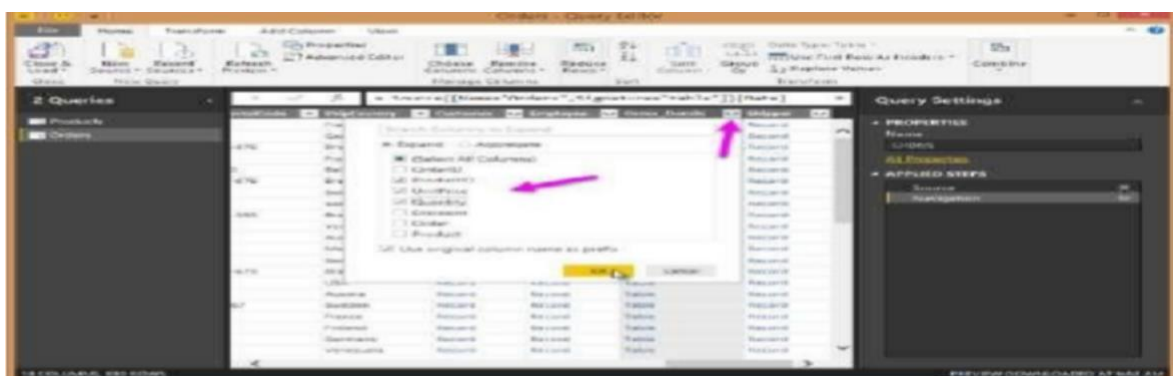
Step 2: Expand the Order_Details table



Expand the **Order_Details** table that is related to the **Orders** table, to combine the **ProductID**, **UnitPrice**, and **Quantity** columns from **Order_Details** into the **Orders** table. The **Expand** operation combines columns from a related table into a subject table. When the query runs, rows from the related table (**Order_Details**) are combined into rows from the subject table (**Orders**).

After you expand the **Order_Details** table, three new columns and additional rows are added to the **Orders** table, one for each row in the nested or related table.

1. In the **Query View**, scroll to the **Order_Details** column.
2. In the **Order_Details** column, select the expand icon ().
3. In the **Expand** drop-down: a. Select (**Select All Columns**) to clear all columns. Select **ProductID**, **UnitPrice**, and **Quantity**. click OK.



Step 3: Remove other columns to only display columns of interest

In this step you remove all columns except **OrderDate**, **ShipCity**, **ShipCountry**, **Order_Details.ProductID**, **Order_Details.UnitPrice**, and **Order_Details.Quantity** columns. In the previous task, you used **Remove Other Columns**. For this task, you remove selected columns.

In the **Query View**, select all columns by completing a. a.

Click the first column (**OrderID**).

b. Shift+Click the last column (**Shipper**).

c. Now that all columns are selected, use Ctrl+Click to unselect the following columns:

OrderDate, ShipCity, ShipCountry, Order_Details.ProductID, Order_Details.UnitPrice, and Order_Details.Quantity.

Now that only the columns we want to remove are selected, right-click on any selected column header and click **Remove Columns**.

Step 4: Calculate the line total for each Order_Details row

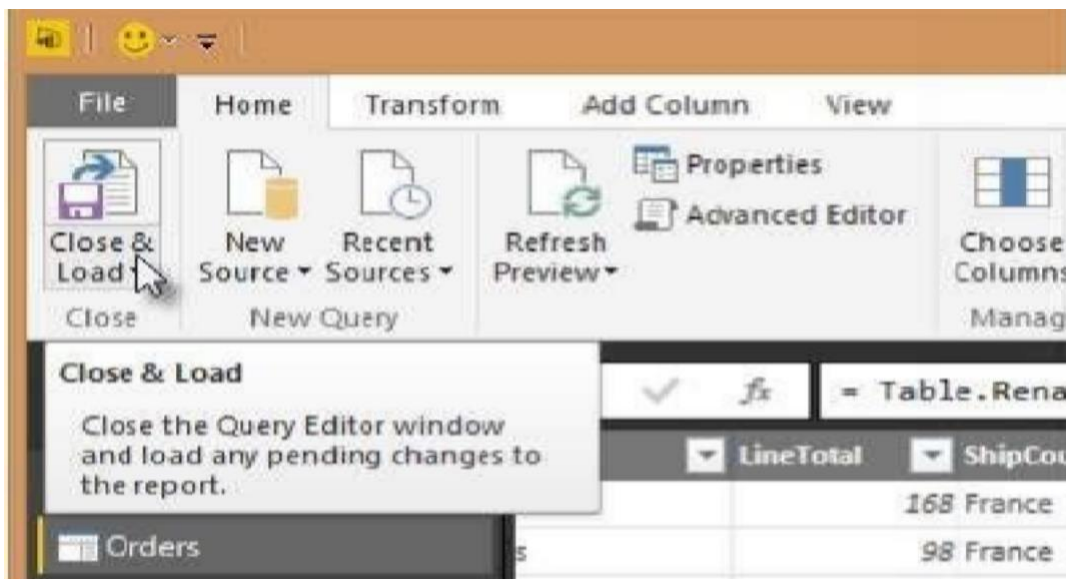
Power BI Desktop lets you to create calculations based on the columns you are importing, so you can enrich the data that you connect to. In this step, you create a **Custom Column** to calculate the line total for each **Order_Details** row.

Calculate the line total for each **Order_Details** row:

1. In the **Add Column** ribbon tab, click **Add Custom Column**.

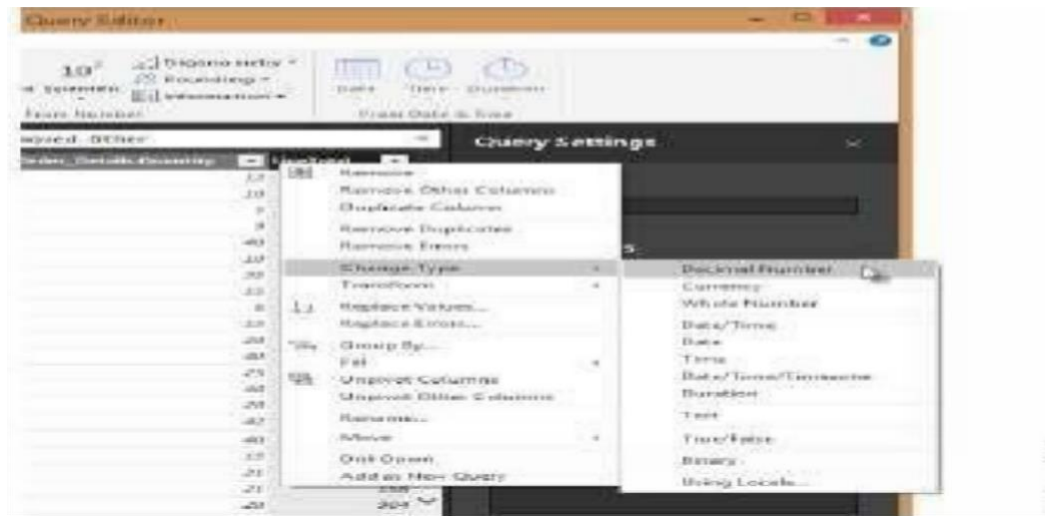
2. In the Add Custom Column dialog box, in the Custom Column Formula textbox, enter **[Order_Details.UnitPrice] * [Order_Details.Quantity]**.

3. In the New column name textbox, enter **LineTotal**.



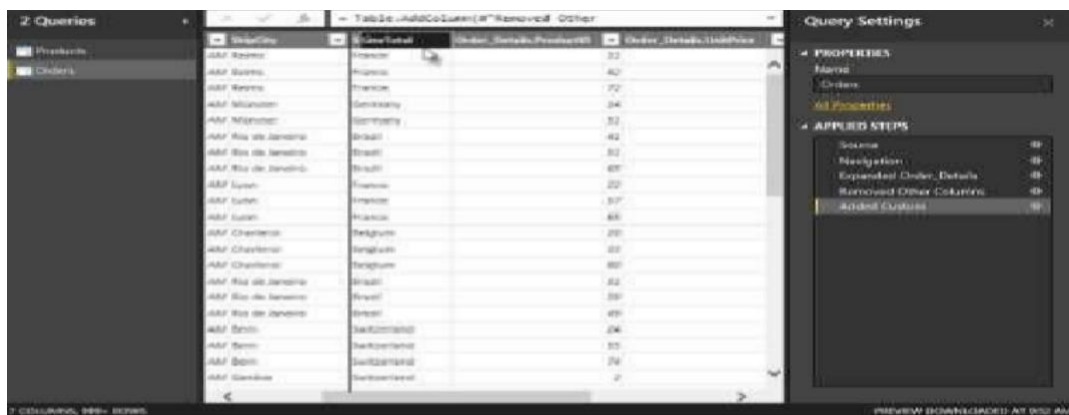
Click **OK**.

- Step 5: Set the datatype of the LineTotal field** 1.
Right click the **LineTotal** column.
2. Select **Change Type** and choose **Decimal Number**.

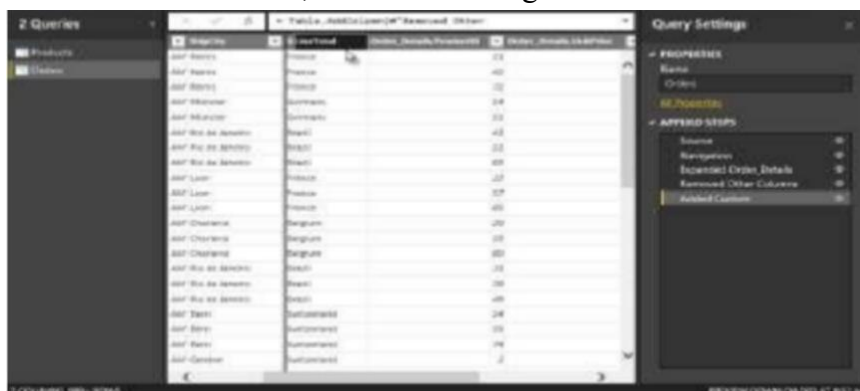


Step 6: Rename and reorder columns in the query

1. In **Query Editor**, drag the **LineTotal** column to the left, after **ShipCountry**.
Remove



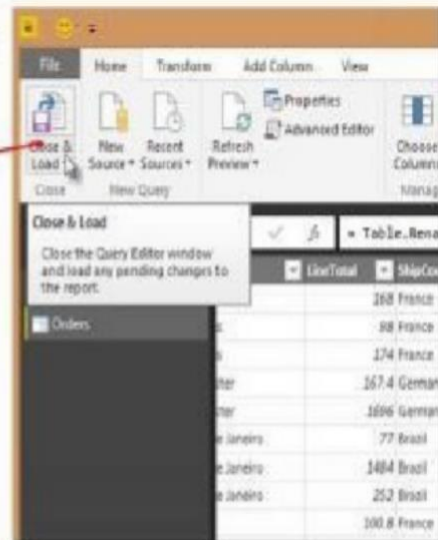
2. Remove the *Order_Details.* prefix from the **Order_Details.ProductID**,
Order_Details.UnitPrice and **Order_Details.Quantity** columns, by double-clicking
on each column header, and then deleting that text from the column name.



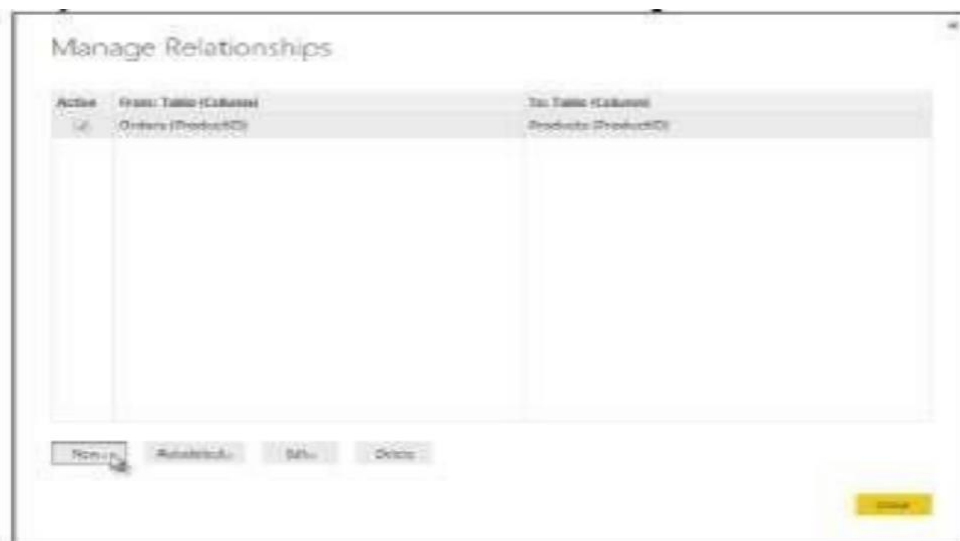
Task 3: Combine the Products and Total Sales queries

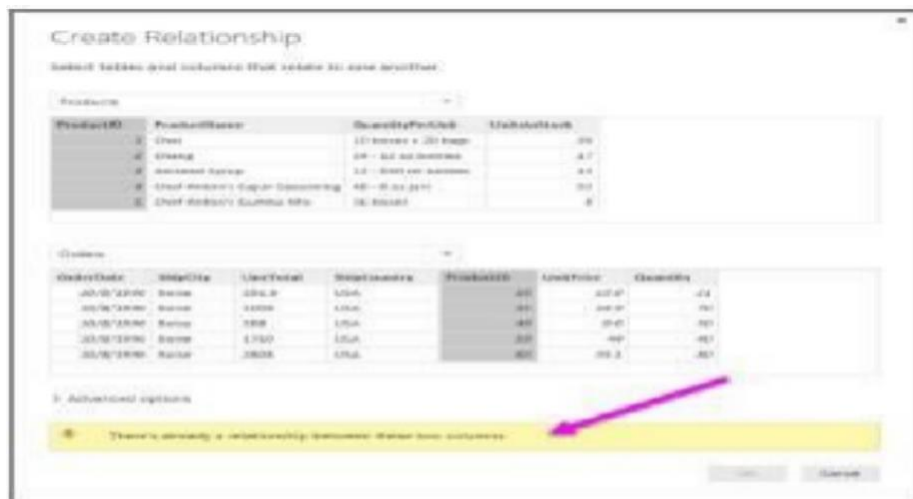
Step 1: Confirm the relationship between Products and Total Sales

1. First, we need to load the model that we created in Query Editor into Power BI Desktop. From the **Home** ribbon of Query Editor, select **Close & Load**.

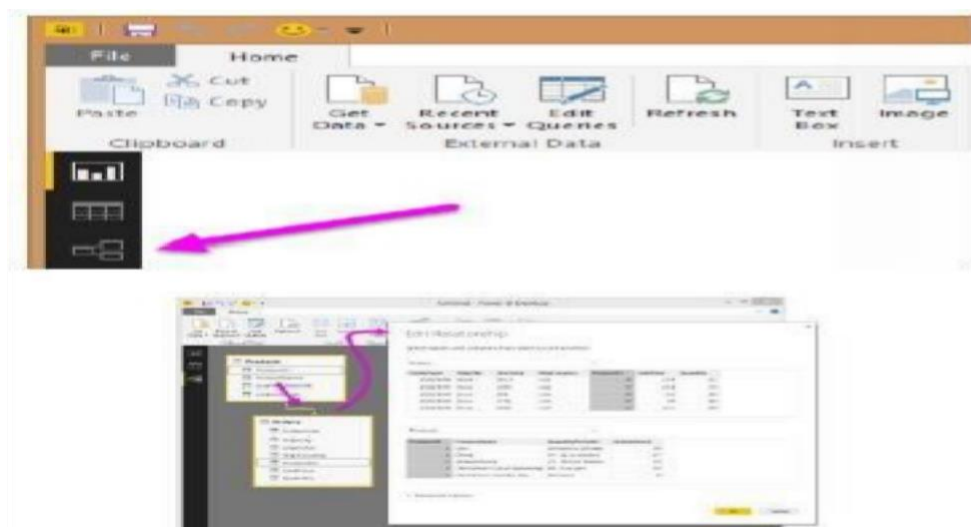


2. Power BI Desktop loads the data from the two queries
3. Once the data is loaded, select the Manage Relationships button Home ribbon
4. Select the New... button
5. When we attempt to create the relationship, we see that one already exists! As shown in the Create Relationship dialog (by the shaded columns), the ProductsID fields in each query already have an established relationship.





6. Select Cancel, and then select Relationship view in Power BI Desktop.

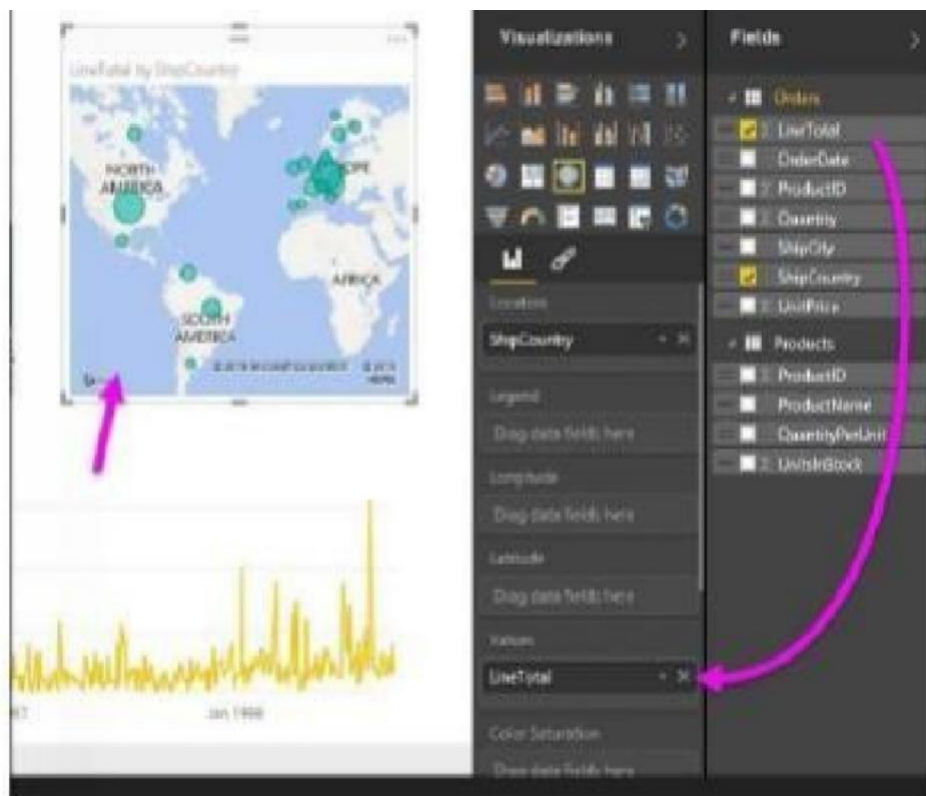


Task 4: Build visuals using your data

Step 1: Create charts showing Units in Stock by Product and Total Sales by Year



3. Next, drag ShipCountry to a space on the canvas in the top right. Because you selected a geographic field, a map was created automatically. Now drag LineTotal to the Values field; the circles on the map for each country are now relative in size to the LineTotal for orders shipped to that country.



Step 2: Interact with your report visuals to analyze further

